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DELAUNAY ASYMPTOTICS FOR THE SINGULAR ESCOBAR–YAMABE EQUATION

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Abstract. In dimensions $n \geq 3$, we prove that near an isolated boundary singularity a positive solution of the Escobar–Yamabe equation for a smooth metric approaches a fixed periodic Delaunay solution or the cylindrical solution if its trace lies between positive multiples of $|\overline{x}|^{-(n-2)/2}$, where $\overline{x}$ is the tangential Fermi coordinate. The relative error decays at a positive power rate, with weighted estimates through two derivatives in Fermi coordinates. The necksize is unique, and the translation parameter is unique modulo the least positive period in the periodic case. The Hamiltonian smoothly parametrizes the periodic family modulo translation. We classify Jacobi fields of polynomial growth at both ends without assuming symmetry. Translation and energy differentiation generate these fields along periodic solutions, and two oscillatory fields generate them along the cylindrical solution. We also exhibit a first-order obstruction to improving these asymptotics by boundary-preserving conformal deformations of the flat models.

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1. Introduction

Let (M, g) be a smooth Riemannian manifold with boundary of dimension $n \geq 3$, and fix $p \in \partial M$. Choose boundary Fermi coordinates $x = (\overline{x}, x_n)$ centered at p, with x_n increasing inward. For sufficiently small $\rho > 0$, these coordinates identify a neighborhood of p with the half-ball $B_\rho^+ = B_\rho(0) \cap \mathbb{R}^n_+$, where $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n : x_n \geq 0\}$ and $B_\rho(0)$ is

the open Euclidean ball of radius ρ. We write $\partial_0 B_\rho^+ = B_\rho^+ \cap \partial\mathbb{R}_+^n$. The coordinate conventions and the expansion of g are given in Section 2.1, following Almaraz [1, Section 2.3] and Marques [17].

For a positive function $U \in C^\infty(B_\rho^+ \setminus \{0\})$, write $u(\overline{x}) = U(\overline{x}, 0)$ for its boundary trace on $\partial_0 B_\rho^+ \setminus \{0\}$. We study the singular Escobar–Yamabe problem

$$\begin{cases} -\Delta_g U + \dfrac{n-2}{4(n-1)} R_g U = 0 & \text{in } \operatorname{int} B_\rho^+, \\ \partial_{\nu_g} U + \dfrac{n-2}{2} h_g U = \dfrac{n-2}{2} u^{n/(n-2)} & \text{on } \partial_0 B_\rho^+ \setminus \{0\}. \end{cases} \tag{1.1}$$

Here R_g is the scalar curvature, ν_g is the outward unit normal, and h_g is the average of the principal curvatures with respect to ν_g. By conformal covariance [9], the metric $U^{4/(n-2)} g$ is scalar flat and has boundary mean curvature 1 on $\partial_0 B_\rho^+ \setminus \{0\}$. We call the singularity at 0 removable if u extends continuously across 0, and nonremovable otherwise.

For the Yamabe equation without boundary, Caffarelli, Gidas and Spruck [4] established radial symmetry for positive singular solutions in punctured Euclidean space and their asymptotic behavior near an isolated singularity. The resulting periodic solutions in Emden–Fowler coordinates are the Fowler solutions. Korevaar, Mazzeo, Pacard and Schoen [14] obtained refined asymptotics using conformal deformations of these solutions. For a smooth metric, Marques [18] proved Fowler asymptotics in dimensions $3 \le n \le 5$, and Xiong and Zhang [22] treated dimension 6. Han, Jiang, Wang, Xiong and Zhang [11] extended the classification of isolated singularities for smooth metrics to dimensions $7 \le n \le 29$.

We refer to (1.1) with the Euclidean metric as the flat Escobar–Yamabe problem. A function U on $\mathbb{R}_+^n \setminus \{0\}$ is *cylindrically symmetric* if $U(\overline{x}, x_n) = U(|\overline{x}|, x_n)$. The extension formulation of Caffarelli and Silvestre [6] relates the flat problem to the critical equation of order $1/2$ on $\mathbb{R}^{n-1}$. Caffarelli, Jin, Sire and Xiong [5, Theorems 1.1 and 1.3] proved the critical two-sided estimates near a nonremovable singularity and cylindrical symmetry for positive global solutions with a nonremovable singularity. DelaTorre and González [8] studied the cylindrical equation and its Hamiltonian. Periodic solutions of the fractional equation were constructed by DelaTorre, del Pino, González and Wei [7].

To describe the flat solutions used below, write $x = r\theta$, where $r = |x| > 0$ and $\theta \in \mathbb{S}_+^{n-1}$, and set $t = -\log r$. The normalized Emden–Fowler transform is $v(t, \theta) = b_n^{-1} r^{(n-2)/2} U(r\theta)$, with b_n and c as in (2.9). For a function v on $\mathbb{R} \times \mathbb{S}_+^{n-1}$, its trace is $\tilde{v}(t, \vartheta) = v(t, (\vartheta, 0))$ for

$\vartheta \in \mathbb{S}^{n-2}$. If U is cylindrically symmetric, v depends only on t and $\beta = \arccos\theta_n$. In this case, we write $\tilde{v}(t) = v(t, L)$, where $L = \pi/2$. The normalized flat equation is

$$\begin{cases} v_{tt} + \Delta_{\mathbb{S}^{n-1}_+} v - \dfrac{(n-2)^2}{4} v = 0 & \text{in } \mathbb{R} \times \operatorname{int} \mathbb{S}^{n-1}_+, \\ \partial_{\nu_{\mathbb{S}^{n-1}_+}} v = c\, \tilde{v}^{n/(n-2)} & \text{on } \mathbb{R} \times \partial\mathbb{S}^{n-1}_+, \end{cases} \tag{1.2}$$

where $\nu_{\mathbb{S}^{n-1}_+}$ is the outward unit conormal to the equator.

A *Delaunay solution* is a positive cylindrically symmetric solution of the flat Escobar–Yamabe problem on $\mathbb{R}^n_+ \setminus \{0\}$ with a nonremovable singularity at 0 whose Emden–Fowler transform is periodic and nonconstant in t. The *necksize* is the minimum of the normalized trace on $\mathbb{R}$. Theorem A of [3] gives, for each $\varepsilon \in (0, 1)$, a unique normalized Delaunay solution v_ε with $\tilde{v}_\varepsilon(0) = \min_{t\in\mathbb{R}} \tilde{v}_\varepsilon(t) = \varepsilon$. We denote its least positive period by T_ε.

A solution of (1.2) is *cylindrical* if it is independent of t. We denote the positive cylindrical solution by $C_n(\theta)$ and set $v_1(t, \theta) = C_n(\theta)$. The boundary trace of C_n is identically 1.

Theorem B of [3] classifies all positive global solutions of the flat Escobar–Yamabe problem with a possible isolated boundary singularity. In the nonremovable case, the normalized transform is either C_n or a translate $v_\varepsilon(t + \tau, \theta)$.

For a smooth background metric in dimensions 3 and 4, Liao and Ma [16, Theorems 1.1 and 1.2] obtained the critical upper and lower bounds near a nonremovable boundary singularity. They also showed that $U(x)$ approaches its average over $\{(\overline{y}, x_n) : |\overline{y}| = |\overline{x}|\}$, with relative error $O(|x|^\alpha)$ for some $\alpha \in (0, 1)$. In their preprint of 5 November 2026, Jin, Liao and Xiong [13, Theorem 1.4] proved that a positive local solution of the flat Escobar–Yamabe problem with a nonremovable boundary singularity approaches a unique global singular solution with a relative error decaying at a positive power rate. Under the critical two-sided bounds on u, we prove convergence to a fixed flat solution in boundary Fermi coordinates for a general smooth metric in dimensions $n \geq 3$. In Emden–Fowler coordinates, the flat solution is either a periodic member of the family or C_n. The estimates include the derivatives of orders 1 and 2.

Main results. For a solution v of (1.2), the Hamiltonian in (2.11) is given by

$$|\mathbb{S}^{n-2}|H(v)(t) = \frac{1}{2}\int_{\mathbb{S}^{n-1}_+}\left(v_t^2 - |\nabla_{\mathbb{S}^{n-1}_+}v|^2 - \frac{(n-2)^2}{4}v^2\right)\,\mathrm{d}\theta + \frac{(n-2)c}{2(n-1)}\int_{\partial\mathbb{S}^{n-1}_+}\tilde{v}^{\frac{2(n-1)}{n-2}}\,\mathrm{d}\sigma.$$

Both integrals are evaluated at t. By (1.2), $H(v)(t)$ is independent of t, and we write $H(v)$ for its value. For the cylindrical solution, $H_c = H(C_n) = -c/(2(n-1))$.

Theorem A. *The map*

$$(0,1) \longrightarrow (H_c, 0), \qquad \varepsilon \longmapsto H(v_\varepsilon)$$

is a smooth diffeomorphism with strictly negative derivative. Two normalized noncylindrical Delaunay solutions have the same Hamiltonian if and only if they differ by a translation in t. The unique normalized singular solution with Hamiltonian H_c is C_n. If $\varepsilon(h)$ is the inverse of this map, then $v_h = v_{\varepsilon(h)}$ has least positive period $T(h) = T_{\varepsilon(h)}$. Let $v_h^{(1)}(s,\theta) = v_h(T(h)s,\theta)$. For every integer $k \geq 0$ and $0 < \gamma < 1$, the map $h \mapsto (T(h), v_h^{(1)})$ is smooth from $(H_c, 0)$ into $(0,\infty) \times C^{k,\gamma}_{1,\mathrm{sym},\mathrm{even}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$.

Idea of the proof. The derivative of $H(v_\varepsilon)$ in ε is the pairing $\Omega(\partial_t v_\varepsilon, \partial_\varepsilon v_\varepsilon)$ in (3.3). Proposition 3.1(iii) of [3] determines an odd cylindrically symmetric Jacobi field from its t-derivative at $(0, L)$. If the pairing were zero, the compatibility condition for a periodic linear problem would hold. A solution of that problem would produce an odd Jacobi field with polynomial growth that changes under translation by T_ε. This contradicts the fact that every such odd field is a multiple of $\partial_t v_\varepsilon$. The limits of $H(v_\varepsilon)$ at $\varepsilon = 0$ and $\varepsilon = 1$ give the sign of its derivative and its range. The proof is given in Section 3.

A Jacobi field along v is a smooth solution Φ of the linearization of (1.2), given in (2.13). The space $\mathcal{K}_{\mathrm{pol}}(v)$ consists of these fields on $\mathbb{R}\times\mathbb{S}^{n-1}_+$ satisfying the polynomial growth condition (2.15) for some $q \geq 0$. The bound is required both as $t \to +\infty$ and as $t \to -\infty$. We write $\mathcal{K}_{\mathrm{bd}}(v)$ for the space of Jacobi fields bounded on $\mathbb{R}\times\mathbb{S}^{n-1}_+$.

Theorem B. *Let $H_c < h < 0$, and set $\Psi_h = \partial_h v_h$, where the derivative is taken at fixed t. Then*

$$\mathcal{K}_{\mathrm{pol}}(v_h) = \mathrm{span}\{\partial_t v_h, \Psi_h\}.$$

For $\Phi = a\partial_t v_h + d\Psi_h$ with $a, d \in \mathbb{R}$, the derivative of the Hamiltonian is $DH_{v_h}[\Phi](t) = d$ for every $t \in \mathbb{R}$. In particular, this derivative vanishes if and only if Φ is a multiple of $\partial_t v_h$.

Idea of the proof. Differentiating tangential translations and the maps obtained by composing a tangential translation with the Kelvin transform on each side gives two positive coefficients z_+ and z_- in the equation for spherical harmonics of degree 1. Their geometric mean satisfies (4.9). Picone's identity (4.10) gives the lower bound (4.8) for degree 1. Since $\lambda_\ell \geq \lambda_1$ for $\ell \geq 1$, the same bound holds in every positive degree. Applying it to cutoffs of a polynomially growing Jacobi field shows that its coefficients of positive degree vanish. The cylindrically symmetric fields are then determined by the parity uniqueness in Proposition 3.1(iii) of [3]. Differentiating $H(v_h) = h$ gives the derivative in the Ψ_h direction. The proof is given in Section 4.1.

Theorem C. *Let $\xi_* > 0$ and let A_{ξ_*} be the positive cylindrically symmetric function determined in (4.19), with $A_{\xi_*}(L) = 1$. Then*

$$\begin{aligned}\mathcal{K}_{\mathrm{pol}}(C_n) &= \mathcal{K}_{\mathrm{bd}}(C_n)\\ &= \operatorname{span}\Big\{A_{\xi_*}(\beta)\cos(\xi_* t),\ A_{\xi_*}(\beta)\sin(\xi_* t)\Big\}.\end{aligned}$$

Idea of the proof. The same estimate in positive spherical harmonic degrees shows that every $\Phi \in \mathcal{K}_{\mathrm{pol}}(C_n)$ is cylindrically symmetric. The equation for A_{ξ_*} gives the cosine and sine fields. Subtracting multiples of them from the even and odd parts of Φ makes the value of the even part and the t-derivative of the odd part vanish at $(0, L)$, respectively. Proposition 3.1(iii) of [3] then implies that both remainders vanish. The proof is given in Section 4.2. At C_n, the derivative of H vanishes, and Proposition 4.8 gives a strictly positive quadratic term on each nonzero field in the theorem.

These classifications identify the directions in which a singular end can be deformed to first order. In the Yamabe problem without boundary, Mazzeo, Pollack and Uhlenbeck [19, Sections IV and V] add the corresponding Jacobi fields on each end to the domain of the linearized operator in weighted spaces. If the global linearized operator has trivial L^2 kernel, the enlarged domain gives surjectivity and permits the use of the implicit function theorem to describe the moduli space locally. Here the two directions along a periodic solution describe changes in its translation and Hamiltonian, and only the translation remains when the Hamiltonian is fixed.

We also consider the self-adjoint operator $\mathcal{A}_v$ on $L^2(\mathbb{R} \times \mathbb{S}^{n-1}_+)$, defined by the interior operator in (2.13) with its boundary condition, and

the self-adjoint operator $\mathcal{J}_v$ on $L^2(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ obtained by subtracting multiplication by $(nc/(n-2))\tilde{v}^{2/(n-2)}$ from the Dirichlet-to-Neumann operator for the same interior equation. For each periodic Delaunay solution and for C_n, both operators have trivial kernel. In both realizations, a kernel element gives a bounded Jacobi field. The classifications make this field periodic, so square integrability forces it to vanish. Cutoffs of a nonzero periodic Jacobi field give a sequence of unit vectors that converges weakly to zero and whose images under $\mathcal{A}_v$ converge strongly to zero. Thus 0 belongs to the essential spectrum of $\mathcal{A}_v$. See Section 4.3 for these statements and the operator domains. For the fractional Yamabe equation on $\mathbb{R}\times\mathbb{S}^{n-1}$, Andrade, DelaTorre, do Ó, Ratzkin and Wei [2, Theorem 3] proved triviality of the L^2 kernel of the linearized operator for orders $s<1$ sufficiently close to 1.

The flat Hamiltonian parametrization and the Jacobi classification in a growth class containing fields of polynomial growth were also obtained by Jin, Liao and Xiong [13, Theorems 1.1 and 1.3 and Proposition A.2]. Our proofs use the flat classification and the parity uniqueness of [3], together with the estimates for the equations in positive spherical harmonic degrees.

For $\varepsilon\in(0,1]$ and $\tau\in\mathbb{R}$, let $U_{\varepsilon,\tau}$ be the flat solution in (5.1). In the normalization of (1.1), it is given by

$$U_{\varepsilon,\tau}(r\theta)=b_n r^{-\frac{n-2}{2}}v_\varepsilon(-\log r+\tau,\theta).$$

Theorem D. *Let $n\geq 3$, and let U be a positive smooth solution of (1.1) on $B^+_\rho\setminus\{0\}$ in boundary Fermi coordinates centered at p. Suppose there are $c_0,C_0>0$ such that, for $0<|\overline{x}|<\rho$,*

$$c_0|\overline{x}|^{-\frac{n-2}{2}}\leq u(\overline{x})\leq C_0|\overline{x}|^{-\frac{n-2}{2}}.$$

Then there exist $\delta>0$, $\rho_0\in(0,\rho)$, $\varepsilon_*\in(0,1]$, $\tau_*\in\mathbb{R}$, and $C>0$ such that, for $x\in B^+_{\rho_0}\setminus\{0\}$,

$$\sum_{j=0}^{2}|x|^{\frac{n-2}{2}+j}|D^j(U-U_{\varepsilon_*,\tau_*})(x)|\leq C|x|^\delta.$$

Here D^j denotes derivatives in the fixed Fermi chart. In particular, $U(x)=(1+O(|x|^\delta))U_{\varepsilon_*,\tau_*}(x)$ as $x\to 0$. The necksize ε_* is unique. If $\varepsilon_*<1$, τ_* is unique modulo T_{ε_*}. If $\varepsilon_*=1$, the flat solution is $b_nr^{-(n-2)/2}C_n(\theta)$ for every τ_*, and its normalized transform is independent of t.

Idea of the proof. The bounds on the trace give uniform positive upper and lower bounds and derivative estimates for the rescalings of U on fixed half-annuli. The equation for its transform w in (2.20) differs

from (1.2) by terms of order e^{-t}. The same bound for the derivative of the Hamiltonian gives convergence of the Pohozaev integral as the radius tends to zero. For any sequence $t_j \to \infty$, the translated functions $w(t_j + \cdot, \cdot)$ have a subsequence converging on compact subsets to a positive global solution of (1.2). Theorem B of [3] identifies that solution as C_n or a translate of a Delaunay solution. The Pohozaev invariant gives the same Hamiltonian for all such subsequential limits, and Theorem A determines the same necksize.

If $\varepsilon_* < 1$, an orthogonality condition on a fixed interval of the cylinder selects the translation there. If the error failed to decrease uniformly on successive intervals, its normalization would give a nonzero Jacobi field of polynomial growth with zero Hamiltonian derivative and orthogonal to $\partial_t v_{\varepsilon_*}$. Theorem B excludes such a field. The resulting exponential estimate gives $|a'(t) - 1| = O(e^{-\delta t})$ for some $\delta > 0$, where $a(t)$ is the translation selected on the interval centered at t. Consequently, $a(t) - t \to \tau_*$. When $\varepsilon_* = 1$, the positive quadratic term of the Hamiltonian on the space in Theorem C gives the corresponding decay estimate for $w - C_n$. Returning to the Fermi chart gives the estimates through order 2. The proof is given in Section 5.1.

Structure of the paper. Section 2 fixes the geometric and cylindrical conventions, establishes the local estimates, and proves the existence of the Pohozaev invariant. Section 3 classifies the cylindrically symmetric Jacobi fields with polynomial growth and proves the Hamiltonian parametrization. Section 4 classifies the Jacobi fields without a symmetry assumption and studies the two self-adjoint operators on the cylinder and its boundary. Section 5 proves the asymptotic theorem and studies the effect of boundary-preserving conformal deformations of the flat solutions on the error in Fermi coordinates. Appendix A gives the computations used in these proofs.

Relation to concurrent work. While this manuscript was being completed, Jin, Liao and Xiong submitted their preprint [13] to arXiv on November 5, 2026. Their Theorem 1.1 and Proposition A.2 give the Hamiltonian parametrization of the flat Delaunay family and its real-analytic relation to the necksize. Their Theorem 1.3 gives the Jacobi field classifications in Theorems B and C for a class that also allows sufficiently slow exponential growth at both ends. Their Theorem 1.4 proves convergence to a unique global singular solution in a Euclidean half-ball, with a relative error that decays at a positive power rate.

In the flat problem, we use the classification and parity uniqueness of [3]. For the positive spherical harmonic degrees, we use the geometric mean of the two positive degree-1 coefficients in Picone's identity to

obtain the coercive estimate in Proposition 4.2. Section 4.3 proves that both L^2 kernels are trivial and that zero belongs to the essential spectrum of the linearized operator on the cylinder.

In the periodic case, both asymptotic arguments use orthogonality to the translation field and the first variation of the Hamiltonian. Under the critical two-sided bounds on the trace, Theorem D extends convergence to a fixed flat solution to smooth background metrics, with weighted estimates through two derivatives in boundary Fermi coordinates. The limit of the Pohozaev integral determines the necksize. Proposition 5.3 shows that the boundary-preserving conformal deformations considered here can leave a first-order error even for a conformally flat background metric.

Statement on the use of AI. The need for Proposition 3.1 of [3] was identified before we used AI. In the mathematical work, we used the model GPT-Sol 6 only to assist with the calculation in that proposition. The proposition is also used in the calculations in the present article. We also used the model to revise the manuscript. All mathematical statements, proofs, and citations were independently checked by the author, who takes full responsibility for the contents of the article. The final manuscript was written and revised by the author.

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2. Preliminaries

We first fix the boundary Fermi coordinates at p for the Escobar–Yamabe equation (2.4). We then pass from the flat Escobar–Yamabe equation (2.5) to its normalized cylindrical form (2.10), recall the results on Delaunay solutions from the companion paper, and give estimates for the linearized equation. In Section 2.5, we establish local estimates for (2.4) and prove the existence of the limiting Pohozaev invariant in (2.30).

2.1. **Notation.** We denote by $\bar{g}$ the metric induced by g on ∂M and by ν_g the outward unit normal to ∂M. Let ∇^g, Δ_g, and R_g denote the Levi-Civita connection, the Laplace–Beltrami operator, and the scalar curvature of g, respectively. For vectors X, Y tangent to ∂M, we set

$$\mathrm{II}_g(X,Y) = g(\nabla^g_X \nu_g, Y), \qquad h_g = \frac{1}{n-1}\,\mathrm{tr}_{\bar{g}}\,\mathrm{II}_g.$$

We use the half-space notation of Almaraz [1, Section 2.1]. Write $x = (\overline{x}, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and let $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n : x_n \geq 0\}$. For $r > 0$, let $B_r(0)$ be the open Euclidean ball of radius r centered at 0 and set $B_r^+ = B_r(0) \cap \mathbb{R}^n_+$. Its flat and spherical faces are denoted by $\partial_0 B_r^+ = B_r^+ \cap \partial\mathbb{R}^n_+$ and $\partial^+ B_r^+ = \partial B_r(0) \cap \mathbb{R}^n_+$, respectively. We write $\mathbb{S}^{n-1}_+ = \mathbb{S}^{n-1} \cap \mathbb{R}^n_+$, so $\partial^+ B_r^+ = r\mathbb{S}^{n-1}_+$. The standard Euclidean metric is denoted by δ, and its Laplacian by $\Delta = \Delta_\delta = \sum_{i=1}^n \partial^2_{x_i}$.

Choose a $\bar{g}$-orthonormal basis $E_1, \dots, E_{n-1}$ of $T_p\partial M$. For $\overline{x}$ near 0, set $q(\overline{x}) = \exp^{\bar{g}}_p(\sum_{a=1}^{n-1} x_a E_a)$. The boundary Fermi map centered at p is

$$\Psi(\overline{x}, x_n) = \exp^g_{q(\overline{x})}\Big(-x_n \nu_g(q(\overline{x}))\Big), \qquad x_n \geq 0. \tag{2.1}$$

See Almaraz [1, Section 2.3] and Marques [17, Section 3]. For sufficiently small $\rho > 0$, the map Ψ identifies B^+_ρ with a neighborhood of p in M and sends 0 to p. We also write g for the pullback $\Psi^* g$ on B^+_ρ, with coefficients

$$g_{ij}(x) = g_{\Psi(x)}(d\Psi_x e_i, d\Psi_x e_j).$$

The Gauss lemma gives

$$g_{nn} = 1, \qquad g_{an} = 0 \tag{2.2}$$

for $1 \leq a \leq n-1$. The coordinate frame at 0 is orthonormal, so $g_{ij}(0) = \delta_{ij}$. Since x_n increases inward, the outward unit normal on the flat face is $-\partial_{x_n}$. In Theorem D, $|x|$ and $|\overline{x}|$ are Euclidean norms, and D^j denotes derivatives of order j with respect to the Fermi coordinates x.

The tangential derivatives of g_{ab} vanish at 0 because $\overline{x}$ consists of geodesic normal coordinates for $\bar{g}$. With the outward normal convention above, $\partial_{x_n} g_{ab}(0) = -2\mathrm{II}_{ab}(p)$, where $\mathrm{II}_{ab}(p) = \mathrm{II}_g(E_a, E_b)$. Thus

$$g_{ab}(x) = \delta_{ab} - 2\mathrm{II}_{ab}(p)x_n + O(|x|^2), \tag{2.3}$$

for $1 \leq a, b \leq n-1$. After writing $x = rz$ on a fixed half-annulus, the remainder is $O(r^2)$ in every fixed C^k norm in z.

The boundary is *umbilic at* p if $\mathrm{II}_g(p) = h_g(p)\bar{g}_p$. In that case,

$$g_{ab}(x) = (1 - 2h_g(p)x_n)\delta_{ab} + O(|x|^2).$$

Definition 2.1. A smooth metric g is *conformally flat* if every point has a neighborhood on which $g = \varphi^{4/(n-2)} g_0$ for some positive smooth function φ and some flat metric g_0.

The conformal Laplacian and the boundary operator are

$$L_g = -\Delta_g + \frac{n-2}{4(n-1)} R_g, \qquad B_g = \partial_{\nu_g} + \frac{n-2}{2} h_g.$$

Let U be a positive smooth function on $B^+_\rho \setminus \{0\}$, and write $u(\overline{x}) = U(\overline{x}, 0)$ for its trace on $\partial_0 B^+_\rho \setminus \{0\}$. For $\kappa > 0$, the singular Escobar–Yamabe equation in this chart is

$$\begin{cases} L_g U = 0 & \text{in } \operatorname{int}(B^+_\rho), \\ B_g U = \kappa u^{n/(n-2)} & \text{on } \partial_0 B^+_\rho \setminus \{0\}. \end{cases} \tag{2.4}$$

By conformal covariance [9], the metric $U^{4/(n-2)} g$ is scalar flat and has constant boundary mean curvature $2\kappa/(n-2)$ on $\partial_0 B^+_\rho \setminus \{0\}$. Multiplying U by $(2\kappa/(n-2))^{(n-2)/2}$ changes the boundary coefficient to $(n-2)/2$ and gives (1.1), with boundary mean curvature one. We call the singularity of the trace removable if u extends continuously to $\overline{x} = 0$.

For a nonnegative integer k and $0 < \gamma < 1$, we use $C^k(M)$ for the space of functions with continuous derivatives through order k and $C^{k,\gamma}(M)$ when the derivatives of order k are also γ-Hölder continuous. We write $C^\infty(M)$ for smooth functions and $C^\infty_c(M)$ for smooth functions of compact support. The subscript loc denotes regularity on compact subsets. On a closed half-annulus or cylinder, the indicated norms include the boundary. We use $L^q(M)$ for the Lebesgue spaces and $W^{k,q}(M)$ for the Sobolev spaces with weak derivatives through order k in $L^q(M)$. The notation $H^s(M)$ denotes the $L^2(M)$-based Sobolev space of order s. In particular, $H^{-1/2}(M)$ is the dual of $H^{1/2}(M)$ under the $L^2(M)$ pairing. The spaces are real unless we take their complexifications for Fourier analysis. The periodic, uniformly local, and weighted spaces on the cylinder are specified in Section 2.3.

2.2. **The flat equation and the cylindrical normalization.** We write the flat Escobar–Yamabe equation in Emden–Fowler coordinates and describe the action of dilations and Kelvin inversion. We then normalize the cylindrical solution to have trace one and use the same normalization to define the Hamiltonian.

We use the flat Escobar–Yamabe equation in the normalization of the companion paper [3],

$$\begin{cases} \Delta U = 0 & \text{for } x_n > 0, \\ \partial_{x_n} U + \dfrac{n-2}{2} u^{n/(n-2)} = 0 & \text{on } \partial\mathbb{R}^n_+ \setminus \{0\}, \end{cases} \tag{2.5}$$

where $u(\overline{x}) = U(\overline{x}, 0)$. A classical global solution of (2.5) is a positive function $U \in C^\infty(\mathbb{R}^n_+ \setminus \{0\})$ satisfying both equations. This smoothness includes $\partial\mathbb{R}^n_+ \setminus \{0\}$. We impose no growth bound on U as $|x| \to \infty$ and assume cylindrical symmetry only where stated. The metric $U^{4/(n-2)}\delta$ is scalar flat and has mean curvature one on $\partial\mathbb{R}^n_+ \setminus \{0\}$.

Write $x = r\theta$, with $r = |x| > 0$ and $\theta \in \mathbb{S}^{n-1}_+$, and set $t = -\log r$. We equip $\mathbb{R} \times \mathbb{S}^{n-1}_+$ with the product metric and volume element $\mathrm{d}\mu = \mathrm{d}t\ \mathrm{d}\theta$, where $\mathrm{d}\theta$ is the round measure on $\mathbb{S}^{n-1}_+$. We write $\mathrm{d}\sigma$ for the round measure on $\partial\mathbb{S}^{n-1}_+ = \mathbb{S}^{n-2}$. For a smooth function f on $\mathbb{R} \times \mathbb{S}^{n-1}_+$, we denote its trace on $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$ by $\tilde{f}$. Thus $\tilde{f}(t, \vartheta) = f(t, (\vartheta, 0))$ for $\vartheta \in \mathbb{S}^{n-2}$.

Let $\beta = \arccos \theta_n$ and $L = \pi/2$. We write

$$\theta = (\sin\beta\, \vartheta, \cos\beta), \qquad 0 \le \beta \le L, \qquad \vartheta \in \mathbb{S}^{n-2}.$$

The outward unit conormal to $\partial\mathbb{S}^{n-1}_+$ is $\nu_{\mathbb{S}^{n-1}_+}$, with $\partial_{\nu_{\mathbb{S}^{n-1}_+}} = \partial_\beta$ at $\beta = L$. We define the inward unit conormal $\eta_{\mathbb{S}^{n-1}_+}$ as $-\nu_{\mathbb{S}^{n-1}_+}$. For $0 < \beta < L$, the round Laplacian is

$$\Delta_{\mathbb{S}^{n-1}_+} = \partial_{\beta\beta} + (n-2)\cot\beta\, \partial_\beta + \sin^{-2}\beta\, \Delta_{\mathbb{S}^{n-2}}.$$

The round measure is $\mathrm{d}\theta = \omega(\beta)\ \mathrm{d}\beta\ \mathrm{d}\sigma(\vartheta)$, where $\omega(\beta) = \sin^{n-2}\beta$.

For a solution U of (2.5), set $V(t, \theta) = r^{(n-2)/2} U(r\theta)$ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$. The Euclidean polar formula gives the interior equation for V. On $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$, the outward conormal derivative of V is $-r^{n/2}\partial_{x_n} U(r\vartheta, 0)$, and $\tilde{V}(t, \vartheta) = r^{(n-2)/2} u(r\vartheta)$. Hence (2.5) becomes

$$\begin{cases} V_{tt} + \Delta_{\mathbb{S}^{n-1}_+} V - \dfrac{(n-2)^2}{4} V = 0 & \text{in } \mathbb{R} \times \operatorname{int} \mathbb{S}^{n-1}_+, \\ \partial_{\nu_{\mathbb{S}^{n-1}_+}} V = \dfrac{n-2}{2} \tilde{V}^{n/(n-2)} & \text{on } \mathbb{R} \times \partial\mathbb{S}^{n-1}_+. \end{cases} \tag{2.6}$$

The dilation $U_\lambda(x) = \lambda^{(n-2)/2} U(\lambda x)$, for $\lambda > 0$, corresponds to

$$V_\lambda(t, \theta) = V(t - \log\lambda, \theta). \tag{2.7}$$

For $y = (\overline{y}, 0)$ and $\rho > 0$, the Kelvin transform is

$$U_{y,\rho}(x) = \left(\frac{\rho}{|x-y|}\right)^{n-2} U\left(y + \frac{\rho^2 (x-y)}{|x-y|^2}\right). \tag{2.8}$$

It preserves (2.5) on the inverted domain. When $y = 0$, its action is $t \mapsto -t - 2\log\rho$.

Following Caffarelli, Jin, Sire and Xiong [5, Theorem 1.3], we call U *cylindrically symmetric* if $U(\overline{x}, x_n) = U(|\overline{x}|, x_n)$. In this case the Emden–Fowler transform is independent of ϑ and is given by

$$V(t,\beta) = e^{-\frac{n-2}{2}t} U(e^{-t}\sin\beta, e^{-t}\cos\beta).$$

Its trace is $\tilde{V}(t) = V(t, L)$. The angular part of the equation reduces to $\omega^{-1}(\omega V_\beta)_\beta$ for $0 < \beta < L$. Smoothness in spherical charts and evenness along meridians give $V_\beta(t, 0) = 0$.

To normalize (2.6), let C_n be the solution of

$$\begin{cases} \left(-\Delta_{\mathbb{S}^{n-1}_+} + \dfrac{(n-2)^2}{4}\right) C_n = 0 & \text{in } \operatorname{int}\mathbb{S}^{n-1}_+, \\ C_n = 1 & \text{on } \partial\mathbb{S}^{n-1}_+. \end{cases}$$

The Dirichlet problem has a unique smooth solution. The maximum principle gives $C_n > 0$, and uniqueness gives cylindrical symmetry. Thus $C_n = C_n(\beta)$, with $C_n'(0) = 0$ and $C_n(L) = 1$. Following [3, Section 3.1], we set

$$c = C_n'(L) = 2\left(\frac{\Gamma(n/4)}{\Gamma((n-2)/4)}\right)^2, \qquad b_n = \left(\frac{2c}{n-2}\right)^{\frac{n-2}{2}}. \tag{2.9}$$

Then $v = b_n^{-1} V$ satisfies

$$\begin{cases} v_{tt} + \Delta_{\mathbb{S}^{n-1}_+} v - \dfrac{(n-2)^2}{4} v = 0 & \text{in } \mathbb{R}\times\operatorname{int}\mathbb{S}^{n-1}_+, \\ \partial_{\nu_{\mathbb{S}^{n-1}_+}} v = c\tilde{v}^{n/(n-2)} & \text{on } \mathbb{R}\times\partial\mathbb{S}^{n-1}_+. \end{cases} \tag{2.10}$$

A solution is *cylindrical* if it is independent of t and *noncylindrical* otherwise. By [3, Theorem B], C_n is the unique positive smooth cylindrical solution of (2.10). The corresponding solution of (2.5) is $b_n r^{-(n-2)/2} C_n(\theta)$.

For a cylindrically symmetric v, we write $\tilde{v}(t) = v(t, L)$. A two-sided bound on the trace means that $m \le \tilde{v}(t) \le M$ for all $t \in \mathbb{R}$, for some $M, m > 0$. If v solves (2.10), then $\widehat{v} = c^{(n-2)/2} v$ satisfies the same interior equation with boundary coefficient one.

Let $I \subset \mathbb{R}$ be an open interval and let $v \in C^\infty(I \times \mathbb{S}^{n-1}_+)$ satisfy $\tilde{v} > 0$ on $I \times \partial\mathbb{S}^{n-1}_+$. For $a > 0$ and $t \in I$, define the Hamiltonian per unit

area of $\mathbb{S}^{n-2}$ by

$$
\begin{aligned}
|\mathbb{S}^{n-2}| H_a(v)(t) = \frac{1}{2}\int_{\mathbb{S}^{n-1}_+} \left(v_t^2 - |\nabla_{\mathbb{S}^{n-1}_+} v|^2 - \frac{(n-2)^2}{4} v^2 \right) \mathrm{d}\theta \\
+ \frac{(n-2)a}{2(n-1)} \int_{\partial \mathbb{S}^{n-1}_+} \tilde{v}^{\frac{2(n-1)}{n-2}} \, \mathrm{d}\sigma.
\end{aligned} \tag{2.11}
$$

The integrands are evaluated at t. Suppose v solves (2.10) with boundary coefficient a in place of c. Differentiating in t and integrating by parts on $\mathbb{S}^{n-1}_+$ give

$$
\begin{aligned}
|\mathbb{S}^{n-2}| \frac{\mathrm{d}}{\mathrm{d}t} H_a(v)(t) = \int_{\mathbb{S}^{n-1}_+} v_t \left(v_{tt} + \Delta_{\mathbb{S}^{n-1}_+} v - \frac{(n-2)^2}{4} v \right) \mathrm{d}\theta \\
+ \int_{\partial \mathbb{S}^{n-1}_+} \partial_t \tilde{v} \left(a \tilde{v}^{n/(n-2)} - \partial_{\nu_{\mathbb{S}^{n-1}_+}} v \right) \mathrm{d}\sigma.
\end{aligned}
$$

Both integrals vanish by the equation for v. Thus $H_a(v)(t)$ is independent of t, and we write $H_a(v)$ for its value. When $a = c$, we abbreviate the Hamiltonian by $H(v)(t)$ and its constant value along a solution by $H(v)$. For a cylindrically symmetric solution of (2.10), integration in ϑ gives

$$
\begin{aligned}
H(v) = \frac{1}{2}\int_0^L \omega(\beta) \left(v_t^2 - v_\beta^2 - \frac{(n-2)^2}{4} v^2 \right) \mathrm{d}\beta \\
+ \frac{(n-2)c}{2(n-1)} \tilde{v}(t)^{\frac{2(n-1)}{n-2}}.
\end{aligned} \tag{2.12}
$$

Multiplying the angular equation for C_n by C_n and integrating by parts gives

$$
\int_0^L \omega(\beta) \left((C_n')^2 + \frac{(n-2)^2}{4} C_n^2 \right) \mathrm{d}\beta = [\omega C_n C_n']_0^L = c.
$$

It follows that $H(C_n) = -c/(2(n-1))$. We denote this value by H_c.

The identity $((n-2)/2) b_n^{2/(n-2)} = c$ gives

$$
H_{(n-2)/2}(b_n v)(t) = b_n^2 H(v).
$$

Section 3.2 uses this relation to classify the dilation orbits of singular solutions of (2.5). The relation between the Hamiltonian and the Pohozaev integral for the geometric equation is given in Section 2.5.

2.3. **Linearized spaces and local estimates.** We give local estimates for the normalized equation and its linearization. We then compare the growth conditions used to classify Jacobi fields and prove the periodic Robin solvability result used in Section 3.1.

Let $k \geq 0$ be an integer and let $0 < \gamma < 1$. The space $C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ consists of functions with $C^{k,\gamma}$ regularity on $I \times \mathbb{S}^{n-1}_+$ for every compact interval $I \subset \mathbb{R}$. Near the equator, write $s = L - \beta$. Then

$$\theta = (\cos s\, \vartheta, \sin s), \qquad \vartheta \in \mathbb{S}^{n-2}, \quad s \geq 0.$$

Thus $s = 0$ on the equator, and s measures distance from the equator along the inward meridian. A local chart on the cylinder uses t, local coordinates for ϑ on $\mathbb{S}^{n-2}$, and s. In these coordinates, regularity up to the equator means that all derivatives through order k extend continuously to $s = 0$ and that the derivatives of order k are Hölder continuous with exponent γ up to and including $s = 0$. We write $C^{\infty}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ for the space of functions that are smooth in these charts on every product $I \times \mathbb{S}^{n-1}_+$ with $I \subset \mathbb{R}$ compact.

The space $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ consists of functions in $C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ with finite norm

$$\|z\|_{C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)} = \sup_{\tau\in\mathbb{R}} \|z\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)}.$$

The boundary space $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R} \times \partial\mathbb{S}^{n-1}_+)$ is defined using $[\tau - 1, \tau + 1] \times \partial\mathbb{S}^{n-1}_+$.

Let v be a smooth solution of (2.10). If $0 < c_0 \leq v \leq C_0$ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$, then

$$\|v\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \leq C(n, c_0, C_0, k, \gamma) \qquad \text{for all } \tau \in \mathbb{R}.$$

The estimates follow from the local Neumann and Schauder estimates on nested intervals inside $[\tau-2, \tau+2]$, as in [3, Proposition 2.1]. Translation in t makes the constants independent of τ. A diagonal argument gives a subsequence converging in $C^{\infty}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ for any sequence of solutions with the same bounds.

Under the bound $0 \leq v \leq C_0$, bounded Neumann data give a uniform local Hölder estimate for v. Since $s \mapsto cs^{n/(n-2)}$ is Lipschitz on $[0, C_0]$, the Neumann data are also uniformly Hölder continuous. The Neumann Schauder estimates then give a uniform bound in $C^{1,\delta_0}([\tau - 1, \tau + 1] \times \mathbb{S}^{n-1}_+)$ for some $0 < \delta_0 < 1$. We use this bound in Section 3.2 when the necksize tends to zero.

For a positive smooth solution v of (2.10), a smooth function Φ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ is a *Jacobi field* along v if

$$\begin{aligned} \mathcal{A}\Phi &= -\Phi_{tt} - \Delta_{\mathbb{S}^{n-1}_+}\Phi + \frac{(n-2)^2}{4}\Phi = 0 \quad \text{in } \mathbb{R} \times \operatorname{int}\mathbb{S}^{n-1}_+, \\ \mathcal{B}_v\Phi &= \partial_{\nu_{\mathbb{S}^{n-1}_+}}\Phi - \frac{nc}{n-2}\tilde{v}^{2/(n-2)}\tilde{\Phi} = 0 \qquad \text{on } \mathbb{R} \times \partial\mathbb{S}^{n-1}_+. \end{aligned} \tag{2.13}$$

Smoothness at the pole is understood in spherical charts.

For $q \geq 0$, set $\langle t \rangle = (1+t^2)^{1/2}$. The space $C^{k,\gamma}_{\mathrm{pol},q}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ consists of functions in $C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ with finite norm

$$\|z\|_{C^{k,\gamma}_{\mathrm{pol},q}(\mathbb{R}\times\mathbb{S}^{n-1}_+)} = \sup_{\tau\in\mathbb{R}} \langle \tau \rangle^{-q} \|z\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)}. \tag{2.14}$$

We define $C^{k,\gamma}_{\mathrm{pol},q}(\mathbb{R} \times \partial\mathbb{S}^{n-1}_+)$ using $[\tau-1,\tau+1] \times \partial\mathbb{S}^{n-1}_+$ in the norm. For $q = 0$ the corresponding spaces are the uniformly local ones.

A Jacobi field has polynomial growth of order at most q if there is a constant $C > 0$ such that

$$|\Phi(t,\theta)| \leq C\langle t \rangle^q \qquad \text{for all } (t,\theta) \in \mathbb{R} \times \mathbb{S}^{n-1}_+. \tag{2.15}$$

The following lemma allows us to use the same growth class in Section 3.1 and Section 4.

Lemma 2.2. *Let v be a positive smooth periodic solution of (2.10), and let Φ be a Jacobi field along v. For each $q \geq 0$, the following are equivalent.*

(i) *The pointwise bound (2.15) holds.*

(ii) *There is a constant $C > 0$ such that*

$$\|\Phi\|_{L^2([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)} \leq C\langle \tau \rangle^q \qquad \textit{for all } \tau \in \mathbb{R}.$$

(iii) *$\Phi \in C^{k,\gamma}_{\mathrm{pol},q}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ for every integer $k \geq 0$ and $0 < \gamma < 1$.*

Proof. On $[\tau-1,\tau+1]$, we have $\langle t \rangle \leq 2\langle \tau \rangle$. The measure of $[\tau-1,\tau+1] \times \mathbb{S}^{n-1}_+$ is independent of τ, so (i) implies (ii).

For every $\delta > 0$ there is $C_\delta > 0$ such that, for any compact interval $I \subset \mathbb{R}$ and smooth z on $I \times \mathbb{S}^{n-1}_+$,

$$\begin{aligned} \int_I \int_{\partial\mathbb{S}^{n-1}_+} |\tilde{z}|^2 \, \mathrm{d}\sigma \, \mathrm{d}t \leq \delta \int_I \int_{\mathbb{S}^{n-1}_+} |\nabla_{\mathbb{S}^{n-1}_+} z|^2 \, \mathrm{d}\theta \, \mathrm{d}t \\ + C_\delta \int_I \int_{\mathbb{S}^{n-1}_+} |z|^2 \, \mathrm{d}\theta \, \mathrm{d}t. \end{aligned} \tag{2.16}$$

To see this, write $z(t,L,\vartheta)$ as $z(t,\beta,\vartheta)$ plus the integral of $\partial_\beta z$ from β to L. Apply Cauchy–Schwarz and average in β over a collar of the equator. The coefficient of the derivative term is proportional to the width of the collar, which can be chosen to give δ.

Fix $\tau \in \mathbb{R}$ and let $I_\tau = [\tau-2,\tau+2]$. Choose a smooth cutoff χ supported in the interior of I_τ, equal to one on $[\tau-3/2,\tau+3/2]$, with derivative bounded independently of τ. Multiplying the interior equation in (2.13) by $\chi^2\Phi$, integrating by parts and using the boundary

equation give

$$\int_{I_\tau\times\mathbb{S}^{n-1}_+}\left(|\nabla(\chi\Phi)|^2+\frac{(n-2)^2}{4}|\chi\Phi|^2\right)\,\mathrm{d}\mu$$
$$=\int_{I_\tau\times\mathbb{S}^{n-1}_+}|\chi'|^2|\Phi|^2\,\mathrm{d}\mu+\frac{nc}{n-2}\int_{I_\tau}\int_{\partial\mathbb{S}^{n-1}_+}\tilde{v}^{2/(n-2)}|\chi\tilde{\Phi}|^2\,\mathrm{d}\sigma\,\mathrm{d}t,$$

where ∇ is the product gradient. Since $\tilde{v}$ is bounded, we can choose δ sufficiently small in (2.16) to move the angular derivative term to the left-hand side. The remaining terms are bounded by a constant times $\|\Phi\|^2_{L^2(I_\tau\times\mathbb{S}^{n-1}_+)}$. This bounds $\|\Phi\|_{H^1([\tau-3/2,\tau+3/2]\times\mathbb{S}^{n-1}_+)}$ by $C\|\Phi\|_{L^2(I_\tau\times\mathbb{S}^{n-1}_+)}$.

By periodicity, the coefficient in the boundary condition and all its derivatives are uniformly bounded. Local elliptic estimates up to the equator on smaller intervals, followed by Sobolev embedding, then give

$$\|\Phi\|_{C^{k,\gamma}([\tau-1,\tau+1]\times\mathbb{S}^{n-1}_+)}\le C_{v,k,\gamma}\|\Phi\|_{L^2(I_\tau\times\mathbb{S}^{n-1}_+)}.$$

The two intervals of length two centered at $\tau-1$ and $\tau+1$ cover I_τ. Since $\langle\tau\pm1\rangle\le 2\langle\tau\rangle$, condition (ii) bounds the last norm by $C'\langle\tau\rangle^q$. This proves (iii). Taking $k=0$ and $\tau=t$ in (2.14) gives (i). □

For $T>0$, let $C^{k,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ be the subspace of T-periodic functions in $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$, with the norm on $[0,T]\times\mathbb{S}^{n-1}_+$. We define $C^{k,\gamma}_T(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ using $[0,T]\times\partial\mathbb{S}^{n-1}_+$. The subscripts sym, even and odd indicate cylindrical symmetry, evenness in t and oddness in t, respectively.

If $v=v(t,\beta)$ and $z=z(t,\beta)$, then $\mathcal{A}z$ depends only on t and β, and $\mathcal{B}_v z$ depends only on t. If $v(-t,\theta)=v(t,\theta)$ and $z(-t,\theta)=\pm z(t,\theta)$, then $\mathcal{A}z$ and $\mathcal{B}_v z$ have the same parity in t as z. Thus the linearized operators preserve cylindrical symmetry when v is cylindrically symmetric, and preserve both parities when v is even in t.

For the periodic weak formulation, write $H^1_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ for the space of T-periodic functions in $H^1_{\mathrm{loc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$, with the $H^1([0,T]\times\mathbb{S}^{n-1}_+)$ norm. We also use $H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ in Section 4.3. In a boundary chart, let h be a term localized by a smooth partition of unity, expressed in the chart and extended by zero to $\mathbb{R}^{n-1}$. Its fractional seminorm is defined by

$$[h]^2_{H^{1/2}(\mathbb{R}^{n-1})}=\int_{\mathbb{R}^{n-1}}\int_{\mathbb{R}^{n-1}}\frac{|h(x)-h(y)|^2}{|x-y|^n}\,\mathrm{d}x\,\mathrm{d}y.$$

The squared norm on $H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ is the sum of $\|h\|^2_{L^2(\mathbb{R}^{n-1})}+[h]^2_{H^{1/2}(\mathbb{R}^{n-1})}$ over the members of a smooth partition of unity subordinate

to a fixed product atlas of uniformly bounded overlap. The trace theorem gives a bounded map

$$H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)\longrightarrow H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+),$$
$$z\longmapsto \tilde z,$$

and a bounded right inverse. See [20, Theorem 3.37]. We write $H^{-1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ for the dual of $H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ with respect to the $L^2(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ pairing.

Proposition 2.3. *Let v be a positive smooth T-periodic solution of (2.10) and let $0<\gamma<1$. The map*

$$(\mathcal{A},\mathcal{B}_v): C^{2,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)\longrightarrow C^{0,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)\times C^{1,\gamma}_T(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$$

is Fredholm of index zero. Let $f\in C^{0,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ and $g\in C^{1,\gamma}_T(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$. The equations $\mathcal{A}y=f$ and $\mathcal{B}_v y=g$ have a solution $y\in C^{2,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ if and only if

$$\int_0^T\int_{\mathbb{S}^{n-1}_+} f\Phi\,\mathrm{d}\theta\,\mathrm{d}t+\int_0^T\int_{\partial\mathbb{S}^{n-1}_+} g\tilde\Phi\,\mathrm{d}\sigma\,\mathrm{d}t=0 \tag{2.17}$$

for every T-periodic Jacobi field Φ along v. When this condition holds, there is a unique solution orthogonal to these fields in $L^2([0,T]\times\mathbb{S}^{n-1}_+)$.

The same assertions hold in the cylindrically symmetric subspaces if v is cylindrically symmetric, and in either parity subspace if v is even in t. In each case the fields in (2.17) belong to the chosen subspace.

Proof. For $y,z\in H^1_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$, define

$$Q_{v,T}(y,z)=\int_0^T\int_{\mathbb{S}^{n-1}_+}\left(y_tz_t+\langle\nabla_{\mathbb{S}^{n-1}_+}y,\nabla_{\mathbb{S}^{n-1}_+}z\rangle+\frac{(n-2)^2}{4}yz\right)\mathrm{d}\theta\,\mathrm{d}t$$
$$-\frac{nc}{n-2}\int_0^T\int_{\partial\mathbb{S}^{n-1}_+}\tilde v^{2/(n-2)}\tilde y\,\tilde z\,\mathrm{d}\sigma\,\mathrm{d}t. \tag{2.18}$$

Given $y\in H^1_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$, choose smooth T-periodic functions y_j converging to y in $H^1_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. Applying (2.16) on $[0,T]$ to y_j-y_ℓ gives

$$\|\tilde y_j-\tilde y_\ell\|^2_{L^2([0,T]\times\partial\mathbb{S}^{n-1}_+)}\le C\|y_j-y_\ell\|^2_{H^1([0,T]\times\mathbb{S}^{n-1}_+)}.$$

The right-hand side tends to zero as $j,\ell\to\infty$. Thus the traces converge in $L^2([0,T]\times\partial\mathbb{S}^{n-1}_+)$ to $\tilde y$. Passing to the limit in (2.16) for y_j gives the same inequality for y. It follows that $Q_{v,T}$ is bounded below and closed. Indeed, for $\lambda>0$ sufficiently large, the quadratic form

$$Q_{v,T}(y,y)+\lambda\|y\|^2_{L^2([0,T]\times\mathbb{S}^{n-1}_+)}$$

defines a norm squared equivalent to $\|y\|^2_{H^1_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)}$. The compact embedding into $L^2([0,T]\times\mathbb{S}^{n-1}_+)$ gives compact resolvent for the associated self-adjoint Robin operator.

The Lax–Milgram theorem gives a unique periodic weak solution for the shifted pair $(\mathcal{A}+\lambda,\mathcal{B}_v)$ with data (f,g). The right-hand side is the continuous functional

$$z \longmapsto \int_0^T \int_{\mathbb{S}^{n-1}_+} fz \,\mathrm{d}\theta\,\mathrm{d}t + \int_0^T \int_{\partial\mathbb{S}^{n-1}_+} g\tilde{z}\,\mathrm{d}\sigma\,\mathrm{d}t,$$

with $z \in H^1_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. Robin Schauder regularity gives $y \in C^{2,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. Thus the shifted map is an isomorphism. The inclusion $C^{2,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+) \hookrightarrow C^{0,\gamma}_T(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ is compact, so removing the shift leaves a Fredholm map of index zero.

Integrating the interior terms in $Q_{v,T}(y,z)$ by parts in t and in the angular variables gives

$$\begin{aligned} Q_{v,T}(y,z) = &\int_0^T \int_{\mathbb{S}^{n-1}_+} (\mathcal{A}y)z \,\mathrm{d}\theta\,\mathrm{d}t \\ &+ \int_0^T \int_{\partial\mathbb{S}^{n-1}_+} (\mathcal{B}_v y)\tilde{z}\,\mathrm{d}\sigma\,\mathrm{d}t + \left[\int_{\mathbb{S}^{n-1}_+} y_t z\,\mathrm{d}\theta\right]_{t=0}^{t=T}. \end{aligned}$$

The last term vanishes because y_t and z are T-periodic. If $\mathcal{A}y = f$ and $\mathcal{B}_v y = g$, we therefore obtain

$$Q_{v,T}(y,z) = \int_0^T \int_{\mathbb{S}^{n-1}_+} fz \,\mathrm{d}\theta\,\mathrm{d}t + \int_0^T \int_{\partial\mathbb{S}^{n-1}_+} g\tilde{z}\,\mathrm{d}\sigma\,\mathrm{d}t.$$

For a T-periodic Jacobi field Φ, the same integration by parts gives $Q_{v,T}(\Phi,y) = 0$, since $\mathcal{A}\Phi = 0$ and $\mathcal{B}_v\Phi = 0$. By symmetry, $Q_{v,T}(y,\Phi) = 0$. Taking $z = \Phi$ in the preceding identity gives (2.17).

Let $\Phi_1,\dots,\Phi_d$ be an orthonormal basis of the periodic kernel in $L^2([0,T]\times\mathbb{S}^{n-1}_+)$. For each basis field Φ_j, condition (2.17) takes the form

$$\int_0^T \int_{\mathbb{S}^{n-1}_+} f\Phi_j \,\mathrm{d}\theta\,\mathrm{d}t + \int_0^T \int_{\partial\mathbb{S}^{n-1}_+} g\tilde{\Phi}_j\,\mathrm{d}\sigma\,\mathrm{d}t = 0, \qquad j = 1,\dots,d.$$

By linearity in Φ, these d equations suffice to impose (2.17) on the entire periodic kernel. To prove that the equations are independent, suppose that a linear combination of their left-hand sides, with coefficients $a_1,\dots,a_d$, vanishes for every pair (f,g). Set $\Psi = \sum_{j=1}^d a_j\Phi_j$. Choosing

$f = \Psi$ and $g = 0$ in that linear combination gives

$$0 = \int_0^T \int_{\mathbb{S}^{n-1}_+} |\Psi|^2 \, \mathrm{d}\theta \, \mathrm{d}t = \sum_{j=1}^{d} |a_j|^2.$$

These d independent equations define a subspace of codimension d consisting of the pairs (f, g) that satisfy (2.17). The preceding integration by parts shows that the range lies in this subspace. Since the operator has index zero and its kernel has dimension d, the range has the same codimension. Hence the inclusion is an equality. Given a solution y, set

$$y_\perp = y - \sum_{j=1}^{d} \left(\int_0^T \int_{\mathbb{S}^{n-1}_+} y\Phi_j \, \mathrm{d}\theta \, \mathrm{d}t \right) \Phi_j.$$

Each term subtracted from y is a Jacobi field, so $y_\perp$ satisfies the same equations and is orthogonal to the kernel. The difference of two orthogonal solutions is a Jacobi field orthogonal to every field in the kernel, and hence vanishes.

If v, f and g are cylindrically symmetric, rotating the angular variable of a solution of the shifted problem gives a solution of the same problem. Uniqueness of the shifted solution therefore gives cylindrical symmetry. If v is even and f and g are both even or both odd in t, then $y(-t, \theta)$ or $-y(-t, \theta)$, respectively, solves the same shifted problem. Uniqueness again gives the required parity. Thus the shifted map is an isomorphism on each of the stated subspaces. Removing the shift is still a compact perturbation, and the preceding argument applies with the kernel taken in the chosen subspace. □

2.4. **The normalized flat family.** We recall the classification and estimates for normalized flat solutions from [3] used in Sections 3 and 5. We use the function spaces and growth conditions introduced in Section 2.3.

For $n \geq 3$, let $\mathcal{B}_n$ be the set of positive, cylindrically symmetric functions $v \in C^\infty(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ that solve (2.10) and satisfy

$$m \leq \tilde{v}(t) \leq M \qquad \text{for all } t \in \mathbb{R}$$

for some $m, M > 0$.

By [3, Theorem A], for each $\varepsilon \in (0, 1)$ there is a unique $v_\varepsilon \in \mathcal{B}_n$ such that

$$\tilde{v}_\varepsilon(0) = \min_{t \in \mathbb{R}} \tilde{v}_\varepsilon(t) = \varepsilon.$$

The minimum ε is the *necksize*. The solution v_ε is even in t and has least positive period T_ε. On $[0, T_\varepsilon)$, its trace attains its minimum only at 0 and its maximum only at $T_\varepsilon/2$. Moreover, $\tilde{v}_\varepsilon''(0) > 0$ and $\tilde{v}_\varepsilon''(T_\varepsilon/2) < 0$. Every noncylindrical $v \in \mathcal{B}_n$ has the form $v(t, \theta) = v_\varepsilon(t + \tau, \theta)$ for a

unique $\varepsilon \in (0,1)$ and some $\tau \in \mathbb{R}$. Two values of τ give the same solution if and only if their difference is an integer multiple of T_ε. We set $v_1 = C_n$.

Set

$$v_\varepsilon^{(1)}(s,\theta) = v_\varepsilon(T_\varepsilon s, \theta), \qquad (s,\theta) \in \mathbb{R} \times \mathbb{S}_+^{n-1}. \tag{2.19}$$

For each integer $k \geq 0$ and $0 < \gamma < 1$, the map

$$\begin{aligned} (0,1) &\longrightarrow (0,\infty) \times C^{k,\gamma}_{1,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \mathbb{S}_+^{n-1}), \\ \varepsilon &\longmapsto (T_\varepsilon, v_\varepsilon^{(1)}) \end{aligned}$$

is smooth by [3, Theorem A]. The estimates in [3, Eqs. (3.8) and (3.27)] give $\varepsilon \leq \tilde{v}_\varepsilon(t) \leq M_n$ for all $t \in \mathbb{R}$ and $\varepsilon C_n(\theta) \leq v_\varepsilon(t,\theta) \leq M_n C_n(\theta)$ on $\mathbb{R} \times \mathbb{S}_+^{n-1}$, where $M_n = ((n-2)/(2c))^{(n-2)/2}$. By [3, Proposition 3.7(i)], $v_\varepsilon \to C_n$ as $\varepsilon \to 1$ in $C^{k,\gamma}_{\mathrm{uloc}}(\mathbb{R} \times \mathbb{S}_+^{n-1})$ for every such k and γ.

Let $v \in \mathcal{B}_n$ be even in t. Let Φ be a cylindrically symmetric Jacobi field along v that is odd in t and satisfies (2.15) for some $q \geq 0$. If v is periodic, we may verify this growth condition using any of the equivalent bounds in Lemma 2.2. By [3, Proposition 3.1(iii)], if $\partial_t \Phi(0, L) = 0$, then $\Phi \equiv 0$.

We use this uniqueness statement to characterize the odd fields along v_ε in Section 3.1.

Theorem B of [3] classifies the positive global solutions of (2.5). If the singularity at 0 is removable, the solution is a standard bubble of Li and Zhu [15, Theorem 1.2].

2.5. **Geometric estimates and the limiting Pohozaev invariant.** We use Harnack's inequality to extend the two-sided trace bounds to closed half-annuli and obtain uniform estimates for the rescaled solutions. In Emden–Fowler coordinates, these estimates give the decay of the error terms in the geometric equation. The resulting Hamiltonian limit determines the Pohozaev invariant used to identify the necksize in Section 5.

For a solution U of (2.4), set $A_\kappa = (\kappa/c)^{(n-2)/2}$ and define

$$w(t,\theta) = A_\kappa e^{-(n-2)t/2} U(e^{-t}\theta), \qquad (t,\theta) \in (-\log\rho, \infty) \times \mathbb{S}_+^{n-1}. \tag{2.20}$$

When $\kappa = (n-2)/2$, we have $A_\kappa^{-1} = b_n$ by (2.9).

Proposition 2.4. *Let U be a positive solution of (2.4) in $C^\infty(B_\rho^+ \setminus \{0\})$, with smoothness up to $\partial_0 B_\rho^+ \setminus \{0\}$. Suppose that its trace $u(\overline{x}) = U(\overline{x}, 0)$ satisfies*

$$c_0 |\overline{x}|^{-\frac{n-2}{2}} \leq u(\overline{x}) \leq C_0 |\overline{x}|^{-\frac{n-2}{2}} \qquad \text{for } 0 < |\overline{x}| < \rho \tag{2.21}$$

for some $C_0, c_0 > 0$. Then w is bounded above and below by positive constants on $[t_0, \infty) \times \mathbb{S}^{n-1}_+$ for some $t_0 > -\log \rho$. For every $\ell > 0$, integer $k \geq 0$ and $0 < \gamma < 1$, the norm of w in $C^{k,\gamma}([t-\ell, t+\ell] \times \mathbb{S}^{n-1}_+)$ is bounded independently of sufficiently large t.

The function w satisfies (2.26), with the error terms in (2.27) satisfying (2.28). The Hamiltonian has the limit in (2.29), and the Pohozaev integral satisfies (2.30).

Proof. Estimates on half-annuli. For $r > 0$, define g_r by $(g_r)_{ij}(z) = g_{ij}(rz)$ and set $v_r(z) = A_\kappa r^{(n-2)/2} U(rz)$ for $z \in \mathbb{R}^n_+$ with $0 < |z| < \rho/r$. On this domain the rescaled equation is

$$\begin{cases} -\Delta_{g_r} v_r + r^2 \dfrac{n-2}{4(n-1)} R_g(rz) v_r = 0 & \text{for } z_n > 0, \\ \partial_{\nu_{g_r}} v_r + \dfrac{n-2}{2} r h_g(r\overline{z}, 0) v_r = c\, v_r(\overline{z}, 0)^{n/(n-2)} & \text{for } z_n = 0. \end{cases} \tag{2.22}$$

Here we used $\kappa A_\kappa^{-2/(n-2)} = c$. For fixed $0 < a < b$, the trace bounds give uniform positive upper and lower bounds for $v_r(\overline{z}, 0)$ when $a/2 \leq |\overline{z}| \leq 2b$ and r is sufficiently small. Thus the coefficient of v_r in the boundary condition written in the form

$$\partial_{\nu_{g_r}} v_r + \left(\frac{n-2}{2} r h_g(r\overline{z}, 0) - c\, v_r(\overline{z}, 0)^{2/(n-2)} \right) v_r = 0$$

is uniformly bounded on this set.

Let B be a ball centered on $\partial \mathbb{R}^n_+$ whose closure lies in $\{z \in \mathbb{R}^n : a/2 < |z| < 2b\}$. For $z = (\overline{z}, z_n) \in B$, define

$$\widehat{v}_r(\overline{z}, z_n) = v_r(\overline{z}, |z_n|), \qquad (\widehat{g}_r)_{ij}(\overline{z}, z_n) = (g_r)_{ij}(\overline{z}, |z_n|), \quad 1 \leq i, j \leq n.$$

By the Fermi identities (2.2), $(g_r)_{nn} = 1$ and $(g_r)_{an} = 0$ for $1 \leq a < n$. Consequently, the surface measure induced by g_r on $B \cap \{z_n = 0\}$ is

$$\mathrm{d}S_{g_r} = \sqrt{\det g_r(\overline{z}, 0)}\ \mathrm{d}\overline{z}.$$

To express the boundary term after reflection, define on B

$$F_r(z) = \sqrt{\det g_r(\overline{z}, 0)} \left(\frac{n-2}{2} r h_g(r\overline{z}, 0) - c\, v_r(\overline{z}, 0)^{2/(n-2)} \right) \operatorname{sgn}(z_n) e_n.$$

The factor multiplying $\operatorname{sgn}(z_n) e_n$ in F_r is independent of z_n. Integrating in z_n on either side of $z_n = 0$ therefore writes the boundary term in the reflected weak equation as $-\int_B F_r \cdot \nabla(\widehat{v}_r \psi)\ \mathrm{d}z$. Integrating by parts in the two half-balls gives, for every $\psi \in C^\infty_c(B)$,

$$\begin{aligned} 0 = &\int_B \left(\sqrt{\det \widehat{g}_r}\, \widehat{g}_r^{-1} \nabla \widehat{v}_r - F_r \widehat{v}_r \right) \cdot \nabla \psi\ \mathrm{d}z \\ &+ \int_B r^2 \frac{n-2}{4(n-1)} R_g(r\overline{z}, r|z_n|) \sqrt{\det \widehat{g}_r}\, \widehat{v}_r \psi\ \mathrm{d}z - \int_B (F_r \cdot \nabla \widehat{v}_r) \psi\ \mathrm{d}z. \end{aligned}$$

The gradients in this identity are Euclidean. Since $g_{ij}(0) = \delta_{ij}$, the coefficients of $\widehat{g}_r$ converge uniformly on B to those of the Euclidean metric as $r \to 0$. Thus, for all sufficiently small r,

$$\frac{1}{2}|\xi|^2 \leq \sqrt{\det \widehat{g}_r(z)}\left(\widehat{g}_r(z)^{-1}\xi\right) \cdot \xi \leq 2|\xi|^2$$

for all $z \in B$ and $\xi \in \mathbb{R}^n$. The trace bounds (2.21) give a uniform bound for F_r on B. The coefficient of $\widehat{v}_r\psi$ in the scalar curvature term is also uniformly bounded by the smoothness of g in the fixed Fermi chart. The two pieces of $\widehat{v}_r$ are smooth up to $z_n = 0$ and agree there, so $\widehat{v}_r \in H^1(B)$.

We may therefore apply the Harnack inequality of Gilbarg and Trudinger [10, Theorem 8.20] to the positive weak solution $\widehat{v}_r$. For each ball $B_R(z_0)$ such that the closure of $B_{4R}(z_0)$ is contained in B, we obtain

$$\sup_{B_R(z_0)} \widehat{v}_r \leq C \inf_{B_R(z_0)} \widehat{v}_r,$$

where C depends on n, R, and the uniform coefficient bounds, and is independent of r. In balls contained in $\{z_n > 0\}$, the same theorem applies to v_r directly through the interior equation in (2.22).

Choose a finite cover of $\{z \in \mathbb{R}^n_+ : a \leq |z| \leq b\}$ by interior balls and half-balls centered on $\{z_n = 0\}$. Choose the cover so that the larger concentric balls required for Harnack's inequality lie in $\{z \in \mathbb{R}^n : a/2 < |z| < 2b\}$, and so that the larger balls used for the interior equation also lie in $\{z_n > 0\}$. The closed half-annulus is connected, so each member of the cover can be joined to a member containing $(a+b)e_1/2$ by a finite sequence of members with consecutive intersections.

At the boundary point $(a+b)e_1/2$, (2.21) gives

$$A_\kappa c_0 \left(\frac{a+b}{2}\right)^{-(n-2)/2} \leq v_r\left(\frac{a+b}{2}e_1\right) \leq A_\kappa C_0 \left(\frac{a+b}{2}\right)^{-(n-2)/2}.$$

On each member of the cover, Harnack's inequality bounds the value of v_r at any point above and below by fixed multiples of its value at any other point. At an intersection of two consecutive members, the bounds obtained in the first therefore give bounds at a point of the second. Repeating this argument along the sequence gives, for all sufficiently small r,

$$0 < c_{a,b} \leq v_r(z) \leq C_{a,b} \tag{2.23}$$

for $z \in \mathbb{R}^n_+$ with $a \leq |z| \leq b$. The cover is independent of r, so the constants are uniform in r.

Apply the interior Hölder estimate [10, Theorem 8.24] to v_r on interior balls and to $\widehat{v}_r$ on balls centered on $\{z_n = 0\}$. The coefficient bounds

are uniform in r, and (2.23) on a larger closed half-annulus bounds the solutions on these balls. Restricting the estimates for $\widehat{v}_r$ to $z_n \geq 0$ gives a uniform bound for v_r in $C^{0,\alpha}(\{z \in \mathbb{R}^n_+ : a \leq |z| \leq b\})$ for some $0 < \alpha < 1$.

We next use the boundary condition in (2.22) to gain one derivative. The bound just obtained makes $\partial_{\nu_{g_r}} v_r$ uniformly α-Hölder continuous on $\{(\overline{z}, 0) : a \leq |\overline{z}| \leq b\}$. On a half-ball $B \cap \mathbb{R}^n_+$ with B centered on $\{z_n = 0\}$, the vector field used in the boundary Schauder estimate is

$$-v_r(\overline{z}, 0) \lim_{s \to 0^+} F_r(\overline{z}, s) = \sqrt{\det g_r(\overline{z}, 0)}\, \partial_{\nu_{g_r}} v_r(\overline{z}, 0) e_n, \tag{2.24}$$

where the equality follows from (2.22). This field points in the e_n direction and is independent of z_n, so its divergence vanishes. On $z_n = 0$, its component in the outward direction $-e_n$ cancels the normal component of $\sqrt{\det g_r}\, g_r^{-1} \nabla_\delta v_r$.

The vector field in (2.24) is uniformly α-Hölder continuous on the half-ball, the coefficients of $\sqrt{\det g_r}\, g_r^{-1}$ are uniformly smooth. The scalar term in the divergence equation is uniformly α-Hölder continuous by the preceding estimate for v_r and the smoothness of the coefficients. The boundary Schauder estimate [21, Theorem 1.1 with weight exponent zero] therefore gives a uniform $C^{1,\alpha}$ estimate on a smaller half-ball. Together with the interior Schauder estimates, this gives a uniform bound for v_r in $C^{1,\alpha}(\{z \in \mathbb{R}^n_+ : a \leq |z| \leq b\})$.

The first derivative estimates give uniform local Lipschitz bounds. Hence, for any $0 < \gamma < 1$, the vector field in (2.24) is uniformly γ-Hölder continuous on each of the fixed half-balls. Applying the same Schauder estimate with exponent γ gives a uniform bound for v_r in $C^{1,\gamma}(\{z \in \mathbb{R}^n_+ : a \leq |z| \leq b\})$. By (2.23), the range of v_r on each fixed closed half-annulus lies in a compact subset of $(0, \infty)$. Thus, if the derivatives of v_r through order j are uniformly bounded and those of order j are uniformly γ-Hölder continuous on a larger half-annulus, the same bounds hold for the vector field in (2.24) and for the scalar term on the half-balls where they are used. Applying the Schauder estimates on smaller balls and half-balls gives the bounds through order $j+1$, with the same exponent γ. Iteration gives, for every integer $k \geq 0$ and $0 < \gamma < 1$,

$$\|v_r\|_{C^{k,\gamma}(\{z \in \mathbb{R}^n_+ : a \leq |z| \leq b\})} \leq C_{a,b,k,\gamma} \tag{2.25}$$

for all sufficiently small r.

Equation in Emden–Fowler coordinates. Fix $\ell > 0$ and take $t_0 > -\log \rho + \ell$. Write $r_0 = e^{-t_0}$ and $t = t_0 + s$, with $|s| \leq \ell$. By the

definitions of w and v_{r_0},

$$w(t_0+s,\theta) = e^{-(n-2)s/2} v_{r_0}(e^{-s}\theta).$$

As (s,θ) varies in $[-\ell,\ell]\times\mathbb{S}^{n-1}_+$, the point $e^{-s}\theta$ lies in the fixed closed half-annulus $\{z\in\mathbb{R}^n_+ : e^{-\ell}\le|z|\le e^{\ell}\}$. Since $e^{-(n-2)s/2}$ is bounded above and below by positive constants for $|s|\le\ell$, (2.23) gives uniform positive lower and upper bounds for w on $[t_0-\ell,t_0+\ell]\times\mathbb{S}^{n-1}_+$. The same change of variables and (2.25) give the uniform bounds in $C^{k,\gamma}([t_0-\ell,t_0+\ell]\times\mathbb{S}^{n-1}_+)$.

For $r=e^{-t}$, the Euclidean polar formula gives

$$\left(w_{tt}+\Delta_{\mathbb{S}^{n-1}_+}w-\frac{(n-2)^2}{4}w\right)(t,\theta) = A_\kappa r^{(n+2)/2}\Delta_\delta U(r\theta).$$

At $\theta=(\vartheta,0)$, with $\vartheta\in\mathbb{S}^{n-2}$, the outward conormal derivative is

$$\partial_{\nu_{\mathbb{S}^{n-1}_+}} w(t,(\vartheta,0)) = -A_\kappa r^{n/2}\partial_{x_n}U(r\vartheta,0).$$

In the Fermi chart, $\nu_g=-\partial_{x_n}$ on $\partial_0 B^+_\rho$. Substituting the two equations in (2.4) and using $\kappa A_\kappa^{-2/(n-2)}=c$, we obtain

$$\begin{cases} w_{tt}+\Delta_{\mathbb{S}^{n-1}_+}w-\dfrac{(n-2)^2}{4}w = f_g, \\ \partial_{\nu_{\mathbb{S}^{n-1}_+}} w = c\,\tilde{w}^{n/(n-2)} + f_{\partial,g}. \end{cases} \tag{2.26}$$

The first equation holds in $(-\log\rho,\infty)\times \operatorname{int}\mathbb{S}^{n-1}_+$ and the second on $(-\log\rho,\infty)\times\partial\mathbb{S}^{n-1}_+$. The interior and boundary error terms are, respectively,

$$\begin{aligned} f_g(t,\theta) &= A_\kappa r^{(n+2)/2}\left[(\Delta_\delta-\Delta_g)U+\frac{n-2}{4(n-1)}R_gU\right](r\theta), \\ f_{\partial,g}(t,\vartheta) &= -\frac{n-2}{2}rh_g(r\vartheta,0)\tilde{w}(t,\vartheta). \end{aligned} \tag{2.27}$$

To estimate f_g on $[t_0-\ell,t_0+\ell]\times\mathbb{S}^{n-1}_+$, put $z=e^{-s}\theta$, so that $r\theta=r_0z$. In these coordinates, the first identity in (2.27) becomes

$$f_g(t_0+s,\theta) = e^{-(n+2)s/2}\Bigg([(\Delta_\delta-\Delta_{g_{r_0}})v_{r_0}](z) + \frac{n-2}{4(n-1)}r_0^2R_g(r_0z)v_{r_0}(z)\Bigg).$$

The coefficients of $\Delta_\delta-\Delta_{g_{r_0}}$ are $O(r_0)$ on the fixed half-annulus by (2.3) and the coordinate formula for the Laplace–Beltrami operator. See Section A.1 for the explicit first-order expansion. The scalar coefficient $r_0^2R_g(r_0z)$ is $O(r_0^2)$. These bounds also hold for derivatives of every fixed order in z. On the boundary, $r=r_0e^{-s}$ with $|s|\le\ell$, so $rh_g(r\vartheta,0)$ and

its derivatives in (s,ϑ) are $O(r_0)$. Combining these bounds with (2.25) gives

$$\begin{aligned}\|f_g\|_{C^{k,\gamma}([t_0-\ell,t_0+\ell]\times\mathbb{S}^{n-1}_+)}\\ +\|f_{\partial,g}\|_{C^{k,\gamma}([t_0-\ell,t_0+\ell]\times\partial\mathbb{S}^{n-1}_+)}\le C_{\ell,k,\gamma}e^{-t_0}\end{aligned}\tag{2.28}$$

for all sufficiently large t_0.

The limiting Pohozaev invariant. Differentiate the Hamiltonian (2.11) with $a=c$ and integrate by parts on $\mathbb{S}^{n-1}_+$. The angular gradient term contributes the integral of $w_t\Delta_{\mathbb{S}^{n-1}_+}w$ in the hemisphere and the integral of $-\partial_t\tilde w\,\partial_{\nu_{\mathbb{S}^{n-1}_+}}w$ on the equator. Combining these terms with the other terms in the derivative gives

$$\begin{aligned}|\mathbb{S}^{n-2}|\frac{\mathrm{d}}{\mathrm{d}t}H(w)(t)&=\int_{\mathbb{S}^{n-1}_+}w_t\left(w_{tt}+\Delta_{\mathbb{S}^{n-1}_+}w-\frac{(n-2)^2}{4}w\right)\mathrm{d}\theta\\&\quad+\int_{\partial\mathbb{S}^{n-1}_+}\partial_t\tilde w\left(c\,\tilde w^{n/(n-2)}-\partial_{\nu_{\mathbb{S}^{n-1}_+}}w\right)\mathrm{d}\sigma\\&=\int_{\mathbb{S}^{n-1}_+}w_tf_g\,\mathrm{d}\theta-\int_{\partial\mathbb{S}^{n-1}_+}\partial_t\tilde w\,f_{\partial,g}\,\mathrm{d}\sigma,\end{aligned}$$

where the last equality follows from (2.26). The uniform bounds for w_t and $\partial_t\tilde w$, together with (2.28), bound the absolute value of this derivative by Ce^{-t} for all sufficiently large t. Since this bound is integrable, $|\mathbb{S}^{n-2}|H(w)(t)$ has a finite limit as $t\to\infty$. We denote the limit by H_0. Integrating the derivative bound from t to infinity gives

$$|\mathbb{S}^{n-2}|H(w)(t)=H_0+O(e^{-t}).\tag{2.29}$$

For $0<r<\rho$, define the Pohozaev integral in the fixed Fermi chart by

$$\begin{aligned}P_\kappa(r,U)&=\int_{\partial^+B_r^+}\left(\frac{n-2}{2}U\partial_rU-\frac{r}{2}|\nabla_\delta U|^2+r(\partial_rU)^2\right)\mathrm{d}S\\&\quad+\frac{(n-2)r\kappa}{2(n-1)}\int_{r\mathbb{S}^{n-2}}u(\overline{x})^{\frac{2(n-1)}{n-2}}\,\mathrm{d}\overline{S},\end{aligned}$$

where $\mathrm{d}S$ and $\mathrm{d}\overline{S}$ are the Euclidean surface measures on $\partial^+B_r^+\subset\mathbb{R}^n$ and $r\mathbb{S}^{n-2}\subset\mathbb{R}^{n-1}$, respectively.

Set $t=-\log r$. Differentiating (2.20) at fixed θ and decomposing the Euclidean gradient into its radial and angular parts give

$$\begin{aligned}r\partial_rU(r\theta)&=-A_\kappa^{-1}r^{-(n-2)/2}\left(w_t+\frac{n-2}{2}w\right)(t,\theta),\\|\nabla_\delta U(r\theta)|^2&=(\partial_rU(r\theta))^2+A_\kappa^{-2}r^{-n}|\nabla_{\mathbb{S}^{n-1}_+}w(t,\theta)|^2.\end{aligned}$$

The terms containing ww_t cancel upon substitution into the hemispherical integral. Since $\mathrm{d}S = r^{n-1}\,\mathrm{d}\theta$, this integral gives A_κ^{-2} times the interior term in (2.11).

On the equator, $u(r\vartheta) = A_\kappa^{-1} r^{-(n-2)/2}\tilde{w}(t,\vartheta)$ and $\mathrm{d}\overline{S} = r^{n-2}\,\mathrm{d}\sigma$. The power of u in P_κ contributes $r^{-(n-1)}$, which cancels the factor r^{n-1} from $r\,\mathrm{d}\overline{S}$. The identity $\kappa A_\kappa^{-2/(n-2)} = c$ then gives A_κ^{-2} times the boundary term in (2.11) with $a = c$. Adding the two integrals and using (2.29) gives

$$P_\kappa(r,U) = A_\kappa^{-2}|\mathbb{S}^{n-2}|H(w)(-\log r) = A_\kappa^{-2}H_0 + O(r). \tag{2.30}$$

The Pohozaev invariant of U at the origin is the limit of $P_\kappa(r,U)$ as $r \to 0$, namely $A_\kappa^{-2}H_0$. □

3. Cylindrically symmetric Jacobi fields and the Hamiltonian

Let $n \geq 3$. We use the family v_ε from Section 2.4 to classify the cylindrically symmetric Jacobi fields with polynomial growth. We then show that the derivative of the Hamiltonian with respect to the necksize never vanishes. The endpoint limits determine the range of the Hamiltonian and give the classification by energy in the half-space.

3.1. Jacobi fields and the first variation of the Hamiltonian.

Differentiation of v_ε with respect to t and ε gives the translation field and the necksize derivative. The uniqueness statements for odd and even fields in [3, Proposition 3.1(iii)] show that these two fields span the space of cylindrically symmetric Jacobi fields with polynomial growth. We then use a periodic problem to show that their pairing is nonzero and compute the first variation of the Hamiltonian.

Fix $\varepsilon \in (0,1)$. With the operators $\mathcal{A}$ and $\mathcal{B}_v$ defined in (2.13), the equations for a Jacobi field along v_ε are

$$\begin{cases} \mathcal{A}\Phi = 0 & \text{in } \mathbb{R}\times \operatorname{int}\mathbb{S}^{n-1}_+, \\ \mathcal{B}_{v_\varepsilon}\Phi = 0 & \text{on } \mathbb{R}\times\partial\mathbb{S}^{n-1}_+. \end{cases} \tag{3.1}$$

For $v \in \mathcal{B}_n$, write $\mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v)$ for the set of $\Phi \in C^\infty_{\mathrm{sym}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ satisfying $\mathcal{A}\Phi = 0$, $\mathcal{B}_v\Phi = 0$ and (2.15) for some $q \geq 0$. By Lemma 2.2, the growth condition is equivalent to membership in $C^{k,\gamma}_{\mathrm{pol},q}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ for every integer $k \geq 0$ and $0 < \gamma < 1$.

We use the smooth period-one family $\varepsilon \mapsto (T_\varepsilon, v_\varepsilon^{(1)})$ from Section 2.4. Set $F_\varepsilon = \partial_t v_\varepsilon$ and $G_\varepsilon = \partial_\varepsilon v_\varepsilon$, with the derivative in ε taken at fixed t.

Differentiating $v_\varepsilon(t,\theta) = v_\varepsilon^{(1)}(t/T_\varepsilon,\theta)$, as given by (2.19), yields

$$G_\varepsilon(t,\theta) = (\partial_\varepsilon v_\varepsilon^{(1)})(t/T_\varepsilon,\theta) - \frac{T_\varepsilon'}{T_\varepsilon}\, tF_\varepsilon(t,\theta). \tag{3.2}$$

Here $\partial_\varepsilon v_\varepsilon^{(1)}$ is taken at fixed s.

For $y_1, y_2 \in C^\infty_{\mathrm{sym}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$, define the skew-symmetric pairing

$$\Omega(y_1,y_2)(t) = \int_0^L \omega(\beta)(y_1\partial_t y_2 - (\partial_t y_1)y_2)(t,\beta)\,\mathrm{d}\beta. \tag{3.3}$$

If y_1 and y_2 satisfy (3.1), differentiation in t and integration by parts in β give

$$\frac{\mathrm{d}}{\mathrm{d}t}\Omega(y_1,y_2)(t) = -\Big[\omega(y_1\partial_\beta y_2 - (\partial_\beta y_1)y_2)\Big]_0^L.$$

At $\beta = L$, the equations $\mathcal{B}_{v_\varepsilon} y_i = 0$ give $\partial_\beta y_i(t,L) = (nc/(n-2))\tilde v_\varepsilon(t)^{2/(n-2)} y_i(t,L)$ for $i = 1,2$. The two products at this endpoint are therefore equal. At $\beta = 0$, the functions and their first derivatives are bounded and $\omega(0) = 0$. Thus the derivative vanishes, and we write $\Omega(y_1,y_2)$ for the constant value.

Proposition 3.1. *For every $\varepsilon \in (0,1)$, $F_\varepsilon, G_\varepsilon \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$. They satisfy*

$$\begin{aligned} F_\varepsilon(-t,\theta) &= -F_\varepsilon(t,\theta), \qquad G_\varepsilon(-t,\theta) = G_\varepsilon(t,\theta),\\ \partial_t F_\varepsilon(0,L) &= \tilde v_\varepsilon''(0) > 0, \qquad G_\varepsilon(0,L) = 1. \end{aligned} \tag{3.4}$$

The field F_ε is T_ε-periodic, and G_ε satisfies (2.15) with $q = 1$ and

$$G_\varepsilon(t+T_\varepsilon,\theta) = G_\varepsilon(t,\theta) - T_\varepsilon' F_\varepsilon(t,\theta). \tag{3.5}$$

Moreover,

$$\mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon) = \mathrm{span}\{F_\varepsilon, G_\varepsilon\}. \tag{3.6}$$

The odd and even subspaces are spanned by F_ε and G_ε, respectively, and $\Omega(F_\varepsilon, G_\varepsilon) \neq 0$.

Proof. Differentiating (2.10) in t and in ε gives (3.1) for F_ε and G_ε. The differentiation in ε at fixed t is justified on each compact interval by (3.2). Since v_ε is even in t, its t-derivative is odd and its ε-derivative is even. The evaluations in (3.4) follow from $F_\varepsilon = \partial_t v_\varepsilon$, $\tilde v_\varepsilon''(0) > 0$ and $\tilde v_\varepsilon(0) = \varepsilon$.

Both F_ε and $(\partial_\varepsilon v_\varepsilon^{(1)})(t/T_\varepsilon,\theta)$ are T_ε-periodic in t. Formula (3.2) therefore gives the pointwise bound (2.15) with $q = 1$ for G_ε. This proves that both fields belong to $\mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$. Differentiating $v_\varepsilon(t+T_\varepsilon,\theta) = v_\varepsilon(t,\theta)$ in ε and using the periodicity of F_ε gives (3.5).

Let $\Phi \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$. If Φ is odd in t, the field

$$\Phi - \frac{\partial_t \Phi(0,L)}{\tilde{v}_\varepsilon''(0)} F_\varepsilon$$

belongs to $\mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$, is odd in t and has vanishing t-derivative at $(0,L)$. If Φ is even in t, the field $\Phi - \Phi(0,L)G_\varepsilon$ belongs to the same space, is even in t and vanishes at $(0,L)$. In both cases the difference is zero by [3, Proposition 3.1(iii)]. This result applies for $n \geq 3$, with $n = 3, 4$ treated in [3, Theorem C.1]. Since v_ε is even, the odd and even parts of every $\Phi \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$ belong to that space. This proves (3.6).

Suppose that $\Omega(F_\varepsilon, G_\varepsilon) = 0$. Set $d = -T_\varepsilon'/T_\varepsilon$ and define

$$E_\varepsilon = G_\varepsilon - dtF_\varepsilon.$$

By (3.5), $E_\varepsilon(t + T_\varepsilon, \theta) = E_\varepsilon(t, \theta)$. Since G_ε and tF_ε are even in t, we have

$$E_\varepsilon \in C^{k,\gamma}_{T_\varepsilon,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$$

for every integer $k \geq 0$ and $0 < \gamma < 1$. The boundary operator $\mathcal{B}_{v_\varepsilon}$ commutes with multiplication by t, while $\mathcal{A}(tF_\varepsilon) = t\mathcal{A}F_\varepsilon - 2\partial_t F_\varepsilon$. Consequently,

$$\mathcal{A}E_\varepsilon = 2d\,\partial_t F_\varepsilon, \qquad \mathcal{B}_{v_\varepsilon} E_\varepsilon = 0.$$

Substituting $G_\varepsilon = E_\varepsilon + dtF_\varepsilon$ into (3.3) cancels the two terms containing $dtF_\varepsilon \partial_t F_\varepsilon$. Integrate the resulting identity over $[0, T_\varepsilon]$. Integration by parts in t turns the integral of $-(\partial_t F_\varepsilon)E_\varepsilon$ into that of $F_\varepsilon \partial_t E_\varepsilon$, because $F_\varepsilon E_\varepsilon$ takes the same values at 0 and T_ε. Thus

$$\int_0^{T_\varepsilon} \int_0^L \omega F_\varepsilon (2\partial_t E_\varepsilon + dF_\varepsilon)\,\mathrm{d}\beta\,\mathrm{d}t = T_\varepsilon \Omega(F_\varepsilon, G_\varepsilon) = 0. \tag{3.7}$$

We now apply Proposition 2.3 with right-hand side $2\partial_t E_\varepsilon + dF_\varepsilon$ and zero boundary data. The right-hand side belongs to $C^{0,\gamma}_{T_\varepsilon,\mathrm{sym},\mathrm{odd}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. The compatibility condition (2.17) requires its $L^2([0,T_\varepsilon]\times\mathbb{S}^{n-1}_+)$ pairing with each odd, cylindrically symmetric T_ε-periodic Jacobi field to vanish. The term involving the boundary data is zero. Every such field is a multiple of F_ε by (3.6) and the parity of G_ε. It therefore suffices to verify

$$\begin{aligned}\int_0^{T_\varepsilon} \int_{\mathbb{S}^{n-1}_+} & F_\varepsilon (2\partial_t E_\varepsilon + dF_\varepsilon)\,\mathrm{d}\theta\,\mathrm{d}t \\ &= |\mathbb{S}^{n-2}| \int_0^{T_\varepsilon} \int_0^L \omega F_\varepsilon (2\partial_t E_\varepsilon + dF_\varepsilon)\,\mathrm{d}\beta\,\mathrm{d}t = 0.\end{aligned}$$

The factor $|\mathbb{S}^{n-2}|$ comes from integration in ϑ, since both functions depend only on t and β. The last equality is (3.7). We obtain $y \in$

$C^{2,\gamma}_{T_\varepsilon,\mathrm{sym,odd}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ satisfying

$$\mathcal{A}y = 2\partial_t E_\varepsilon + dF_\varepsilon, \qquad \mathcal{B}_{v_\varepsilon} y = 0.$$

Elliptic regularity gives $y \in C^\infty(\mathbb{R}\times\mathbb{S}^{n-1}_+)$.

The product rule gives

$$\mathcal{A}(tE_\varepsilon) = 2dt\,\partial_t F_\varepsilon - 2\partial_t E_\varepsilon,$$
$$\mathcal{A}\Big((d/2)t^2 F_\varepsilon\Big) = -2dt\,\partial_t F_\varepsilon - dF_\varepsilon.$$

Thus the function

$$\Phi = y + tE_\varepsilon + (d/2)t^2 F_\varepsilon$$

satisfies $\mathcal{A}\Phi = 0$. It also satisfies $\mathcal{B}_{v_\varepsilon}\Phi = 0$, since $\mathcal{B}_{v_\varepsilon}$ commutes with multiplication by functions of t. The three summands are odd in t and have at most quadratic growth, so $\Phi \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$. By (3.6) and the parity of G_ε, we have $\Phi \in \mathrm{span}\{F_\varepsilon\}$. In particular, Φ is T_ε-periodic.

On the other hand, $y(T_\varepsilon, L) = y(0, L)$ by periodicity, $F_\varepsilon(T_\varepsilon, L) = F_\varepsilon(0, L) = 0$ by periodicity and oddness, and $E_\varepsilon(T_\varepsilon, L) = E_\varepsilon(0, L) = G_\varepsilon(0, L) = 1$. Hence

$$\begin{aligned}\Phi(T_\varepsilon, L) - \Phi(0, L) &= y(T_\varepsilon, L) - y(0, L) + T_\varepsilon E_\varepsilon(T_\varepsilon, L) + \frac{d}{2}T_\varepsilon^2 F_\varepsilon(T_\varepsilon, L)\\ &= T_\varepsilon > 0.\end{aligned}$$

This contradicts the periodicity of Φ and proves $\Omega(F_\varepsilon, G_\varepsilon) \neq 0$. □

Set $h(\varepsilon) = H(v_\varepsilon)$ for $0 < \varepsilon < 1$, with H as in (2.12).

Lemma 3.2. *Let $\Phi \in C^\infty_{\mathrm{sym}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ satisfy (3.1). At each fixed t, the first variation of H at v_ε in the direction Φ is*

$$DH_{v_\varepsilon}[\Phi](t) = \Omega(F_\varepsilon, \Phi). \tag{3.8}$$

In particular,

$$h'(\varepsilon) = \Omega(F_\varepsilon, G_\varepsilon) \neq 0. \tag{3.9}$$

Proof. Fix $t \in \mathbb{R}$. At (t, L), the boundary value of $v_\varepsilon + s\Phi$ is $\tilde{v}_\varepsilon(t) + s\tilde{\Phi}(t)$. Since $\tilde{v}_\varepsilon(t) > 0$, this value remains positive for sufficiently small $|s|$. Holding t fixed, the chain rule gives the derivative of the boundary term in (2.12):

$$\frac{\mathrm{d}}{\mathrm{d}s}\bigg|_{s=0}\left[\frac{(n-2)c}{2(n-1)}\Big(\tilde{v}_\varepsilon(t) + s\tilde{\Phi}(t)\Big)^{\frac{2(n-1)}{n-2}}\right] = c\,\tilde{v}_\varepsilon(t)^{n/(n-2)}\tilde{\Phi}(t).$$

Differentiating the full expression gives

$$\begin{aligned}DH_{v_\varepsilon}[\Phi](t) = &\int_0^L \omega\left(\partial_t v_\varepsilon\,\partial_t\Phi - \partial_\beta v_\varepsilon\,\partial_\beta\Phi - \frac{(n-2)^2}{4}v_\varepsilon\Phi\right)\mathrm{d}\beta\\ &+ c\,\tilde{v}_\varepsilon(t)^{n/(n-2)}\tilde{\Phi}(t).\end{aligned}$$

Integration by parts in β gives

$$-\int_0^L \omega\,\partial_\beta v_\varepsilon\,\partial_\beta\Phi\,\mathrm{d}\beta = -\Big[\omega\,\partial_\beta v_\varepsilon\,\Phi\Big]_0^L + \int_0^L (\omega\,\partial_\beta v_\varepsilon)_\beta\Phi\,\mathrm{d}\beta.$$

The term at $\beta = 0$ vanishes because $\omega(0) = 0$. At $\beta = L$, we have $\omega(L) = 1$ and $\partial_\beta v_\varepsilon(t, L) = c\,\tilde{v}_\varepsilon(t)^{n/(n-2)}$. The endpoint term is therefore $-c\,\tilde{v}_\varepsilon(t)^{n/(n-2)}\tilde{\Phi}(t)$, which cancels the contribution from differentiating the boundary term in (2.12). The interior equation for v_ε gives

$$(\omega\,\partial_\beta v_\varepsilon)_\beta - \frac{(n-2)^2}{4}\omega v_\varepsilon = -\omega\,\partial_t^2 v_\varepsilon.$$

Substitution into the formula for the first variation yields

$$DH_{v_\varepsilon}[\Phi](t) = \int_0^L \omega(\partial_t v_\varepsilon\,\partial_t\Phi - \partial_t^2 v_\varepsilon\,\Phi)\,\mathrm{d}\beta = \Omega(F_\varepsilon, \Phi).$$

Taking $\Phi = G_\varepsilon = \partial_\varepsilon v_\varepsilon$ gives the derivative of h. Its nonvanishing follows from Proposition 3.1. □

For $\Phi \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_\varepsilon)$, write $\Phi = aF_\varepsilon + bG_\varepsilon$ with $a, b \in \mathbb{R}$. By (3.8) and the skew-symmetry of Ω,

$$DH_{v_\varepsilon}[\Phi] = b\,h'(\varepsilon). \tag{3.10}$$

Thus the first variation vanishes if and only if $\Phi \in \operatorname{span}\{F_\varepsilon\}$.

3.2. **The range of the Hamiltonian and classification by energy.** We compute the limits of h as $\varepsilon \to 0$ and $\varepsilon \to 1$. Together with (3.9), they show that the energy determines the necksize. We use the energy as a parameter to describe the bounded Jacobi fields and then return to the original normalization of the half-space equation.

Lemma 3.3. *The centered family satisfies*

$$\lim_{\varepsilon\to 0} h(\varepsilon) = 0, \qquad \lim_{\varepsilon\to 1} h(\varepsilon) = H_c. \tag{3.11}$$

Moreover,

$$v_\varepsilon \to 0 \quad \text{in } C^1_{\mathrm{loc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+) \quad \text{as } \varepsilon \to 0. \tag{3.12}$$

Proof. By [3, Proposition 3.7(i)], $v_\varepsilon \to C_n$ in $C^{1,\gamma}_{\mathrm{uloc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ for $0 < \gamma < 1$ as $\varepsilon \to 1$. Evaluating (2.12) at $t = 0$ gives $h(\varepsilon) \to H_c$.

For the other limit, let (ε_j) be a sequence in $(0, 1)$ tending to zero. The bound $v_{\varepsilon_j} \le M_n C_n$ in Section 2.4 and the local estimates of Section 2.3 give uniform $C^{1,\delta_0}([-R, R]\times\mathbb{S}^{n-1}_+)$ bounds for each $R > 0$, for some $0 < \delta_0 < 1$. A diagonal subsequence therefore converges in $C^1_{\mathrm{loc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ to a nonnegative function v_∞. Passing to the limit in the interior equation in the sense of distributions, and in the boundary

condition by C^1 convergence, gives the normalized equation for v_∞. Interior elliptic regularity makes the limit smooth in $\mathbb{R} \times \operatorname{int} \mathbb{S}^{n-1}_+$.

At $(0, L)$, the trace and the outward conormal derivative of v_∞ both vanish, since their values for v_{ε_j} are ε_j and $c\varepsilon_j^{n/(n-2)}$. If v_∞ were nonzero, the strong maximum principle would give positivity in the interior. The boundary point lemma at $(0, L)$ would then give a strictly negative outward conormal derivative there. It follows that $v_\infty = 0$. Since every sequence tending to zero admits such a subsequence, this proves (3.12). Evaluating (2.12) at $t = 0$ and using this convergence for the first derivatives and the trace gives $h(\varepsilon) \to 0$ as $\varepsilon \to 0$. □

Theorem A. *The map*

$$h : (0, 1) \longrightarrow (H_c, 0), \qquad h(\varepsilon) = H(v_\varepsilon), \tag{3.13}$$

is a smooth diffeomorphism with $h'(\varepsilon) < 0$. Two noncylindrical solutions in $\mathcal{B}_n$ have the same Hamiltonian if and only if they differ by a translation in t. The only solution in $\mathcal{B}_n$ with Hamiltonian H_c is C_n.

Let $\varepsilon(h)$ denote the inverse of (3.13) and set

$$v_h = v_{\varepsilon(h)}, \qquad T(h) = T_{\varepsilon(h)}, \qquad \Psi_h = \partial_h v_h, \tag{3.14}$$

where the derivative is taken at fixed t. The map $h \mapsto (T(h), v_h^{(1)})$, where $v_h^{(1)}(s, \theta) = v_h(T(h)s, \theta)$, is smooth from $(H_c, 0)$ into $(0, \infty) \times C^{k,\gamma}_{1,\mathrm{sym},\mathrm{even}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ for every integer $k \geq 0$ and $0 < \gamma < 1$. The field Ψ_h is even in t, satisfies (2.15) with $q = 1$, and obeys

$$\Omega(\partial_t v_h, \Psi_h) = 1, \qquad \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_h) = \operatorname{span}\{\partial_t v_h, \Psi_h\}. \tag{3.15}$$

Moreover,

$$\Psi_h(t + T(h), \theta) = \Psi_h(t, \theta) - T'(h)\partial_t v_h(t, \theta). \tag{3.16}$$

Proof. The smooth dependence of the family on ε gives smoothness of h, and (3.9) gives $h'(\varepsilon) \neq 0$ for every $\varepsilon \in (0, 1)$. If h' took both positive and negative values, its continuity would give a zero between them. Thus h' has the same sign throughout $(0, 1)$. If it were positive, h would be strictly increasing, and the endpoint limits in Lemma 3.3 would give $0 \leq H_c$. Since $H_c < 0$, we have $h'(\varepsilon) < 0$. Strict monotonicity and the endpoint limits give a bijection onto $(H_c, 0)$. The inverse function theorem gives smoothness of $\varepsilon(h)$.

By Theorem A of [3], every noncylindrical member of $\mathcal{B}_n$ is a translate of a centered solution with a unique necksize. Translating in t leaves H unchanged, so two such solutions have the same energy if and only if their centered representatives have the same necksize. Their energies lie in $(H_c, 0)$, and C_n has energy H_c. Thus C_n is the only member of $\mathcal{B}_n$ with that energy.

The period-one family parametrized by h is the composition

$$h \longmapsto \varepsilon(h) \longmapsto \left(T_{\varepsilon(h)}, v^{(1)}_{\varepsilon(h)}\right) = \left(T(h), v^{(1)}_h\right).$$

Both maps in the composition are smooth. At fixed t, the chain rule gives

$$\Psi_h = \frac{G_{\varepsilon(h)}}{h'(\varepsilon(h))}.$$

By Proposition 3.1, Ψ_h is even in t and has at most linear growth. Since $h'(\varepsilon(h)) \neq 0$ and $\partial_t v_h = F_{\varepsilon(h)}$, the same proposition gives $\mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_h) = \mathrm{span}\{\partial_t v_h, \Psi_h\}$. By (3.9),

$$\Omega(\partial_t v_h, \Psi_h) = \frac{\Omega(F_{\varepsilon(h)}, G_{\varepsilon(h)})}{h'(\varepsilon(h))} = 1.$$

Finally, differentiating $v_h(t + T(h), \theta) = v_h(t, \theta)$ in h at fixed t gives

$$\Psi_h(t + T(h), \theta) + T'(h)\partial_t v_h(t + T(h), \theta) = \Psi_h(t, \theta).$$

The periodicity of $\partial_t v_h$ gives (3.16). □

For $H_c < h < 0$, define

$$\mathcal{K}^{\mathrm{sym}}_{\mathrm{bd}}(v_h) = \left\{ \Phi \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_h) : \sup_{(t,\theta)\in\mathbb{R}\times\mathbb{S}^{n-1}_+} |\Phi(t, \theta)| < \infty \right\}.$$

Corollary 3.5. *For $H_c < h < 0$,*

$$\mathcal{K}^{\mathrm{sym}}_{\mathrm{bd}}(v_h) = \begin{cases} \mathrm{span}\{\partial_t v_h\}, & T'(h) \neq 0, \\ \mathrm{span}\{\partial_t v_h, \Psi_h\}, & T'(h) = 0. \end{cases}$$

Proof. Let $\Phi \in \mathcal{K}^{\mathrm{sym}}_{\mathrm{bd}}(v_h)$. By (3.15), write $\Phi = a\partial_t v_h + b\Psi_h$ with $a, b \in \mathbb{R}$. If $T'(h) = 0$, (3.16) gives $\Psi_h(t + T(h), \theta) = \Psi_h(t, \theta)$. Thus both $\partial_t v_h$ and Ψ_h are bounded, since each is continuous on $[0, T(h)] \times \mathbb{S}^{n-1}_+$ and $T(h)$-periodic in t.

Suppose that $T'(h) \neq 0$. Applying (3.16) at $t, t + T(h), \ldots, t + (k - 1)T(h)$ and using the periodicity of $\partial_t v_h$ gives

$$\Psi_h(t + kT(h), \theta) = \Psi_h(t, \theta) - kT'(h)\partial_t v_h(t, \theta)$$

for every integer $k \geq 1$. It follows that

$$\Phi(t + kT(h), \theta) - \Phi(t, \theta) = -kbT'(h)\partial_t v_h(t, \theta).$$

Since v_h is noncylindrical, choose (t_0, θ_0) with $\partial_t v_h(t_0, \theta_0) \neq 0$. The boundedness of Φ gives

$$\begin{aligned} k|bT'(h)\partial_t v_h(t_0, \theta_0)| &= |\Phi(t_0 + kT(h), \theta_0) - \Phi(t_0, \theta_0)| \\ &\leq 2 \sup_{(t,\theta)\in\mathbb{R}\times\mathbb{S}^{n-1}_+} |\Phi(t, \theta)|. \end{aligned}$$

The right-hand side is finite and independent of k, so $b = 0$. Every multiple of $\partial_t v_h$ is bounded by periodicity. □

For a positive classical solution U of (2.5) on $\mathbb{R}^n_+ \setminus \{0\}$, set $v_U(t,\theta) = e^{-(n-2)t/2}U(e^{-t}\theta)$ for $(t,\theta) \in \mathbb{R}\times\mathbb{S}^{n-1}_+$. Formula (2.11) with $a = (n-2)/2$ gives

$$\begin{aligned} &|\mathbb{S}^{n-2}|H_{(n-2)/2}(v_U)(t) \\ &\quad = \frac{1}{2}\int_{\mathbb{S}^{n-1}_+}\left((\partial_t v_U)^2 - |\nabla_{\mathbb{S}^{n-1}_+} v_U|^2 - \frac{(n-2)^2}{4}v_U^2\right)\mathrm{d}\theta \\ &\qquad + \frac{(n-2)^2}{4(n-1)}\int_{\partial\mathbb{S}^{n-1}_+} \tilde{v}_U^{\frac{2(n-1)}{n-2}}\,\mathrm{d}\sigma. \end{aligned} \tag{3.17}$$

Since v_U solves (2.6), Section 2.2 shows that $H_{(n-2)/2}(v_U)(t)$ is independent of t. We denote its value by $H_{(n-2)/2}(v_U)$. If $v_U = b_n v$ for a solution v of (2.10), then $((n-2)/2)b_n^{2/(n-2)} = c$ gives

$$H_{(n-2)/2}(v_U) = b_n^2 H(v). \tag{3.18}$$

Corollary 3.6. *For every positive classical solution U of (2.5) on $\mathbb{R}^n_+ \setminus \{0\}$,*

$$b_n^2 H_c \le H_{(n-2)/2}(v_U) \le 0.$$

(i) *$H_{(n-2)/2}(v_U) = 0$ if and only if the trace singularity at 0 is removable.*
(ii) *$H_{(n-2)/2}(v_U) = b_n^2 H_c$ if and only if $U(r\theta) = b_n r^{-(n-2)/2}C_n(\theta)$.*
(iii) *Each value in $[b_n^2 H_c, 0)$ is attained by a singular solution. Two singular solutions have the same value if and only if they differ by a dilation $U(x) \mapsto \lambda^{(n-2)/2}U(\lambda x)$ with $\lambda > 0$.*

Proof. By Theorem B of [3], a solution with removable trace singularity is a standard bubble and extends smoothly across 0. The definition of v_U and the chain rule give

$$|v_U| + |\partial_t v_U| + |\nabla_{\mathbb{S}^{n-1}_+} v_U| = O(e^{-(n-2)t/2})$$

uniformly on $\mathbb{S}^{n-1}_+$ as $t \to \infty$. Each integral in (3.17) therefore tends to zero. Since $H_{(n-2)/2}(v_U)(t)$ is independent of t, we obtain $H_{(n-2)/2}(v_U) = 0$.

If the trace singularity is nonremovable, Theorem B of [3] gives $v_U(t,\theta) = b_n v_\varepsilon(t+\tau,\theta)$ for a unique $\varepsilon \in (0,1]$ and some $\tau \in \mathbb{R}$, with $v_1 = C_n$. By (3.18), its energy is $b_n^2 H(v_\varepsilon)$. Theorem A gives $b_n^2 H_c < b_n^2 H(v_\varepsilon) < 0$ for $0 < \varepsilon < 1$, while $\varepsilon = 1$ gives $b_n^2 H_c$. This proves (i) and (ii), since the representative $b_n C_n$ corresponds to $U(r\theta) = b_n r^{-(n-2)/2}C_n(\theta)$.

For an energy in $(b_n^2 H_c, 0)$, division by b_n^2 gives a number in $(H_c, 0)$. The inverse of (3.13) then gives exactly one $\varepsilon \in (0,1)$. At energy $b_n^2 H_c$, the only choice is $\varepsilon = 1$. For each such ε, the representative $b_n v_\varepsilon$ gives a singular solution with the prescribed energy by Theorem B of [3].

If two singular solutions U_1 and U_2 have the same energy, their representatives therefore have the form $b_n v_\varepsilon(t+\tau_1, \theta)$ and $b_n v_\varepsilon(t+\tau_2, \theta)$ with the same ε. By (2.7), the representative of $\lambda^{(n-2)/2} U_1(\lambda x)$ is $v_{U_1}(t - \log\lambda, \theta)$. Taking $\lambda = e^{\tau_1 - \tau_2}$ gives $v_{U_2}(t, \theta)$ and hence $U_2(x) = \lambda^{(n-2)/2} U_1(\lambda x)$. Conversely, such a dilation replaces t by $t - \log\lambda$ in (3.17). Since $H_{(n-2)/2}(v_{U_1})(t)$ is independent of t, the two solutions have the same energy. □

4. Jacobi fields without a symmetry assumption

Let $n \geq 3$. We use Jacobi fields obtained from tangential translations and the Kelvin transform to prove that every field with polynomial growth on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ is cylindrically symmetric. The classification along the noncylindrical solutions then follows from Section 3. For the cylindrical solution, we solve the eigenvalue problem on $\mathbb{S}^{n-1}_+$ and compute the quadratic term of the Hamiltonian. We also determine the L^2 kernels of the operators on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ and $\mathbb{R} \times \partial\mathbb{S}^{n-1}_+$.

Throughout this section, v is either v_h from (3.14), with $H_c < h < 0$, or the cylindrical solution C_n of (2.10). In the first case let $T = T(h)$. In the second case choose any $T > 0$. Then $v(t+T, \theta) = v(t, \theta)$ on $\mathbb{R} \times \mathbb{S}^{n-1}_+$. By (2.13), if Φ is a Jacobi field along v, then $\Phi(t+\tau, \theta)$ is a Jacobi field along $v(t+\tau, \theta)$ for any $\tau \in \mathbb{R}$.

We use the spaces $\mathcal{K}_{\mathrm{pol}}(v)$ and $\mathcal{K}_{\mathrm{bd}}(v)$ defined in the introduction. Lemma 2.2 characterizes polynomial growth by L^2 bounds on translated intervals. The spaces $\mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v)$ and $\mathcal{K}^{\mathrm{sym}}_{\mathrm{bd}}(v)$ from Section 3 are their respective intersections with $C^\infty_{\mathrm{sym}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$.

4.1. Angular estimates and the noncylindrical classification.

We prove a positive lower bound for the quadratic form associated with each spherical harmonic of positive degree. The bound follows from two positive functions obtained by differentiating families of solutions of (2.5) generated by tangential translations and the Kelvin transform. We then apply the bound to the spherical harmonic coefficients of a Jacobi field with polynomial growth.

Fix a real spherical harmonic $S_\ell \in C^\infty(\mathbb{S}^{n-2})$ of degree ℓ, normalized by $\|S_\ell\|_{L^2(\mathbb{S}^{n-2})} = 1$. Its eigenvalue is

$$-\Delta_{\mathbb{S}^{n-2}} S_\ell = \lambda_\ell S_\ell, \qquad \lambda_\ell = \ell(\ell + n - 3).$$

For a Jacobi field Φ along v as in (2.13), define

$$y_\ell(t,\beta) = \int_{\mathbb{S}^{n-2}} \Phi(t,\beta,\vartheta) S_\ell(\vartheta)\, \mathrm{d}\sigma(\vartheta),$$
$$Z_\ell(t,\beta) = \sin^{(n-2)/2}\beta\, y_\ell(t,\beta).$$

The formula for $\Delta_{\mathbb{S}^{n-1}_+}$ in Section 2.2 gives the equation for y_ℓ. Multiplying y_ℓ by $\sin^{(n-2)/2}\beta$ removes the first derivative in β from that equation. At $\beta = L$, the multiplier equals 1 and its β-derivative vanishes. Thus the boundary condition in (2.13) is preserved under multiplication. The first line of the system below holds on $\mathbb{R}\times(0,L)$, and the second holds at $\beta = L$ for every $t\in\mathbb{R}$.

$$\begin{cases} \mathcal{L}_\ell Z_\ell = 0, \\ \partial_\beta Z_\ell(t,L) = \dfrac{nc}{n-2}\tilde{v}(t)^{2/(n-2)} Z_\ell(t,L), \end{cases} \tag{4.1}$$

where

$$\mathcal{L}_\ell = -\partial_t^2 - \partial_\beta^2 + \left(\frac{(n-2)(n-4)}{4} + \lambda_\ell\right)\csc^2\beta.$$

For $\ell \geq 1$, we describe the behavior as $\beta \to 0$. The value $\beta = 0$ corresponds to the point $(0,\dots,0,1) \in \mathbb{S}^{n-1}_+$. In geodesic normal coordinates centered at this point, the vector $\beta\vartheta \in \mathbb{R}^{n-1}$ represents $\theta = (\sin\beta\,\vartheta, \cos\beta)$. The restriction to $\mathbb{S}^{n-2}$ of any polynomial of degree less than ℓ is orthogonal to S_ℓ in $L^2(\mathbb{S}^{n-2})$. Thus, after expanding Φ in these coordinates through degree $\ell-1$, only the remainder contributes to y_ℓ. The remainder is $O(\beta^\ell)$, and its derivative along each ray $\beta\vartheta$ is $O(\beta^{\ell-1})$. Applying the same expansion to $\partial_t\Phi$ gives

$$y_\ell(t,\beta) = O(\beta^\ell),$$
$$\partial_t y_\ell(t,\beta) = O(\beta^\ell),$$
$$\partial_\beta y_\ell(t,\beta) = O(\beta^{\ell-1}).$$

These estimates are uniform on compact intervals in t as $\beta \to 0$.

We next construct two positive solutions of the system (4.1) with $\ell = 1$. The function $c^{(n-2)/2}v$ solves the equation with boundary coefficient 1 and has trace bounded above and below by positive constants on $\mathbb{R}$. Proposition A.2 and equation (A.2) of [3] therefore give

$$\left(\frac{\tilde{v}'(t)}{\tilde{v}(t)}\right)^2 + c^2\tilde{v}(t)^{4/(n-2)} \leq \frac{(n-2)^2}{4}.$$

The second term on the left is positive. It follows that, for every $t\in\mathbb{R}$,

$$\left|\frac{\tilde{v}'(t)}{\tilde{v}(t)}\right| < \frac{n-2}{2}. \tag{4.2}$$

For $(t,\beta)\in\mathbb{R}\times(0,L]$, define

$$\begin{aligned} h_\pm(t,\beta) &= \frac{n-2}{2}v(t,\beta) - \cot\beta\,\partial_\beta v(t,\beta) \pm \partial_t v(t,\beta),\\ z_\pm(t,\beta) &= e^{\pm t}\sin^{n/2}\beta\, h_\pm(t,\beta). \end{aligned} \tag{4.3}$$

Proposition 4.1. *The functions $z_\pm$ in (4.3) satisfy the system (4.1) with $\ell=1$. Each $h_\pm$ extends to a T-periodic function in $C^\infty_{\mathrm{sym}}(\mathbb{R}\times\mathbb{S}^{n+1}_+)$. There are constants $C_v, c_v>0$ such that, for every $(t,\beta)\in\mathbb{R}\times[0,L]$,*

$$\begin{aligned} c_v \le h_+(t,\beta) \le C_v,\\ c_v \le h_-(t,\beta) \le C_v. \end{aligned} \tag{4.4}$$

In particular, $z_\pm>0$ on $\mathbb{R}\times(0,L]$.

Proof. Define U on $\mathbb{R}^n_+\setminus\{0\}$ by $U(r\theta)=b_n r^{-(n-2)/2}v(-\log r,\theta)$. By (2.6) and (2.10), U solves (2.5). For $i\in\{1,\dots,n-1\}$, the chain rule at $x=r\theta$ gives

$$b_n^{-1}r^{(n-2)/2}\partial_{x_i}U(r\theta) = -e^t\sin\beta\, h_+(t,\beta)\vartheta_i.$$

Indeed, $\partial_{x_i}r = \sin\beta\,\vartheta_i$ and $\partial_{x_i}\beta = r^{-1}\cos\beta\,\vartheta_i$, while $\partial_{x_i}t = -r^{-1}\sin\beta\,\vartheta_i$. Cylindrical symmetry eliminates derivatives in ϑ.

For each $x\neq 0$, the translates $U(x+ae_i)$ satisfy (2.5) in a neighborhood of x for sufficiently small $|a|$. Differentiating at $a=0$ and passing to Emden–Fowler coordinates therefore gives a Jacobi field along v. The factor ϑ_i is the restriction to $\mathbb{S}^{n-2}$ of the linear harmonic function $\overline{x}\mapsto x_i$ on $\mathbb{R}^{n-1}$. Thus it is a spherical harmonic with $\ell=1$ and $\lambda_1=n-2$. Multiplying its coefficient by $\sin^{(n-2)/2}\beta$ gives $-z_+$, which consequently satisfies (4.1).

By (2.8), the Kelvin transform centered at 0 with radius 1 replaces $v(t,\beta)$ by $v(-t,\beta)$. Apply the translation calculation to this solution and then replace t by $-t$ in the resulting field. This changes e^t to e^{-t} and reverses the sign of $\partial_t v$. The coefficient of ϑ_i, multiplied by $\sin^{(n-2)/2}\beta$, is $-z_-$. The equations for both coefficients also follow from [3, Eqs. (A.9) and (A.10)] by multiplication by $\sin^{(n-2)/2}\beta$.

To verify smoothness of $h_\pm$ at $\beta=0$, fix $\vartheta\in\mathbb{S}^{n-2}$ and consider the map

$$\beta \longmapsto v(t,(\sin\beta\,\vartheta,\cos\beta))$$

for β near zero, including negative β. Replacing β by $-\beta$ has the same effect as replacing ϑ by $-\vartheta$. Cylindrical symmetry therefore makes this smooth function even in β. In particular, $\partial_\beta v(t,0)=0$, and

$$\cot\beta\,\partial_\beta v(t,\beta) = (\beta\cot\beta)\int_0^1 \partial_\beta^2 v(t,a\beta)\,\mathrm{d}a.$$

Both factors on the right are smooth and even in β near zero, with $\beta\cot\beta\to 1$. Hence $h_\pm$ extends smoothly and evenly in β. In geodesic normal coordinates $\overline{x}\in\mathbb{R}^{n+1}$ centered at $(0,\dots,0,1)\in\mathbb{S}^{n+1}_+$, we have $\beta=|\overline{x}|$. The extension is therefore given by $(t,\overline{x})\mapsto h_\pm(t,|\overline{x}|)$. It is smooth because the Taylor expansion of $h_\pm$ in β to any order contains only even powers, which become polynomials in the coordinates of $\overline{x}$ after setting $\beta=|\overline{x}|$. Periodicity follows from the formula for $h_\pm$ and the periodicity of v.

For $0<\beta<L$, substitute $z_\pm=e^{\pm t}\sin^{n/2}\beta\,h_\pm$ into $\mathcal{L}_1 z_\pm=0$. Differentiating first the factor $e^{\pm t}$ gives

$$(\mathcal{L}_1-1)(\sin^{n/2}\beta\,h_\pm)=\pm 2\sin^{n/2}\beta\,\partial_t h_\pm. \tag{4.5}$$

Since

$$\frac{(\sin^{n/2}\beta)''}{\sin^{n/2}\beta}=\frac{n(n-2)}{4}\csc^2\beta-\frac{n^2}{4},$$

the term with $\csc^2\beta$ cancels. On a function depending only on β, the spherical Laplacian on $\mathbb{S}^{n+1}_+$ is $\partial_\beta^2+n\cot\beta\,\partial_\beta$. For $t\in\mathbb{R}$ and $0<\beta<L$, the resulting equation is

$$-\partial_t^2 h_\pm\mp 2\partial_t h_\pm-\Delta_{\mathbb{S}^{n+1}_+}h_\pm+\left(\frac{n^2}{4}-1\right)h_\pm=0. \tag{4.6}$$

The smooth extension just proved makes the equation hold at $\beta=0$ as well. This is the equation in [3, Eq. (A.8)].

At $\beta=L$, the term with $\cot\beta$ in (4.3) vanishes. By (4.2),

$$h_\pm(t,L)=\frac{n-2}{2}\tilde{v}(t)\pm\tilde{v}'(t)>0.$$

For each sign, $h_\pm$ attains a minimum on the compact set $[0,T]\times\mathbb{S}^{n+1}_+$. Periodicity makes it a minimum on $\mathbb{R}\times\mathbb{S}^{n+1}_+$, including when the minimizing point has $t=0$ or $t=T$. If the minimum were nonpositive, its angular coordinate would be in $\operatorname{int}\mathbb{S}^{n+1}_+$ because $h_\pm(t,L)>0$. At a negative minimum, the first derivative in t vanishes and $\partial_t^2 h_\pm+\Delta_{\mathbb{S}^{n+1}_+}h_\pm\geq 0$. Since $n^2/4-1>0$, the left-hand side of (4.6) would be strictly negative. If the minimum were zero, the strong maximum principle applied to $-h_\pm\leq 0$ in the connected interior would give $h_\pm\equiv 0$, contrary to $h_\pm(t,L)>0$.

For c_v, take the smaller of the two positive minima on $[0,T]\times\mathbb{S}^{n+1}_+$. For C_v, take the larger of the two maxima there. Periodicity gives the same bounds for all $t\in\mathbb{R}$, proving (4.4). □

For $\ell \geq 1$, $\lambda_\ell \geq n-2$, so the coefficient of $\csc^2\beta$ in $\mathcal{L}_\ell$ is at least $n(n-2)/4 > 0$. For $I = \mathbb{R}$ or $I = [0,T]$, define

$$\begin{aligned} Q_{\ell,I}[Z] = &\int_I \int_0^L \left(|\partial_t Z|^2 + |\partial_\beta Z|^2\right) \mathrm{d}\beta \, \mathrm{d}t \\ &+ \left(\frac{(n-2)(n-4)}{4} + \lambda_\ell\right) \int_I \int_0^L \frac{Z^2}{\sin^2\beta} \, \mathrm{d}\beta \, \mathrm{d}t \\ &- \frac{nc}{n-2} \int_I \tilde{v}(t)^{2/(n-2)} Z(t,L)^2 \, \mathrm{d}t. \end{aligned} \tag{4.7}$$

We consider this quadratic form on

$$\mathcal{D}_I = \left\{ Z \in H^1(I \times (0,L)) : Z|_{\beta=0} = 0, \quad \frac{Z}{\sin\beta} \in L^2(I \times (0,L)) \right\}.$$

When $I = [0,T]$, we require in addition that the boundary values at $t = 0$ and $t = T$ agree in the Sobolev trace sense. The trace theorem on $I \times (0,L)$ bounds the $L^2(I)$ norm of $Z(\cdot, L)$ by the $H^1(I \times (0,L))$ norm of Z.

Every $Z \in \mathcal{D}_I$ can be approximated by smooth functions that vanish near $\beta = 0$. On $I = \mathbb{R}$, the approximating functions also have compact support in t. On $I = [0,T]$, they are restrictions of smooth functions that are T-periodic in t. The convergence holds in $H^1(I \times (0,L))$ and, after division by $\sin\beta$, in $L^2(I \times (0,L))$.

To see this, let $\chi_\delta(\beta)$ be zero for $\beta \leq \delta$, equal to 1 for $\beta \geq 2\delta$, and satisfy $0 \leq \chi_\delta \leq 1$ and $|\chi_\delta'| \leq C/\delta$. The differences $Z - \chi_\delta Z$, $(1-\chi_\delta)\partial_t Z$, $(1-\chi_\delta)\partial_\beta Z$ and $(1-\chi_\delta)Z/\sin\beta$ tend to zero in $L^2(I \times (0,L))$ by dominated convergence. The other term in the derivative of $Z - \chi_\delta Z$ is $-\chi_\delta' Z$. Its squared $L^2(I \times (0,L))$ norm is bounded by

$$C \int_I \int_\delta^{2\delta} \frac{Z^2}{\sin^2\beta} \, \mathrm{d}\beta \, \mathrm{d}t,$$

since $\sin\beta \leq 2\delta$ on the support of χ_δ'. The integrand is integrable and the integration region shrinks to $\beta = 0$, so the bound tends to zero.

On $I = \mathbb{R}$, multiplication by cutoffs in t that equal 1 on $[-R,R]$, vanish for $|t| \geq 2R$, and have derivatives bounded by C/R gives convergence in the same spaces as $R \to \infty$. On $I = [0,T]$, equality of the traces at 0 and T gives a periodic extension in $H^1_{\mathrm{loc}}(\mathbb{R} \times (0,L))$. In both cases, extension by reflection across $\beta = L$ and smoothing by convolution give smooth approximations in $H^1(I \times (0,L))$. Choose the mollifier with support small enough to keep the approximations zero near $\beta = 0$. With δ fixed, $1/\sin\beta$ is bounded on their support, so the convergence after division by $\sin\beta$ follows as well.

Proposition 4.2. *For the quadratic forms in (4.7), there is $\lambda_v > 0$ independent of ℓ such that, for every $\ell \geq 1$ and $Z \in \mathcal{D}_I$,*

$$Q_{\ell,I}[Z] \geq \lambda_v \|Z\|^2_{L^2(I\times(0,L))}. \tag{4.8}$$

Proof. Let $\phi_0 = \sqrt{z_+ z_-} = \sin^{n/2}\beta\,\sqrt{h_+ h_-}$. It is positive on $\mathbb{R}\times(0,L]$ and T-periodic in t by Proposition 4.1. Differentiating the product at $\beta = L$ and using (4.1) gives $\partial_\beta\phi_0(t,L) = (nc/(n-2))\tilde{v}(t)^{2/(n-2)}\phi_0(t,L)$.

On $\mathbb{R}\times(0,L)$, the identity $\log\phi_0 = (\log z_+ + \log z_-)/2$ and the equations $\mathcal{L}_1 z_\pm = 0$ give

$$\mathcal{L}_1\phi_0 = \frac{\phi_0}{4}\left|\nabla_{t,\beta}\log\frac{z_+}{z_-}\right|^2. \tag{4.9}$$

Here $\nabla_{t,\beta} = (\partial_t, \partial_\beta)$. Indeed, for each positive smooth function z,

$$\frac{(\partial_t^2+\partial_\beta^2)z}{z} = (\partial_t^2+\partial_\beta^2)\log z + |\nabla_{t,\beta}\log z|^2.$$

Applying this first to $z_\pm$ and then to ϕ_0, the terms linear in $\log z_\pm$ cancel and the remaining difference is $|\nabla_{t,\beta}(\log z_+ - \log z_-)|^2/4$.

Take a smooth Z from the approximating functions described after (4.7) and put $w = Z/\phi_0$. Expanding the derivative of $Z = \phi_0 w$ gives

$$|\nabla_{t,\beta}Z|^2 - \phi_0^2|\nabla_{t,\beta}w|^2 = \nabla_{t,\beta}\phi_0\cdot\nabla_{t,\beta}(\phi_0 w^2).$$

Integrating by parts and using (4.9) gives Picone's identity

$$\begin{aligned} Q_{1,I}[Z] = {} & \int_I\int_0^L \phi_0^2|\nabla_{t,\beta}w|^2\,\mathrm{d}\beta\,\mathrm{d}t \\ & + \frac{1}{4}\int_I\int_0^L \left|\nabla_{t,\beta}\log\frac{z_+}{z_-}\right|^2 Z^2\,\mathrm{d}\beta\,\mathrm{d}t. \end{aligned} \tag{4.10}$$

At $\beta = L$, the term from integration by parts is

$$\int_I \frac{\partial_\beta\phi_0(t,L)}{\phi_0(t,L)}Z(t,L)^2\,\mathrm{d}t.$$

By the boundary condition for ϕ_0, it cancels the last term in (4.7). The vanishing of Z near $\beta = 0$ eliminates the term there. The terms at $t = 0$ and $t = T$ cancel when $I = [0,T]$ by periodicity. When $I = \mathbb{R}$, Z has compact support in t.

Fix $0 < \beta \leq L$ and an interval $J = [a, a+T]$. By (4.3), $z_+/z_- = e^{2t}h_+/h_-$, and h_+/h_- is T-periodic. Thus

$$\int_J \partial_t\log(z_+/z_-)\,\mathrm{d}t = 2T.$$

Cauchy–Schwarz gives $\int_J |\nabla_{t,\beta} \log(z_+/z_-)|^2 \, \mathrm{d}t \geq 4T$. For $t, s \in J$, the fundamental theorem of calculus and Cauchy–Schwarz give

$$|w(t,\beta)|^2 \leq 2|w(s,\beta)|^2 + 2T \int_J |\partial_t w(r,\beta)|^2 \, \mathrm{d}r.$$

Multiply by $|\nabla_{s,\beta} \log(z_+/z_-)(s,\beta)|^2$ and integrate in s on J. Divide by the integral of this weight, which is at least $4T$, and integrate in t on J. The resulting inequality is

$$\int_J |w(t,\beta)|^2 \, \mathrm{d}t \leq \frac{1}{2} \int_J \left| \nabla_{t,\beta} \log \frac{z_+}{z_-} \right|^2 |w(t,\beta)|^2 \, \mathrm{d}t + 2T^2 \int_J |\partial_t w(t,\beta)|^2 \, \mathrm{d}t.$$

By (4.4), the ratio $\phi_0^2 / \sin^n \beta$ lies between c_v^2 and C_v^2. Multiplying the last inequality by $\sin^n \beta$ and integrating in β therefore gives

$$\int_J \int_0^L Z^2 \, \mathrm{d}\beta \, \mathrm{d}t \leq C \int_J \int_0^L \phi_0^2 |\partial_t w|^2 \, \mathrm{d}\beta \, \mathrm{d}t + C \int_J \int_0^L \left| \nabla_{t,\beta} \log \frac{z_+}{z_-} \right|^2 Z^2 \, \mathrm{d}\beta \, \mathrm{d}t,$$

where C depends on c_v, C_v, T. For $I = [0,T]$, take $J = [0,T]$. For $I = \mathbb{R}$, sum over $J = [jT, (j+1)T]$ for $j \in \mathbb{Z}$. In either case, (4.10) gives (4.8) with $\ell = 1$ for the smooth approximating functions.

For $Z \in \mathcal{D}_I$, the approximation following (4.7) gives convergence of the derivatives in $L^2(I \times (0,L))$, of $Z/\sin\beta$ in the same space, and of the boundary values at $\beta = L$ in $L^2(I)$. Since $\tilde{v}$ is bounded, every term in (4.7) converges along this approximation. This proves the inequality for $Z \in \mathcal{D}_I$ when $\ell = 1$. Finally, for $\ell \geq 1$, the boundary terms for $Q_{\ell,I}$ and $Q_{1,I}$ agree, and

$$Q_{\ell,I}[Z] - Q_{1,I}[Z] = (\lambda_\ell - (n-2)) \int_I \int_0^L \frac{Z^2}{\sin^2 \beta} \, \mathrm{d}\beta \, \mathrm{d}t \geq 0.$$

□

Proposition 4.3. *Every $\Phi \in \mathcal{K}_{\mathrm{pol}}(v)$ is cylindrically symmetric.*

Proof. Fix a spherical harmonic of degree $\ell \geq 1$ and consider its coefficient Z_ℓ in (4.1). Choose $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ that equals 1 on $[-1,1]$ and vanishes for $|t| \geq 2$. For $R \geq 1$, write $\chi_R(t) = \chi(t/R)$. Near $\beta = 0$, the estimates for y_ℓ and the product rule for $Z_\ell = \sin^{(n-2)/2} \beta \, y_\ell$

give

$$Z_\ell(t,\beta) = O\Big(\beta^{(n-2)/2+\ell}\Big),$$
$$\partial_t Z_\ell(t,\beta) = O\Big(\beta^{(n-2)/2+\ell}\Big),$$
$$\partial_\beta Z_\ell(t,\beta) = O\Big(\beta^{(n-4)/2+\ell}\Big).$$

The constants are uniform for $t \in [-2R, 2R]$. In the last line, the two terms come from $(\sin^{(n-2)/2}\beta)' = O(\beta^{(n-4)/2})$ and $\partial_\beta y_\ell = O(\beta^{\ell-1})$. Since $(n-4)/2 + \ell \geq 1/2$, both $\chi_R \partial_\beta Z_\ell$ and $\chi_R Z_\ell / \sin\beta$ are square integrable near $\beta = 0$, and $\chi_R Z_\ell$ vanishes there. Thus $\chi_R Z_\ell \in \mathcal{D}_{\mathbb{R}}$.

Multiply $\mathcal{L}_\ell Z_\ell = 0$ by $\chi_R^2 Z_\ell$ and integrate on $\mathbb{R}\times[\delta, L]$, with $0 < \delta < L$. The integration by parts in β gives the terms

$$\int_{\mathbb{R}} \chi_R^2 Z_\ell(t,\delta)\partial_\beta Z_\ell(t,\delta)\ \mathrm{d}t - \int_{\mathbb{R}} \chi_R^2 Z_\ell(t,L)\partial_\beta Z_\ell(t,L)\ \mathrm{d}t.$$

The first is $O(\delta^{n+2\ell-3})$ for fixed R, by the estimates just proved and the support of χ_R. It tends to zero as $\delta \to 0$. By (4.1), the second equals the boundary term in (4.7) with $Z = \chi_R Z_\ell$. Expanding $|\partial_t(\chi_R Z_\ell)|^2$ and letting $\delta \to 0$ now gives

$$Q_{\ell,\mathbb{R}}[\chi_R Z_\ell] = \int_{\mathbb{R}}\int_0^L |\chi_R'|^2 Z_\ell^2\ \mathrm{d}\beta\ \mathrm{d}t. \tag{4.11}$$

By (4.8),

$$\begin{aligned}\int_{-R}^{R}\int_0^L Z_\ell^2\ \mathrm{d}\beta\ \mathrm{d}t &\leq \lambda_v^{-1} Q_{\ell,\mathbb{R}}[\chi_R Z_\ell] \\ &\leq \frac{\|\chi'\|^2_{L^\infty(\mathbb{R})}}{\lambda_v R^2}\int_{-2R}^{2R}\int_0^L Z_\ell^2\ \mathrm{d}\beta\ \mathrm{d}t.\end{aligned} \tag{4.12}$$

For each t, the normalization of S_ℓ in $L^2(\mathbb{S}^{n-2})$ and Cauchy–Schwarz in ϑ give

$$\begin{aligned}\int_0^L Z_\ell(t,\beta)^2\ \mathrm{d}\beta &= \int_0^L \omega(\beta) y_\ell(t,\beta)^2\ \mathrm{d}\beta \\ &\leq \int_{\mathbb{S}^{n-1}_+} |\Phi(t,\theta)|^2\ \mathrm{d}\theta.\end{aligned}$$

The polynomial growth bound for Φ and Lemma 2.2 therefore give

$$\int_{-R}^{R}\int_0^L Z_\ell^2\ \mathrm{d}\beta\ \mathrm{d}t \leq C R^{2N+1}$$

for $R \geq 1$ and some integer $N \geq 0$. Indeed, $[-R, R]$ is covered by at most $2R + 2$ intervals of length 2, and the squared local L^2 norm of Φ on the corresponding products with $\mathbb{S}^{n-1}_+$ is bounded by CR^{2N}. Applying (4.12) at $R, 2R, \dots, 2^{j-1}R$ gives

$$\begin{aligned}\int_{-R}^{R}\int_0^L Z_\ell^2 \, \mathrm{d}\beta \, \mathrm{d}t &\leq C_j R^{-2j} \int_{-2^j R}^{2^j R}\int_0^L Z_\ell^2 \, \mathrm{d}\beta \, \mathrm{d}t \\ &\leq C_j' R^{2N+1-2j}.\end{aligned}$$

Choose an integer $j > N + 1/2$ and let $R \to \infty$. The integral on the left is nonnegative and nondecreasing in R, so it must vanish for every R. This proves $Z_\ell = 0$. Since the argument applies to every spherical harmonic of degree $\ell \geq 1$, completeness in $L^2(\mathbb{S}^{n-2})$ and smoothness imply that $\Phi(t, \beta, \vartheta)$ is independent of ϑ. □

For $y \in C^\infty(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ and fixed $t \in \mathbb{R}$, the Hamiltonian in (2.11) satisfies

$$|\mathbb{S}^{n-2}| DH_v[y](t) = \int_{\mathbb{S}^{n-1}_+} \Big(v_t y_t - v_{tt} y \Big) \, \mathrm{d}\theta. \tag{4.13}$$

This is the same integration by parts as in the proof of Lemma 3.2, now on $\mathbb{S}^{n-1}_+$. The boundary term cancels by (2.10), and the interior equation replaces $\Delta_{\mathbb{S}^{n-1}_+} v - (n-2)^2 v/4$ by $-v_{tt}$. The differentiation at $s = 0$ is valid because the trace of $v + sy$ stays positive near the fixed section for small $|s|$. Indeed, $\tilde{v}$ has a positive lower bound by the definition of $\mathcal{B}_n$ in Section 2.4, and $\tilde{y}$ is bounded on each compact product of a closed interval with $\partial\mathbb{S}^{n-1}_+$.

If $y = \Phi$ satisfies (2.13), the right-hand side of (4.13) is independent of t. Its derivative, by Green's identity, is the integral over $\partial\mathbb{S}^{n-1}_+$ of

$$\tilde{\Phi} \, \partial_{\nu_{\mathbb{S}^{n-1}_+}} v_t - \partial_t \tilde{v} \, \partial_{\nu_{\mathbb{S}^{n-1}_+}} \Phi.$$

This integrand vanishes because v_t and Φ satisfy the same boundary condition in (2.13).

Theorem B. *Let $H_c < h < 0$, and let $\Psi_h = \partial_h v_h$ be the field in (3.14). Then*

$$\mathcal{K}_{\mathrm{pol}}(v_h) = \mathrm{span}\{\partial_t v_h, \Psi_h\}. \tag{4.14}$$

For $\Phi = a\partial_t v_h + d\Psi_h$, with $a, d \in \mathbb{R}$, the derivative of H at v_h in the direction Φ satisfies

$$DH_{v_h}[\Phi](t) = d \tag{4.15}$$

for every $t \in \mathbb{R}$. In particular, the derivative vanishes if and only if $\Phi \in \mathrm{span}\{\partial_t v_h\}$.

Proof. By Proposition 4.3, $\mathcal{K}_{\mathrm{pol}}(v_h) = \mathcal{K}^{\mathrm{sym}}_{\mathrm{pol}}(v_h)$. The description of the latter space in (3.15) gives (4.14). By Lemma 3.2 and (3.15),

$$DH_{v_h}[\Phi](t) = a\,\Omega(\partial_t v_h, \partial_t v_h) + d\,\Omega(\partial_t v_h, \Psi_h) = d.$$

The first term is zero by skew-symmetry, and the pairing in the second term is equal to 1. □

Corollary 4.5. *For $H_c < h < 0$,*

$$\mathcal{K}_{\mathrm{bd}}(v_h) = \begin{cases} \mathrm{span}\{\partial_t v_h\}, & T'(h) \neq 0, \\ \mathrm{span}\{\partial_t v_h, \Psi_h\}, & T'(h) = 0. \end{cases} \tag{4.16}$$

Every field in $\mathcal{K}_{\mathrm{bd}}(v_h)$ is $T(h)$-periodic in t.

Proof. Proposition 4.3 gives $\mathcal{K}_{\mathrm{bd}}(v_h) = \mathcal{K}^{\mathrm{sym}}_{\mathrm{bd}}(v_h)$. Corollary 3.5 then gives (4.16). The field $\partial_t v_h$ is $T(h)$-periodic, and when $T'(h) = 0$, (3.16) gives the same periodicity for Ψ_h. □

4.2. **The cylindrical solution.** We use the eigenvalue problem on $\mathbb{S}^{n-1}_+$ with the boundary condition obtained by linearizing (2.10) at C_n. Its negative eigenvalue gives two bounded Jacobi fields. The uniqueness results for even and odd fields then give the classification of $\mathcal{K}_{\mathrm{pol}}(C_n)$. We also compute the quadratic term in the expansion of the Hamiltonian at C_n.

On $L^2((0,L),\omega(\beta)\,\mathrm{d}\beta)$, let $\mathcal{A}_{nc/(n-2)}$ be the self-adjoint operator associated with the following quadratic form, defined on $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$,

$$\int_0^L \omega(\beta)\left(|f'(\beta)|^2 + \frac{(n-2)^2}{4}|f(\beta)|^2\right)\mathrm{d}\beta - \frac{nc}{n-2}|f(L)|^2.$$

This is the operator of [3, Eqs. (A.21)–(A.23)] with the boundary coefficient $nc/(n-2)$ from (2.13) at C_n. It has compact resolvent, and its eigenfunctions are smooth on $\mathbb{S}^{n-1}_+$. The operator acts on $0 < \beta < L$ by the first line below, and its domain imposes the boundary condition on the second line.

$$\begin{aligned} \mathcal{A}_{nc/(n-2)} f &= -\frac{1}{\omega}(\omega f')' + \frac{(n-2)^2}{4} f, \\ f'(L) &= \frac{nc}{n-2} f(L). \end{aligned} \tag{4.17}$$

At the point $(0,\dots,0,1)$, corresponding to $\beta = 0$, we use the $H^1_{\mathrm{sym}}(\mathbb{S}^{n-1}_+)$ condition on the hemisphere itself. The weighted integral includes smooth functions with $f(0) \neq 0$ also when $n = 3$.

Proposition 4.6. *Let μ_0 and μ_1 be the first two eigenvalues of the operator in (4.17). Then*

$$\mu_0 = -\xi_*^2 < 0 < \frac{n^2}{4} < \mu_1, \tag{4.18}$$

where $\xi_* > 0$ is the solution of $\Theta_n(\xi_*) = nc/(n-2)$ in [3, Eq. (3.51)]. Here $\Theta_n(\xi) = A'_\xi(L)$, with A_ξ defined in [3, Eq. (3.47)]. The eigenfunction for μ_0 with value 1 at $\beta = L$ is A_{ξ_*}, which is positive on $\mathbb{S}^{n-1}_+$. Its equation on $0 < \beta < L$ and its boundary values at $\beta = L$ are

$$\begin{aligned} -\frac{1}{\omega}(\omega A'_{\xi_*})' + \frac{(n-2)^2}{4} A_{\xi_*} &= -\xi_*^2 A_{\xi_*}, \\ A_{\xi_*}(L) &= 1, \\ A'_{\xi_*}(L) &= \frac{nc}{n-2} A_{\xi_*}(L). \end{aligned} \tag{4.19}$$

Proof. Equations (3.47) and (3.51) of [3] give (4.19). The Dirichlet problem for A_{ξ_*} on $\mathbb{S}^{n-1}_+$ has boundary value 1 and zeroth-order coefficient $(n-2)^2/4 + \xi_*^2 > 0$. The maximum principle therefore gives $A_{\xi_*} > 0$. By [3, Lemma A.6], $\mu_1 > n^2/4$ for $n \geq 3$. Thus $-\xi_*^2$ is the only negative eigenvalue and equals μ_0. □

Theorem C. *Let A_{ξ_*} be as in (4.19). For each $\Phi \in \mathcal{K}_{\mathrm{pol}}(C_n)$ there are $a, d \in \mathbb{R}$ such that, on $\mathbb{R} \times \mathbb{S}^{n-1}_+$,*

$$\Phi(t,\theta) = A_{\xi_*}(\beta)\Big(a\cos(\xi_* t) + d\sin(\xi_* t)\Big). \tag{4.20}$$

In particular,

$$\begin{aligned} \mathcal{K}_{\mathrm{pol}}(C_n) &= \mathcal{K}_{\mathrm{bd}}(C_n) \\ &= \mathrm{span}\Big\{A_{\xi_*}(\beta)\cos(\xi_* t), A_{\xi_*}(\beta)\sin(\xi_* t)\Big\}. \end{aligned}$$

Proof. By (4.19), both functions in the span satisfy (2.13) at C_n and are bounded (the cosine field is constructed after equation (3.52) in [3]).

By Proposition 4.3, each $\Phi \in \mathcal{K}_{\mathrm{pol}}(C_n)$ is cylindrically symmetric. Its even and odd parts in t, denoted by Φ_{even} and Φ_{odd}, belong to $\mathcal{K}_{\mathrm{pol}}(C_n)$ because the coefficients in (2.13) at C_n are independent of t. Since $A_{\xi_*}(L) = 1$, the even field

$$\Phi_{\mathrm{even}}(t,\theta) - \Phi(0,L)A_{\xi_*}(\beta)\cos(\xi_* t)$$

vanishes at $(0, L)$, and the odd field

$$\Phi_{\mathrm{odd}}(t,\theta) - \frac{\partial_t \Phi(0,L)}{\xi_*} A_{\xi_*}(\beta)\sin(\xi_* t)$$

has zero t-derivative at $(0, L)$. The cylindrical solution C_n is even in t and has trace 1. Thus [3, Proposition 3.1(iii)] applies to both fields and forces them to vanish. This proves (4.20). □

Fix $y \in C^\infty(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ and $t \in \mathbb{R}$. The trace of $C_n + sy$ on $\{t\} \times \partial\mathbb{S}^{n-1}_+$ is $1 + s\tilde{y}(t, \vartheta)$, which is positive for small $|s|$ because $\tilde{y}(t, \cdot)$ is bounded on $\partial\mathbb{S}^{n-1}_+$. By (4.13) and $\partial_t C_n = 0$,

$$DH_{C_n}[y](t) = 0. \tag{4.21}$$

Define the quadratic term for $|\mathbb{S}^{n-2}|H$ by

$$\begin{aligned} \mathcal{Q}_c[y](t) = \frac{1}{2} \int_{\mathbb{S}^{n-1}_+} \left((\partial_t y)^2 - |\nabla_{\mathbb{S}^{n-1}_+} y|^2 - \frac{(n-2)^2}{4} y^2 \right) \mathrm{d}\theta \\ + \frac{nc}{2(n-2)} \int_{\partial\mathbb{S}^{n-1}_+} \tilde{y}^2 \, \mathrm{d}\sigma. \end{aligned} \tag{4.22}$$

Proposition 4.8. *Let $\Phi \in \mathcal{K}_{\mathrm{pol}}(C_n)$, and let a, d be its coefficients in (4.20). For every $t \in \mathbb{R}$,*

$$\mathcal{Q}_c[\Phi](t) = \frac{\xi_*^2}{2}(a^2 + d^2) \int_{\mathbb{S}^{n-1}_+} A_{\xi_*}^2 \, \mathrm{d}\theta. \tag{4.23}$$

In particular, $\mathcal{Q}_c[\Phi](t) = 0$ at any t if and only if $\Phi = 0$.

Proof. Multiply the equation for A_{ξ_*} in (4.19) by A_{ξ_*} and integrate by parts on $\mathbb{S}^{n-1}_+$. Using the conormal derivative at $\beta = L$ from the same equation gives

$$\begin{aligned} \int_{\mathbb{S}^{n-1}_+} \left(|\nabla_{\mathbb{S}^{n-1}_+} A_{\xi_*}|^2 + \frac{(n-2)^2}{4} A_{\xi_*}^2 \right) \mathrm{d}\theta \\ - \frac{nc}{n-2} \int_{\partial\mathbb{S}^{n-1}_+} A_{\xi_*}^2 \, \mathrm{d}\sigma = -\xi_*^2 \int_{\mathbb{S}^{n-1}_+} A_{\xi_*}^2 \, \mathrm{d}\theta. \end{aligned}$$

Substitute (4.20) in (4.22) and use this identity. The result is

$$\begin{aligned} \mathcal{Q}_c[\Phi](t) = \frac{\xi_*^2}{2} \int_{\mathbb{S}^{n-1}_+} A_{\xi_*}^2 \, \mathrm{d}\theta \\ \times \left[\Big(-a \sin(\xi_* t) + d \cos(\xi_* t) \Big)^2 + \Big(a \cos(\xi_* t) + d \sin(\xi_* t) \Big)^2 \right]. \end{aligned}$$

The mixed terms cancel, and $\cos^2(\xi_* t) + \sin^2(\xi_* t) = 1$ gives (4.23). □

To expand $H(C_n + sy)(t)$, substitute $C_n + sy$ in (2.11) with $a = c$. The integral over $\mathbb{S}^{n-1}_+$ is a polynomial in s of degree at most two. The boundary integral contains the factor $(1 + s\tilde{y}(t, \vartheta))^{2(n-1)/(n-2)}$. For $|s| \sup_\vartheta |\tilde{y}(t, \vartheta)| \le 1/2$, its argument lies in $[1/2, 3/2]$ at every ϑ. The

function $x \mapsto x^{2(n-1)/(n-2)}$ has bounded third derivative on this interval. Taylor's formula at $x = 1$ therefore bounds the error in the boundary integral by

$$C_{n,c}|s|^3 \int_{\partial \mathbb{S}^{n-1}_+} |\tilde{y}(t,\vartheta)|^3 \, \mathrm{d}\sigma.$$

The linear term vanishes by (4.21). Combining the quadratic terms gives (4.22). Thus, at a fixed t, as $s \to 0$,

$$\begin{aligned} |\mathbb{S}^{n-2}|H(C_n + sy)(t) = |\mathbb{S}^{n-2}|H(C_n) \\ + s^2 \mathcal{Q}_c[y](t) + O(|s|^3). \end{aligned} \tag{4.24}$$

If $\tilde{y}$ is bounded on $\mathbb{R} \times \partial \mathbb{S}^{n-1}_+$, the same choice of $|s|$ works for all t, and the bound on the remainder is uniform in t. This applies to the Jacobi fields in (4.20), whose traces are linear combinations of $\cos(\xi_* t)$ and $\sin(\xi_* t)$.

4.3. **The L^2 kernels on the cylinder and its boundary.** We define the self-adjoint realizations of the linearized equations on $\mathbb{R} \times \mathbb{S}^{n-1}_+$ and on $\mathbb{R} \times \partial \mathbb{S}^{n-1}_+$. The periodicity of bounded Jacobi fields implies that both L^2 kernels are trivial. A sequence of cutoffs of a periodic Jacobi field then places zero in the essential spectrum of the operator on $\mathbb{R} \times \mathbb{S}^{n-1}_+$.

For $y, z \in H^1(\mathbb{R} \times \mathbb{S}^{n-1}_+)$, we define $Q_v(y,z)$ using the integrands in (2.18), with integration in t over $\mathbb{R}$.

$$\begin{aligned} Q_v(y,z) = \int_{\mathbb{R}} \int_{\mathbb{S}^{n-1}_+} \left(y_t z_t + \langle \nabla_{\mathbb{S}^{n-1}_+} y, \nabla_{\mathbb{S}^{n-1}_+} z \rangle + \frac{(n-2)^2}{4} yz \right) \mathrm{d}\theta \, \mathrm{d}t \\ - \frac{nc}{n-2} \int_{\mathbb{R}} \int_{\partial \mathbb{S}^{n-1}_+} \tilde{v}^{2/(n-2)} \tilde{y} \, \tilde{z} \, \mathrm{d}\sigma \, \mathrm{d}t. \end{aligned} \tag{4.25}$$

Write $Q_0(y,z)$ for the integral over $\mathbb{R} \times \mathbb{S}^{n-1}_+$ in (4.25). We write the forms for real functions and take their sesquilinear extensions on complex functions when using the spectral theorem.

The constants in (2.16) are independent of the interval I. Apply that inequality on $[-R, R]$ and let $R \to \infty$. The resulting inequality holds for all $y \in H^1(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ by the continuity of the trace map in Section 2.3. The trace $\tilde{v}$ is bounded by periodicity. Choose the coefficient of the angular derivative term in (2.16) small enough that the absolute value of the boundary term in $Q_v(y,y)$ is bounded by

$$\frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{S}^{n-1}_+} |\nabla_{\mathbb{S}^{n-1}_+} y|^2 \, \mathrm{d}\theta \, \mathrm{d}t + C \|y\|^2_{L^2(\mathbb{R} \times \mathbb{S}^{n-1}_+)}.$$

For a sufficiently large $\lambda > 0$, the square root of

$$Q_v(y,y) + \lambda \|y\|^2_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)}$$

is thus equivalent to the norm on $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. In particular, Q_v is bounded below, and $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ is complete in this equivalent norm. This proves that Q_v is closed.

Let $\mathcal{A}_v$ be the self-adjoint operator on $L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ associated with Q_v. Its domain consists of the functions $y \in H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ for which there is an $f \in L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ such that

$$Q_v(y,z) = \langle f, z\rangle_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)} \tag{4.26}$$

for every $z \in H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. For such y, we set $\mathcal{A}_v y = f$. By integration by parts, (4.26) is the weak formulation of $\mathcal{A}y = f$ with $\mathcal{B}_v y = 0$, with the operators from (2.13).

Let $f \in H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$. We denote by Ef the function in $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ whose trace on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$ equals f in the Sobolev trace sense and that satisfies

$$Q_0(Ef, z) = 0 \tag{4.27}$$

for every $z \in H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ with $z|_{\mathbb{R}\times\partial\mathbb{S}^{n-1}_+} = 0$. In particular, Ef solves $\mathcal{A}(Ef) = 0$ in the weak sense in $\mathbb{R}\times\operatorname{int}\mathbb{S}^{n-1}_+$.

To construct Ef, use the right inverse of the trace from Section 2.3 to choose $y_f \in H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ whose trace equals f and whose $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ norm is bounded by a constant times $\|f\|_{H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)}$. Since $n \geq 3$, the coefficient $(n-2)^2/4$ is positive, so $Q_0(y,y)$ is bounded below by a positive multiple of $\|y\|^2_{H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)}$. The functions with trace zero on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$ form a closed subspace of $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$. The Lax–Milgram theorem on this subspace gives a unique w with $Q_0(w,z) = -Q_0(y_f, z)$ for all z in the same subspace. Then $Ef = y_f + w$ has trace f and satisfies (4.27). Uniqueness also follows because the difference of two such extensions has trace zero and its Q_0 energy vanishes by (4.27).

For any $y \in H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ with trace f, the difference $y - Ef$ has trace zero on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$. Consequently,

$$Q_0(y,y) = Q_0(Ef, Ef) + Q_0(y - Ef, y - Ef)$$

by (4.27). Thus Ef minimizes the energy among extensions of f. For the upper bound below, use $Q_0(Ef, Ef) \leq Q_0(y_f, y_f)$ and the norm bound for y_f. For the lower bound, continuity of the trace map bounds the $H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ norm of f by a constant times the $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ norm of Ef. The squared $H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ norm of Ef is bounded by a

constant times $Q_0(Ef, Ef)$. This gives

$$\begin{aligned} C^{-1}\|f\|^2_{H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)} &\le Q_0(Ef, Ef) \\ &\le C\|f\|^2_{H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)}. \end{aligned} \tag{4.28}$$

We define the Dirichlet-to-Neumann map

$$P : H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+) \longrightarrow H^{-1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$$

by $\langle Pf, g\rangle = Q_0(Ef, Eg)$ for every $g \in H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$. Here $\langle\cdot,\cdot\rangle$ denotes the dual pairing. By Cauchy–Schwarz for Q_0 and (4.28),

$$|\langle Pf, g\rangle| \le C\|f\|_{H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)}\|g\|_{H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)}.$$

To relate this definition to the conormal derivative, take $z \in H^1(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ and write $z = E(\tilde{z}) + (z - E(\tilde{z}))$. The second summand has trace zero on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$, so (4.27) gives

$$Q_0(Ef, z) = \langle Pf, \tilde{z}\rangle.$$

If f is smooth, integration by parts using $\mathcal{A}(Ef) = 0$ identifies Pf with $\partial_{\nu_{\mathbb{S}^{n-1}_+}}(Ef)$ on $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$.

For $f, g \in H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$, define

$$Q^{\partial}_v(f, g) = \langle Pf, g\rangle - \frac{nc}{n-2}\int_{\mathbb{R}}\int_{\partial\mathbb{S}^{n-1}_+} \tilde{v}^{2/(n-2)} fg \, \mathrm{d}\sigma \, \mathrm{d}t. \tag{4.29}$$

This form is closed and bounded below on $L^2(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ by (4.28) and boundedness of $\tilde{v}$. Its self-adjoint operator $\mathcal{J}_v$ acts by

$$\mathcal{J}_v f = Pf - \frac{nc}{n-2}\tilde{v}^{2/(n-2)} f. \tag{4.30}$$

The right-hand side is first defined in $H^{-1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$. The domain of $\mathcal{J}_v$ consists of the functions $f \in H^{1/2}(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ for which this right-hand side belongs to $L^2(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$. Indeed, this is precisely when the functional $g \mapsto Q^{\partial}_v(f, g)$ is represented by an element of $L^2(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$ under the L^2 pairing.

Corollary 4.9. *For the solutions v considered in this section,*

$$\begin{aligned} \ker\mathcal{A}_v &= \{0\}, \\ \ker\mathcal{J}_v &= \{0\}. \end{aligned} \tag{4.31}$$

The kernels are taken in $L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ and $L^2(\mathbb{R}\times\partial\mathbb{S}^{n-1}_+)$, respectively.

Proof. Let $\Phi \in \ker\mathcal{A}_v$. Equation (4.26) with $f = 0$ is the weak formulation of (2.13). Interior elliptic regularity gives smoothness in $\mathbb{R}\times\operatorname{int}\mathbb{S}^{n-1}_+$. Near $\mathbb{R}\times\partial\mathbb{S}^{n-1}_+$, the conormal derivative is prescribed by $\mathcal{B}_v\Phi = 0$ as a smooth coefficient times the trace of Φ. Starting from

$\Phi \in H^1(\mathbb{R} \times \mathbb{S}^{n-1}_+)$, the trace belongs locally to $H^{1/2}$ on the boundary, so the local estimate for the prescribed conormal derivative gives H^2 regularity on smaller products with $\mathbb{S}^{n-1}_+$. The improved trace regularity then allows the estimate to be applied again, as in the proof of Lemma 2.2. Repeating this step gives $\Phi \in C^\infty(\mathbb{R} \times \mathbb{S}^{n-1}_+)$.

For every $a \in \mathbb{R}$,

$$\|\Phi\|_{L^2([a-1,a+1]\times\mathbb{S}^{n-1}_+)} \leq \|\Phi\|_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)}.$$

By Lemma 2.2, this uniform bound gives $\Phi \in \mathcal{K}_{\mathrm{bd}}(v)$. By Corollary 4.5, Φ is $T(h)$-periodic when $v = v_h$. By Theorem C, Φ is $2\pi/\xi_*$-periodic when $v = C_n$. Denote the corresponding period by τ. If $\Phi \neq 0$, the integral of $|\Phi|^2$ over $[0,\tau] \times \mathbb{S}^{n-1}_+$ is positive. Periodicity gives the same positive integral over every $[k\tau,(k+1)\tau] \times \mathbb{S}^{n-1}_+$ with $k \in \mathbb{Z}$, contrary to $\Phi \in L^2(\mathbb{R} \times \mathbb{S}^{n-1}_+)$. Thus $\Phi = 0$.

Let $f \in \ker \mathcal{J}_v$ and let $z \in H^1(\mathbb{R} \times \mathbb{S}^{n-1}_+)$. Since the trace of Ef equals f, the identity $Q_0(Ef, z) = \langle Pf, \tilde{z}\rangle$ gives

$$Q_v(Ef, z) = Q^\partial_v(f, \tilde{z}) = \langle \mathcal{J}_v f, \tilde{z}\rangle = 0.$$

Equation (4.26) now gives $Ef \in \ker \mathcal{A}_v$. Hence $Ef = 0$, and its trace is $f = 0$. □

We use the Weyl criterion to show that 0 belongs to the essential spectrum of $\mathcal{A}_v$. For a self-adjoint operator, the criterion states that a real number λ belongs to the essential spectrum if and only if there is a sequence of unit vectors in the operator domain that converges weakly to zero and whose residuals at λ converge to zero in norm. In our case, $\lambda = 0$, so it suffices to find unit vectors $e_j \in D(\mathcal{A}_v)$ such that

$$e_j \rightharpoonup 0 \quad \text{and} \quad \|\mathcal{A}_v e_j\|_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)} \to 0.$$

The first convergence is weak in $L^2(\mathbb{R} \times \mathbb{S}^{n-1}_+)$. It matters because, if 0 were an isolated eigenvalue of finite multiplicity, the projections of e_j onto its eigenspace would converge to zero in norm. The components orthogonal to the eigenspace would then have norms tending to 1. On that orthogonal complement, the spectral gap around 0 would give a positive lower bound for $\|\mathcal{A}_v e_j\|_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)}$. If 0 belonged to the resolvent set, the same lower bound would follow from boundedness of $\mathcal{A}_v^{-1}$.

Let $\Phi_0 = \partial_t v_h$ when $v = v_h$ and $\Phi_0(t,\theta) = A_{\xi_*}(\beta)\cos(\xi_* t)$ when $v = C_n$, with A_{ξ_*} as in (4.19). These are nonzero periodic Jacobi fields by Theorems B and C. Choose a nonzero $\chi \in C^\infty_c((1,2))$. For

sufficiently large R, set

$$y_R(t,\theta) = \chi(t/R)\Phi_0(t,\theta),$$
$$e_R = \frac{y_R}{\|y_R\|_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)}}.$$

Since the cutoff depends only on t, $\mathcal{B}_v y_R = \chi(t/R)\mathcal{B}_v\Phi_0 = 0$. Integration by parts in (4.25) shows that y_R satisfies (4.26) with $f = \mathcal{A}y_R$. Thus $y_R \in D(\mathcal{A}_v)$ and

$$\mathcal{A}_v y_R = -\frac{2}{R}\chi'(t/R)\partial_t\Phi_0 - \frac{1}{R^2}\chi''(t/R)\Phi_0.$$

The functions Φ_0 and $\partial_t\Phi_0$ are bounded by periodicity, and the terms on the right are supported in $[R,2R]\times\mathbb{S}^{n-1}_+$. Their $L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ norms are bounded by $CR^{-1/2}$ and $CR^{-3/2}$, respectively.

Choose $[a,b]\subset(1,2)$ and $\eta>0$ such that $|\chi|\geq\eta$ on $[a,b]$. If τ is a period of Φ_0, the interval $[aR,bR]$ contains $\lfloor(b-a)R/\tau\rfloor$ consecutive intervals of length τ. Periodicity gives

$$\|y_R\|^2_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)} \geq \eta^2\left\lfloor\frac{(b-a)R}{\tau}\right\rfloor\int_0^\tau\int_{\mathbb{S}^{n-1}_+}|\Phi_0|^2\,\mathrm{d}\theta\,\mathrm{d}t.$$

The last integral is positive because $\Phi_0\neq 0$. For large R, the lower bound for the norm of y_R is therefore a positive multiple of $R^{1/2}$. Dividing the bounds for $\mathcal{A}_v y_R$ by this norm shows that $\|\mathcal{A}_v e_R\|_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)} = O(R^{-1})$. For $z\in L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)$, Cauchy–Schwarz and the support of e_R give

$$|\langle e_R, z\rangle_{L^2(\mathbb{R}\times\mathbb{S}^{n-1}_+)}| \leq \|z\|_{L^2([R,\infty)\times\mathbb{S}^{n-1}_+)}.$$

The right-hand side tends to zero as $R\to\infty$ by square integrability of z. This proves weak convergence to zero. The Weyl criterion now gives 0 in the essential spectrum of $\mathcal{A}_v$.

5. Asymptotics at an isolated boundary singularity

Section 5.1 proves Theorem D, which gives convergence to a fixed member of the flat family. Section 5.2 studies conformal deformations of the flat solutions and gives a first-order obstruction in boundary Fermi coordinates.

5.1. **Asymptotic convergence.**

For a noncylindrical limiting solution, we select a translation parameter on each fixed interval in t by an orthogonality condition. We use the first variation of the Hamiltonian to estimate the error and show that the parameter minus t converges. At the cylindrical solution, the quadratic term of the Hamiltonian gives the error estimate.

For $0 < \varepsilon \le 1$ and $\tau \in \mathbb{R}$, let $U_{\varepsilon,\tau}$ be the flat solution with boundary coefficient κ obtained from v_ε in Section 2.4. For $r > 0$ and $\theta \in \mathbb{S}^{n-1}_+$, with A_κ as in (2.20), we have

$$U_{\varepsilon,\tau}(r\theta) = A_\kappa^{-1} r^{-\frac{n-2}{2}} v_\varepsilon(-\log r + \tau, \theta). \tag{5.1}$$

Theorem D. *Let $n \ge 3$ and let g be a smooth metric in boundary Fermi coordinates centered at p, which is represented by 0. Let U be a positive smooth solution of (2.4) on $B^+_\rho \setminus \{0\}$. Suppose that there are $c_0, C_0 > 0$ such that the trace $u(\overline{x}) = U(\overline{x}, 0)$ satisfies, for $0 < |\overline{x}| < \rho$,*

$$c_0 |\overline{x}|^{-\frac{n-2}{2}} \le u(\overline{x}) \le C_0 |\overline{x}|^{-\frac{n-2}{2}}. \tag{5.2}$$

Then there are $\delta > 0$, $\rho_0 \in (0, \rho)$, $\varepsilon_* \in (0, 1]$, $\tau_* \in \mathbb{R}$, and $C > 0$ such that $U_* = U_{\varepsilon_*,\tau_*}$ satisfies

$$\sum_{j=0}^{2} |x|^{\frac{n-2}{2}+j} |D^j(U - U_*)(x)| \le C|x|^\delta \tag{5.3}$$

for $x \in \mathbb{R}^n_+$ with $0 < |x| < \rho_0$. In particular, $U(x) = (1 + O(|x|^\delta))U_*(x)$ as $x \to 0$.

The necksize ε_* is unique. If $\varepsilon_* < 1$, the translation parameter τ_* is unique modulo T_{ε_*}. If $\varepsilon_* = 1$, all values of τ_* give the same cylindrical solution. For the same U_*, the estimate holds with every $\delta' \in (0, \delta]$ after adjusting C.

Proof. Let w be the transform of U in (2.20). Proposition 2.4 gives two-sided positive bounds and uniform local smooth estimates for w, the error estimate (2.28), and the Hamiltonian limit (2.29). We prove that there are $\varepsilon_* \in (0, 1]$, $\tau_* \in \mathbb{R}$, and $\delta > 0$ such that, for all sufficiently large t,

$$\|w(t + \cdot, \cdot) - v_{\varepsilon_*}(t + \cdot + \tau_*, \cdot)\|_{C^2([-1,1]\times\mathbb{S}^{n-1}_+)} \le Ce^{-\delta t}. \tag{5.4}$$

Identification of the limiting necksize. Let $t_j \to \infty$. The estimates in Proposition 2.4 give, after passing to a subsequence, convergence in $C^k_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$ for every integer $k \ge 0$,

$$w(t_j + \cdot, \cdot) \longrightarrow v_\infty. \tag{5.5}$$

The convergence holds up to the equator. The positive upper and lower bounds pass to v_∞. By (2.28), the error terms in (2.26) tend to zero on every compact interval in the translated t variable. Thus v_∞ solves (2.10) on the whole cylinder. For each $s \in \mathbb{R}$, the convergence on the section at s and (2.29) give

$$|\mathbb{S}^{n-2}| H(v_\infty)(s) = H_0. \tag{5.6}$$

By the normalization in Section 2.2, $b_n r^{-(n-2)/2} v_\infty(-\log r, \theta)$ defines a positive solution of (2.5) on $\mathbb{R}^n_+ \setminus \{0\}$. The lower bound for the trace of v_∞ makes the singularity at 0 nonremovable. Theorem B(ii) of [3] therefore gives $v_\infty = v_\varepsilon(\cdot + \tau, \cdot)$ for some $\varepsilon \in (0, 1]$ and $\tau \in \mathbb{R}$. By Theorem A, $H(v_\varepsilon)$ decreases strictly from 0 to H_c as ε increases from 0 to 1, with $v_1 = C_n$ and $H(C_n) = H_c$. It follows from (5.6) that every translated limit has the same necksize $\varepsilon_* \in (0, 1]$, determined by

$$H(v_{\varepsilon_*}) = \frac{H_0}{|\mathbb{S}^{n-2}|}. \tag{5.7}$$

Write $v_* = v_{\varepsilon_*}$. For every compact interval $I \subset \mathbb{R}$, as $t \to \infty$ we have

$$\inf_{a \in \mathbb{R}} \|w(t + \cdot, \cdot) - v_*(\cdot + a, \cdot)\|_{C^2(I \times \mathbb{S}^{n-1}_+)} \longrightarrow 0. \tag{5.8}$$

Otherwise there is a sequence of translates whose distance from every $v_*(\cdot + a, \cdot)$ stays bounded below by a positive constant. A subsequence converges by (5.5) and its limit is a translate of v_*, a contradiction.

Choice of the translation parameter. Suppose that $0 < \varepsilon_* < 1$ and put $T = T_{\varepsilon_*}$. Choose $a_0 > \max\{T, 1\}$ and set $J = [-a_0, a_0]$. Let $\chi \in C^\infty_c((-a_0, a_0))$ be nonnegative and positive on an interval of length greater than T. In $L^2(J \times \mathbb{S}^{n-1}_+, \chi(s)\, \mathrm{d}s\, \mathrm{d}\theta)$, use

$$\langle F, G \rangle_\chi = \int_J \int_{\mathbb{S}^{n-1}_+} \chi(s) F G \, \mathrm{d}\theta \, \mathrm{d}s,$$
$$\|F\|^2_\chi = \langle F, F \rangle_\chi.$$

Write $\Gamma(a) = v_*(\cdot + a, \cdot)$ for $a \in \mathbb{R}$. This map is smooth and T-periodic. If $\|\Gamma(a) - \Gamma(b)\|_\chi = 0$, the two translates agree on an interval of length greater than T and hence everywhere by periodicity. Since T is the least positive period, $a - b \in T\mathbb{Z}$. Also $\|\Gamma'(a)\|_\chi > 0$ for every $a \in \mathbb{R}$. If this norm vanished, $\partial_t v_*$ would vanish on an interval of length greater than T and hence on the whole cylinder, contrary to $0 < \varepsilon_* < 1$.

For $F \in L^2(J \times \mathbb{S}^{n-1}_+, \chi(s)\, \mathrm{d}s\, \mathrm{d}\theta)$, minimize $\|F - \Gamma(a)\|_\chi$ over $a \in [0, T]$. A minimizer exists by compactness and satisfies

$$\langle F - \Gamma(a), \Gamma'(a) \rangle_\chi = 0.$$

Since $a \mapsto \|F - \Gamma(a)\|^2_\chi$ is T-periodic, a minimizer on $[0, T]$ is also a minimizer on $\mathbb{R}$. Thus the orthogonality equation holds even when the minimizer is represented by 0 or T.

Fix $\alpha \in \mathbb{R}$. At $F = \Gamma(\alpha)$ and $a = \alpha$, the derivative of the orthogonality equation with respect to a, holding F fixed, is

$$\frac{\partial}{\partial a} \langle F - \Gamma(a), \Gamma'(a) \rangle_\chi \Big|_{F = \Gamma(\alpha),\, a = \alpha} = -\|\Gamma'(\alpha)\|^2_\chi < 0.$$

The implicit function theorem therefore gives, for F in a neighborhood of $\Gamma(\alpha)$, a unique solution a near α. This solution depends smoothly on F.

We claim that there is $\eta > 0$ such that the minimizing parameter is unique modulo T whenever

$$\inf_{a\in[0,T]} \|F - \Gamma(a)\|_\chi < \eta.$$

Otherwise there are functions F_j and minimizing parameters $a_j, b_j \in [0,T]$ such that $a_j - b_j \notin T\mathbb{Z}$ and

$$\|F_j - \Gamma(a_j)\|_\chi = \|F_j - \Gamma(b_j)\|_\chi \longrightarrow 0.$$

By compactness, after passing to a subsequence we have $a_j \to a_\infty$ and $b_j \to b_\infty$. The triangle inequality gives $\Gamma(a_\infty) = \Gamma(b_\infty)$. Hence $b_\infty - a_\infty = kT$ for some $k \in \mathbb{Z}$. By periodicity, $b_j - kT$ is also a minimizing parameter, and both a_j and $b_j - kT$ tend to a_∞. Since $F_j \to \Gamma(a_\infty)$ in $\|\cdot\|_\chi$, the local uniqueness given by the implicit function theorem implies $a_j = b_j - kT$ for all sufficiently large j. This contradicts $a_j - b_j \notin T\mathbb{Z}$.

By (5.8) with $I = J$, and since the weighted norm is bounded by a constant times the C^0 norm on $J \times \mathbb{S}^{n-1}_+$,

$$\inf_{a\in[0,T]} \|w(t+\cdot,\cdot) - \Gamma(a)\|_\chi \longrightarrow 0 \qquad \text{as } t \to \infty.$$

Thus, for all sufficiently large t, the minimizing parameter is unique modulo T.

At a minimizing parameter a, the derivative with respect to a of the orthogonality equation is

$$-\|\Gamma'(a)\|^2_\chi + \langle w(t+\cdot,\cdot) - \Gamma(a), \Gamma''(a)\rangle_\chi.$$

The first term is bounded above by a negative constant, since Γ' never vanishes and is T-periodic. The second tends to zero uniformly over the minimizing parameters, since Γ'' is bounded and the minimizing distance tends to zero. The implicit function theorem therefore gives a local C^1 solution of the orthogonality equation through each minimizing parameter.

These local solutions give the minimizing parameters for nearby times. Indeed, if $t_j \to t$, any limit of minimizing representatives at t_j in $[0,T]$ minimizes the distance at t, by continuity. Uniqueness modulo T then allows the representatives at t_j to be chosen to converge to the representative at t. Local uniqueness in the implicit function theorem identifies them with the local C^1 solution. Let a_1 and a_2 be local representatives defined on intervals I_1 and I_2 with nonempty intersection. By uniqueness modulo T, we have $a_2(t) - a_1(t) \in T\mathbb{Z}$ for

every $t \in I_1 \cap I_2$. Since $a_2 - a_1$ is continuous on the interval $I_1 \cap I_2$ and $T\mathbb{Z}$ is discrete, there is a fixed $k \in \mathbb{Z}$ such that $a_2(t) - a_1(t) = kT$ throughout the intersection. Replacing a_2 by $a_2 - kT$ leaves the minimizing translate unchanged and makes the two representatives agree on $I_1 \cap I_2$. Fix a sufficiently large initial time t_0 and a real minimizing representative at t_0. By compactness, we can subdivide each interval $[t_0 + k, t_0 + k + 1]$, with k a nonnegative integer, into finitely many subintervals, each contained in an open interval carrying a local C^1 minimizing representative. The subdivision points give a sequence $t_0 = s_0 < s_1 < \cdots$ tending to infinity. For each j, let a_j be a local representative on an open interval containing $[s_j, s_{j+1}]$.

Add an integer multiple of T to a_0 so that $a_0(t_0)$ equals the chosen initial value. Having fixed a_j, add the constant $a_j(s_{j+1}) - a_{j+1}(s_{j+1})$ to a_{j+1}. This constant belongs to $T\mathbb{Z}$ by uniqueness modulo T. The adjusted representatives agree at s_{j+1} and therefore agree on a neighborhood of s_{j+1}, since their difference is constant on the overlap.

Define $a(t) = a_j(t)$ for $t \in [s_j, s_{j+1}]$, using the adjusted representatives. Agreement near each junction makes a a real-valued C^1 function on $[t_0, \infty)$, and $a(t)$ is a minimizing representative at every t.

Define E_t on $J \times \mathbb{S}^{n-1}_+$. The minimizing property of $a(t)$ gives the orthogonality to the translation field.

$$\begin{aligned} E_t(s,\theta) &= w(t+s,\theta) - v_*(s + a(t), \theta), \\ 0 &= \int_J \int_{\mathbb{S}^{n-1}_+} \chi(s) E_t(s,\theta)\, \partial_t v_*(s + a(t), \theta)\, \mathrm{d}\theta\, \mathrm{d}s. \end{aligned} \tag{5.9}$$

Define

$$d(t) = \|E_t\|_{C^2(J \times \mathbb{S}^{n-1}_+)}. \tag{5.10}$$

This is a continuous function of t. To show that $d(t) \to 0$, choose parameters $a_1(t)$ for which the error in (5.8) tends to zero on J. The minimizing property of $a(t)$ and the triangle inequality give

$$\|\Gamma(a(t)) - \Gamma(a_1(t))\|_\chi \leq 2\|w(t + \cdot, \cdot) - \Gamma(a_1(t))\|_\chi \longrightarrow 0.$$

By compactness of $[0, T]$ and the characterization of equal translates, the distance from $a(t) - a_1(t)$ to $T\mathbb{Z}$ tends to zero. Smoothness and periodicity of v_* then give convergence of the difference of these translates to zero in $C^2(J \times \mathbb{S}^{n-1}_+)$, so $d(t) \to 0$.

Differentiate the second line of (5.9) in t and use $\partial_t w(t + s, \theta) = \partial_s w(t + s, \theta)$. Writing $w(t + s, \theta) = E_t(s, \theta) + v_*(s + a(t), \theta)$ and

integrating the term with $\partial_s E_t$ by parts gives

$$\begin{aligned}(a'(t)-1)\int_J\int_{\mathbb{S}^{n-1}_+}\chi(s)\Big((\partial_t v_*)^2-E_t\partial_t^2 v_*\Big)\,\mathrm{d}\theta\,\mathrm{d}s\\ =-\int_J\int_{\mathbb{S}^{n-1}_+}\chi'(s)E_t\partial_t v_*\,\mathrm{d}\theta\,\mathrm{d}s.\end{aligned}$$

Here the derivatives of v_* are evaluated at $(s+a(t),\theta)$, and χ vanishes near the endpoints of J. The integral of $\chi(s)(\partial_t v_*)^2$ is $\|\Gamma'(a(t))\|^2_\chi$. The function $a\mapsto\|\Gamma'(a)\|^2_\chi$ has a positive minimum on $[0,T]$. The term with E_t in the same integral is bounded by $Cd(t)$ and tends to zero. The right-hand side is also bounded by $Cd(t)$. Thus there is t_0 such that, for $t\geq t_0$,

$$|a'(t)-1|\leq Cd(t). \tag{5.11}$$

An estimate for the error along a noncylindrical solution. We compare $d(t)$ with its supremum on an interval centered at t. If the estimate below fails, a subsequence of the rescaled errors converges to a Jacobi field. The Hamiltonian and the orthogonality condition then force this field to vanish.

We claim that there are $R,C>0$ and $t_1\geq t_0$ such that, for $t\geq t_1+R$,

$$d(t)\leq\frac{1}{2}\sup_{|s-t|\leq R}d(s)+Ce^{-(t-R)}. \tag{5.12}$$

If the estimate failed, taking $R=C=j$ and $t_1=t_0+j$ would give $t_j\geq t_0+2j$ such that, with $D_j=\sup_{|s-t_j|\leq j}d(s)$,

$$d(t_j)>\frac{1}{2}D_j+je^{-(t_j-j)}. \tag{5.13}$$

Since $t_j-j\to\infty$ and $d(t)\to 0$, we have $D_j\to 0$. Moreover, $D_j\geq d(t_j)>0$, so

$$\frac{d(t_j)}{D_j}>\frac{1}{2},\qquad\frac{e^{-(t_j-j)}}{D_j}<\frac{1}{j}. \tag{5.14}$$

Set $a_j=a(t_j)$ and define

$$y_j(s,\theta)=\frac{w(t_j+s,\theta)-v_*(s+a_j,\theta)}{D_j} \tag{5.15}$$

for $s>-\log\rho-t_j$ and $\theta\in\mathbb{S}^{n-1}_+$. After passing to a subsequence, the representatives of a_j in $[0,T)$ converge to some $a_\infty\in[0,T]$. We continue to use $a_j=a(t_j)$ when estimating differences of the real-valued function a.

Fix $\xi \in \mathbb{R}$. For $j > |\xi|$, the interval with endpoints t_j and $t_j + \xi$ lies in $[t_j - j, t_j + j]$. Integrating (5.11) on this interval gives

$$|a(t_j + \xi) - a_j - \xi| \leq C|\xi| D_j. \tag{5.16}$$

On $J \times \mathbb{S}^{n-1}_+$, the C^2 norm of $w(t_j+\xi+s,\theta) - v_*(s+a(t_j+\xi),\theta)$ is at most D_j by (5.10). By (5.16) and the uniform bounds for derivatives of v_* through order three, the C^2 norm of $v_*(s+a(t_j+\xi),\theta) - v_*(s+\xi+a_j,\theta)$ is at most $C|\xi|D_j$. Both translation parameters are fixed when taking derivatives in (s,θ). Thus

$$\|y_j(\xi + \cdot, \cdot)\|_{C^2(J\times\mathbb{S}^{n-1}_+)} \leq C(1 + |\xi|), \tag{5.17}$$

for $|\xi| < j$, with C independent of ξ and j.

Subtract the equation (2.10) for $v_*(\cdot + a_j, \cdot)$ from (2.26) for $w(t_j + \cdot, \cdot)$. With the operators in (2.13), we obtain

$$\mathcal{A} y_j = -D_j^{-1} f_g(t_j + \cdot, \cdot), \tag{5.18}$$

$$\mathcal{B}_{v_*(\cdot+a_j,\cdot)} y_j = D_j^{-1} f_{\partial,g}(t_j + \cdot, \cdot) + \mathcal{R}_j. \tag{5.19}$$

on the translated cylinder and its lateral boundary, respectively. At a point (s, ϑ) on that boundary, Taylor's formula for the power in (2.10) gives

$$\begin{aligned} \mathcal{R}_j(s,\vartheta) = {} & \frac{2nc}{(n-2)^2} D_j \widetilde{y}_j(s,\vartheta)^2 \\ & \times \int_0^1 (1-\lambda)\Big(\tilde{v}_*(s+a_j) + \lambda D_j \widetilde{y}_j(s,\vartheta)\Big)^{(4-n)/(n-2)} \, \mathrm{d}\lambda. \end{aligned} \tag{5.20}$$

For $0 \leq \lambda \leq 1$, the expression in parentheses in (5.20) is the convex combination

$$(1-\lambda)\tilde{v}_*(s + a_j) + \lambda \tilde{w}(t_j + s, \vartheta).$$

Proposition 2.4 and positivity and periodicity of v_* bound it above and away from zero, uniformly on compact intervals in s. By (5.17), the traces $\widetilde{y}_j$ have bounded C^2 norms on these intervals. Differentiating the products and the power in (5.20) therefore gives a bound CD_j for the $C^{1,\gamma}$ norm of $\mathcal{R}_j$ on each compact product with $\partial\mathbb{S}^{n-1}_+$, for $0 < \gamma < 1$.

By (2.28) and (5.14), the geometric errors in (5.18) and (5.19) tend to zero in $C^{0,\gamma}(I \times \mathbb{S}^{n-1}_+)$ and $C^{1,\gamma}(I \times \partial\mathbb{S}^{n-1}_+)$, respectively, for each compact interval $I \subset \mathbb{R}$. The operator $\mathcal{B}_{v_*(\cdot+a_j,\cdot)}$ in (5.19) is the boundary operator defined in (2.13), evaluated along the translate $v_*(\cdot + a_j, \cdot)$. On $I \times \partial\mathbb{S}^{n-1}_+$ it acts by

$$\mathcal{B}_{v_*(\cdot+a_j,\cdot)} y_j = \partial_{\nu_{\mathbb{S}^{n-1}_+}} y_j - \frac{nc}{n-2} \tilde{v}_*(s + a_j)^{2/(n-2)} \widetilde{y}_j(s,\vartheta).$$

The factor multiplying $\widetilde{y}_j$ after the minus sign is the derivative of the boundary nonlinearity $z \mapsto cz^{n/(n-2)}$ in (2.10), evaluated at the translated trace $z = \tilde{v}_*(s+a_j)$. Since $v_* = v_{\varepsilon_*}$ belongs to the smooth positive periodic family of Section 2.4, the translated traces are bounded away from zero and have uniform bounds through order two. Hence the coefficient is bounded in $C^{1,\gamma}(I \times \partial\mathbb{S}^{n-1}_+)$ independently of j. The local Schauder estimates, applied on nested intervals as in Section 2.3, give uniform $C^{2,\gamma}$ bounds on each compact product with $\mathbb{S}^{n-1}_+$. A diagonal subsequence converges to Φ in $C^2_{\mathrm{loc}}(\mathbb{R} \times \mathbb{S}^{n-1}_+)$, including the boundary. Set $v_\infty(s,\theta) = v_*(s+a_\infty,\theta)$. The convergence of the translated coefficients and the vanishing of the right-hand sides give the interior and boundary equations of (2.13) along v_∞.

$$\begin{cases} \mathcal{A}\Phi = 0, \\ \mathcal{B}_{v_\infty}\Phi = 0. \end{cases} \tag{5.21}$$

Elliptic regularity makes Φ smooth up to the boundary, and (5.17) gives at most linear growth in both directions of s. Thus $\Phi \in \mathcal{K}_{\mathrm{pol}}(v_\infty)$.

We evaluate the Hamiltonian at $s = 0$ in (5.15). The definition (2.11) and Taylor's formula give

$$\begin{aligned} &|\mathbb{S}^{n-2}|\Big(H(w)(t_j) - H(v_*)\Big) \\ &\qquad = D_j|\mathbb{S}^{n-2}|DH_{v_*(\cdot+a_j,\cdot)}[y_j](0) + O(D_j^2). \end{aligned}$$

The remainder is $O(D_j^2)$ uniformly in j. Indeed, (5.17) with $\xi = 0$ bounds y_j and its first derivatives on $\{0\} \times \mathbb{S}^{n-1}_+$. Since the interior integral in (2.11) is quadratic in the function and its first derivatives, its remainder after the linear term is bounded by CD_j^2.

For the boundary integral, fix $\vartheta \in \partial\mathbb{S}^{n-1}_+$. By (5.15), the trace along the interpolation, for $0 \le \lambda \le 1$, is

$$\tilde{v}_*(a_j) + \lambda D_j\widetilde{y}_j(0,\vartheta) = (1-\lambda)\tilde{v}_*(a_j) + \lambda\tilde{w}(t_j,\vartheta).$$

As λ varies, this expression runs through the segment joining $\tilde{v}_*(a_j)$ to $\tilde{w}(t_j,\vartheta)$. By Proposition 2.4 and the positivity and periodicity of v_* from Section 2.4, these segments lie in a fixed compact subinterval of $(0,\infty)$, independently of j and ϑ.

On this interval the boundary integrand in (2.11), as a function of the trace, has bounded second derivative. Taylor's formula therefore bounds the pointwise remainder by $CD_j^2|\widetilde{y}_j(0,\vartheta)|^2$. Integrating over $\partial\mathbb{S}^{n-1}_+$ and using the bound from (5.17) gives the uniform $O(D_j^2)$ estimate for the boundary integral. By (2.29) and (5.7), the left-hand side is $O(e^{-t_j})$.

Dividing by D_j and using (5.14) yields

$$\left|DH_{v_*(\cdot+a_j,\cdot)}[y_j](0)\right| \leq C\left(\frac{e^{-t_j}}{D_j}+D_j\right) \leq C\left(\frac{e^{-j}}{j}+D_j\right) \longrightarrow 0.$$

Passing to the limit in (4.13) at $s=0$ gives

$$DH_{v_\infty}[\Phi](0)=0. \tag{5.22}$$

The first variation in (4.13) is constant in s along a Jacobi field. Theorem B, applied after translating by $-a_\infty$, gives $\Phi = b\,\partial_s v_\infty$ for some $b\in\mathbb{R}$. Dividing the orthogonality condition (5.9) at t_j by D_j and passing to the limit gives

$$0 = b\int_J\int_{\mathbb{S}^{n-1}_+} \chi(s)\big(\partial_s v_\infty(s,\theta)\big)^2\,\mathrm{d}\theta\,\mathrm{d}s.$$

The integral is positive because χ is positive on an interval containing a period and v_∞ depends on s. Hence $\Phi=0$. On the other hand, (5.15) and (5.14) give $\|y_j\|_{C^2(J\times\mathbb{S}^{n-1}_+)} > 1/2$, and convergence in this norm yields $\|\Phi\|_{C^2(J\times\mathbb{S}^{n-1}_+)}\geq 1/2$. This contradiction proves (5.12).

Rate of convergence. We use the following argument here and for the cylindrical solution. Suppose d is continuous and nonnegative on $[t_0,\infty)$, tends to zero, and satisfies, for $t\geq t_0+R$,

$$d(t) \leq \frac{1}{2}\sup_{|s-t|\leq R} d(s) + Ce^{-\sigma(t-R)} \tag{5.23}$$

with $R,\sigma>0$. Choose $0<\delta<\min\{\sigma,(\log 2)/R\}$. For a sufficiently large $B_0>0$, the function $B(t)=B_0e^{-\delta(t-t_0)}$ satisfies $d(t)\leq B(t)$ on $[t_0,t_0+R]$ and

$$\left(1-\frac{1}{2}e^{\delta R}\right)B(t) \geq Ce^{-\sigma(t-R)}$$

for $t\geq t_0+R$. Indeed, the factor on the left is positive, and the ratio of the forcing term to $e^{-\delta(t-t_0)}$ decreases for $t\geq t_0+R$ because $\delta<\sigma$.

If $d-B$ has a positive supremum M, it attains this value at some $t_*>t_0+R$, since $d-B$ is continuous and tends to zero. For $|s-t_*|\leq R$, we have $d(s)\leq M+B(s)\leq M+e^{\delta R}B(t_*)$. Substituting this bound in (5.23) gives

$$M+B(t_*) \leq \frac{1}{2}M+\frac{1}{2}e^{\delta R}B(t_*)+Ce^{-\sigma(t_*-R)} \leq \frac{1}{2}M+B(t_*),$$

which contradicts $M>0$. Thus $d(t)\leq B(t)$.

For the noncylindrical solution, apply this argument to (5.12) with $\sigma=1$ and R as in that estimate, taking t_1 as the initial time. It gives $d(t)\leq Ce^{-\delta t}$ for each $0<\delta<\min\{1,(\log 2)/R\}$. By (5.11), the

derivative of $a(t) - t$ is bounded in absolute value by $Ce^{-\delta t}$ and is therefore integrable on $[t_0, \infty)$. There is a real number τ_* such that $a(t) - t \to \tau_*$ as $t \to \infty$. For all sufficiently large t,

$$|a(t) - t - \tau_*| \leq \int_t^\infty |a'(s) - 1| \, \mathrm{d}s \leq Ce^{-\delta t}. \tag{5.24}$$

By the smoothness and periodicity of v_*, the mean value theorem gives

$$\|v_*(\cdot + a(t), \cdot) - v_*(t + \cdot + \tau_*, \cdot)\|_{C^2([-1,1]\times\mathbb{S}^{n-1}_+)} \leq C|a(t) - t - \tau_*|.$$

Together with (5.10), this proves (5.4) when $\varepsilon_* < 1$.

The cylindrical limiting solution. Suppose $\varepsilon_* = 1$, so that $v_* = C_n$. Fix $a_0 > 1$ and let

$$d(t) = \|w(t + \cdot, \cdot) - C_n\|_{C^2([-a_0,a_0]\times\mathbb{S}^{n-1}_+)}.$$

Equation (5.8) gives $d(t) \to 0$. We prove that, for some $R, C > 0$ and all sufficiently large t,

$$d(t) \leq \frac{1}{2} \sup_{|s-t|\leq R} d(s) + Ce^{-(t-R)/2}. \tag{5.25}$$

By (4.21), the first variation of H at C_n vanishes. The argument therefore compares the quadratic term in (4.24) with the Hamiltonian error of order e^{-t} in (2.29). This accounts for the exponent $1/2$ in (5.25).

Fix t_0 sufficiently large that d is defined on $[t_0, \infty)$. If (5.25) fails, take $R = C = j$ and choose $t_j \geq t_0 + 2j$ such that, with $D_j = \sup_{|s-t_j|\leq j} d(s)$,

$$d(t_j) > \frac{1}{2} D_j + je^{-(t_j - j)/2}.$$

Then $D_j > 0$ and $D_j \to 0$, because $t_j - j \to \infty$. We also have

$$\frac{d(t_j)}{D_j} > \frac{1}{2}, \qquad \frac{e^{-(t_j-j)}}{D_j^2} < \frac{1}{j^2}. \tag{5.26}$$

Set $y_j(s, \theta) = (w(t_j + s, \theta) - C_n(\theta))/D_j$. For each $\xi \in \mathbb{R}$ and all sufficiently large j,

$$\|y_j(\xi + \cdot, \cdot)\|_{C^2([-a_0,a_0]\times\mathbb{S}^{n-1}_+)} = \frac{d(t_j + \xi)}{D_j} \leq 1.$$

The geometric errors divided by D_j tend to zero on each fixed interval in the local Hölder norms of (2.28), since

$$\frac{e^{-t_j}}{D_j} \leq \frac{e^{-j}}{j^2} D_j \longrightarrow 0.$$

The trace of C_n equals 1. Subtracting its equation from (2.26) gives (5.18) and (5.19) with C_n as the solution at which the boundary condition is linearized. The remainder in (5.20), with trace 1 in place of the translated trace, tends to zero in $C^{1,\gamma}$ on each fixed product with $\partial\mathbb{S}^{n-1}_+$, since y_j is bounded there in C^2 and $D_j \to 0$. The same local Schauder estimates as in the noncylindrical case yield, after passing to a subsequence, convergence in $C^2_{\mathrm{loc}}(\mathbb{R}\times\mathbb{S}^{n-1}_+)$ to a Jacobi field Φ along C_n. The bounds on the translated intervals give $\Phi \in \mathcal{K}_{\mathrm{bd}}(C_n)$. By (5.26) and convergence on $[-a_0,a_0]\times\mathbb{S}^{n-1}_+$,

$$\|\Phi\|_{C^2([-a_0,a_0]\times\mathbb{S}^{n-1}_+)} \geq \frac{1}{2}. \tag{5.27}$$

At $s=0$, equation (4.24) gives

$$|\mathbb{S}^{n-2}|\Big(H(w)(t_j) - H(C_n)\Big) = D_j^2\mathcal{Q}_c[y_j](0) + O(D_j^3).$$

Since C_n has trace 1, the trace of y_j at $s=0$ is

$$\widetilde{y}_j(0,\vartheta) = \frac{\tilde{w}(t_j,\vartheta)-1}{D_j}$$

for $\vartheta \in \partial\mathbb{S}^{n-1}_+$. The C^2 estimate for y_j above, taken with $\xi=0$, gives $|\widetilde{y}_j(0,\vartheta)| \leq d(t_j)/D_j \leq 1$. Thus the remainder in (4.24) is uniform in j. By (5.7) with $\varepsilon_*=1$, the left-hand side is $O(e^{-t_j})$ by (2.29). Dividing by D_j^2 and using (5.26), we obtain

$$|\mathcal{Q}_c[y_j](0)| \leq C\left(\frac{e^{-t_j}}{D_j^2} + D_j\right) \leq C\left(\frac{e^{-j}}{j^2} + D_j\right) \longrightarrow 0.$$

The convergence of y_j and its first derivatives on $\{0\}\times\mathbb{S}^{n-1}_+$ gives $\mathcal{Q}_c[\Phi](0)=0$. Since Φ belongs to $\mathcal{K}_{\mathrm{pol}}(C_n)$, Proposition 4.8 implies $\Phi=0$, contradicting (5.27). This proves (5.25).

By (5.25), d satisfies (5.23) with $\sigma=1/2$ for all sufficiently large t. The preceding estimate gives $d(t) \leq Ce^{-\delta t}$ for each $0<\delta<\min\{1/2,(\log 2)/R\}$. Since $a_0>1$ and C_n is independent of t, this proves (5.4) when $\varepsilon_*=1$, for every choice of τ_*.

Estimates in Fermi coordinates and uniqueness. Let $r=e^{-t}$ and let $U_* = U_{\varepsilon_*,\tau_*}$ be defined by (5.1). We work on the closed half-annulus

$$\{z\in\mathbb{R}^n : z_n \geq 0,\ 1/2 \leq |z| \leq 2\}.$$

For all sufficiently large t, we have $2r < \rho$. Equations (2.20) and (5.1) give, on this half-annulus,

$$r^{\frac{n-2}{2}}(U-U_*)(rz) = A_\kappa^{-1}|z|^{-\frac{n-2}{2}}\left[w\left(t-\log|z|,\frac{z}{|z|}\right) - v_{\varepsilon_*}\left(t-\log|z|+\tau_*,\frac{z}{|z|}\right)\right]. \tag{5.28}$$

The factor $A_\kappa^{-1}|z|^{-(n-2)/2}$ comes from undoing the normalization in (2.20) at $x = rz$. The map

$$z \longmapsto \left(-\log|z|,\frac{z}{|z|}\right)$$

gives the cylindrical coordinates relative to t, since $-\log|rz| = t-\log|z|$. Its image is contained in $[-1,1]\times\mathbb{S}^{n-1}_+$ because $\left|\log|z|\right| \le \log 2 < 1$. The scalar factor and the coordinate map are smooth away from $z=0$, with bounded derivatives through order two on the half-annulus. For the spherical component, these bounds can be taken in smooth charts of the hemisphere, including charts at the equator.

The expression in brackets in (5.28) is the difference estimated in (5.4), evaluated at $(-\log|z|, z/|z|)$. The product and chain rules therefore bound the C^2 norm of the left-hand side on the half-annulus by $Ce^{-\delta t} = Cr^\delta$. For a multi-index α with $|\alpha|\le 2$, differentiation at fixed r gives

$$\partial_z^\alpha\left[r^{\frac{n-2}{2}}(U-U_*)(rz)\right] = r^{\frac{n-2}{2}+|\alpha|}(\partial_x^\alpha(U-U_*))(rz).$$

Evaluating at $|z|=1$ gives (5.3) with $|x|=r$. The estimates include $z_n = 0$ because the C^2 bound holds up to the equator.

For $\varepsilon_* < 1$, positivity and periodicity give a positive minimum of v_{ε_*} on $[0,T_{\varepsilon_*}]\times\mathbb{S}^{n-1}_+$. For $\varepsilon_* = 1$, the positive minimum is that of C_n on $\mathbb{S}^{n-1}_+$. Thus (5.1) gives $U_*(x) \ge C^{-1}|x|^{-(n-2)/2}$. Dividing the estimate for $j=0$ in (5.3) by $U_*(x)$ gives

$$\left|\frac{U(x)}{U_*(x)} - 1\right| \le C|x|^\delta.$$

To prove uniqueness, suppose that $U_{\varepsilon'_*,\tau'_*}$ satisfies (5.3) with an exponent $\delta' > 0$. For $s\in[-1,1]$ and $\theta\in\mathbb{S}^{n-1}_+$, put $z = e^{-s}\theta$. The points z range over the closed half-annulus

$$\{z\in\mathbb{R}^n : z_n \ge 0,\ e^{-1}\le|z|\le e\}.$$

For sufficiently small r, the assumed weighted bounds apply at $x = rz$. For $|\alpha| \le 2$, they give

$$\begin{aligned}&\left|\partial_z^\alpha\left[r^{\frac{n-2}{2}}(U - U_{\varepsilon'_*,\tau'_*})(rz)\right]\right| \\ &\qquad \le Cr^{\delta'}|z|^{\delta' - \frac{n-2}{2} - |\alpha|} \le C'r^{\delta'}.\end{aligned}$$

The last bound is uniform on this half-annulus because $e^{-1} \le |z| \le e$. By (2.20) and (5.1),

$$\begin{aligned}&w(t+s,\theta) - v_{\varepsilon'_*}(t+s+\tau'_*,\theta) \\ &\qquad = A_\kappa e^{-\frac{n-2}{2}s} r^{\frac{n-2}{2}}(U - U_{\varepsilon'_*,\tau'_*})(re^{-s}\theta).\end{aligned}$$

The factor $A_\kappa e^{-(n-2)s/2}$ and the map $(s,\theta) \mapsto e^{-s}\theta$ have bounded derivatives through order two on $[-1,1] \times \mathbb{S}^{n-1}_+$. The chain rule thus gives

$$\left\|w(t+\cdot,\cdot) - v_{\varepsilon'_*}(t+\cdot+\tau'_*,\cdot)\right\|_{C^2([-1,1]\times\mathbb{S}^{n-1}_+)} \le Ce^{-\delta' t}.$$

In particular, at $s = 0$, the differences of the functions, their first derivatives and their boundary traces tend uniformly to zero. Substituting in (2.11) and using the constancy of the Hamiltonian of $v_{\varepsilon'_*}$ gives

$$H(w)(t) - H(v_{\varepsilon'_*}) \longrightarrow 0 \qquad \text{as } t \to \infty.$$

By (2.29), $H(v_{\varepsilon'_*}) = H_0/|\mathbb{S}^{n-2}|$. Together with (5.7) and Theorem A, this gives $\varepsilon'_* = \varepsilon_*$.

Suppose $\varepsilon_* < 1$. The two cylindrical estimates imply that the norm of the difference of $v_{\varepsilon_*}(t+\cdot+\tau_*,\cdot)$ and $v_{\varepsilon_*}(t+\cdot+\tau'_*,\cdot)$ in $C^2([-1,1]\times\mathbb{S}^{n-1}_+)$ tends to zero as $t \to \infty$. This norm is T_{ε_*}-periodic in t. For any fixed t, it therefore has the same value at $t + kT_{\varepsilon_*}$ for every positive integer k. Taking $k \to \infty$ shows that this value is zero. The two translates consequently agree on $\mathbb{R} \times \mathbb{S}^{n-1}_+$, so $\tau'_* - \tau_*$ is a period of v_{ε_*}. Since T_{ε_*} is its least positive period, we have $\tau'_* - \tau_* \in T_{\varepsilon_*}\mathbb{Z}$. If $\varepsilon_* = 1$, every translate is C_n. □

5.2. **Conformal deformations of the flat model.** We compose tangential translations with the Kelvin transform in (2.8) and identify the resulting term of order e^{-t} in Emden–Fowler coordinates. After multiplication by $\sin^{(n-2)/2}\beta$, the term has the coefficient z_- from (4.3) in degree 1. We then construct a metric for which the deformed flat solutions leave a nonzero term of order $|x|$ in the relative error in boundary Fermi coordinates. For the corresponding construction in the problem without boundary, see [14, Sections 2.1 and 5] and [18, Theorem 2 and Section 5].

Proposition 5.2. *Let $U_* = U_{\varepsilon,\tau}$ be the flat solution on $(\mathbb{R}^n_+ \setminus \{0\}, \delta)$ defined by (5.1), where $0 < \varepsilon \le 1$ and $\tau \in \mathbb{R}$. Fix $a = (\overline{a}, 0)$ with $\overline{a} \in \mathbb{R}^{n-1}$ and choose $\rho_a > 0$ such that $|a|\rho_a < 1/2$. For $x \in \mathbb{R}^n_+$ with $0 < |x| \le \rho_a$, define*

$$U_{*,a}(x) = \left(1 - 2a \cdot x + |a|^2|x|^2\right)^{-\frac{n-2}{2}} \times U_*\left(\frac{x - |x|^2 a}{1 - 2a \cdot x + |a|^2|x|^2}\right).$$

Let w_a be the Emden–Fowler transform of $U_{*,a}$ as in (2.20), and let h_- and z_- be the functions in (4.3) associated with $v_\varepsilon(t+\tau,\theta)$. Write $\theta = (\bar{\theta}, \theta_n)$. For $t > -\log \rho_a$ and $\theta \in \mathbb{S}^{n-1}_+$, we have

$$w_a(t,\theta) = v_\varepsilon(t+\tau,\theta) + e^{-t}(\overline{a} \cdot \bar{\theta}) h_-(t,\beta) + R_a(t,\theta).$$

For each $J > 0$, there is a constant $C_{\varepsilon,\tau,a,J}$ such that, for all sufficiently large t,

$$\|R_a(t+\cdot,\cdot)\|_{C^2([-J,J]\times\mathbb{S}^{n-1}_+)} \le C_{\varepsilon,\tau,a,J} e^{-2t}.$$

Since $\bar{\theta} = \sin\beta\, \vartheta$, multiplying the term of order e^{-t} by $\sin^{(n-2)/2}\beta$ gives $z_-(t,\beta)(\overline{a} \cdot \vartheta)$.

Proof. By (2.20) and (5.1), we have $w_a(t,\theta) = F(t, e^{-t}, \theta)$, where

$$F(t,r,\theta) = |\theta - ra|^{-\frac{n-2}{2}}\, v_\varepsilon\left(t + \tau + \log|\theta - ra|, \frac{\theta - ra}{|\theta - ra|}\right).$$

We expand F in r at $r = 0$, with t and θ fixed. Its value there is $v_\varepsilon(t+\tau,\theta)$, and

$$\partial_r \log|\theta - ra||_{r=0} = -a \cdot \theta.$$

Since $a_n = 0$, the last coordinate of $(\theta - ra)/|\theta - ra|$ is $\cos\beta/|\theta - ra|$. For $0 < \beta \le L$, this gives

$$\partial_r \arccos\left(\frac{\cos\beta}{|\theta - ra|}\right)\Big|_{r=0} = -(a \cdot \theta)\cot\beta.$$

By cylindrical symmetry, v_ε depends on the spherical variable only through β. The product and chain rules therefore give the following derivative at $r = 0$:

$$(\overline{a} \cdot \bar{\theta})\left(\frac{n-2}{2} v_\varepsilon(t+\tau,\beta) - \partial_t v_\varepsilon(t+\tau,\beta) - \cot\beta\, \partial_\beta v_\varepsilon(t+\tau,\beta)\right).$$

This equals $(\overline{a} \cdot \bar{\theta}) h_-(t,\beta)$ by (4.3). The derivative is smooth in spherical charts, and h_- is smooth up to $\beta = 0$ by Proposition 4.1. Hence the identity holds at $\beta = 0$ as well.

Taylor's formula gives the remainder

$$R_a(t,\theta) = e^{-2t}\int_0^1 (1-\lambda)\partial_r^2 F(t,\lambda e^{-t},\theta)\,\mathrm{d}\lambda.$$

For $0 \le r \le \rho_a$, we have $|\theta - ra| \ge 1 - |a|\rho_a > 1/2$, and $(\theta - ra)/|\theta - ra|$ belongs to $\mathbb{S}^{n-1}_+$ because $a_n = 0$. These facts and the uniform smooth bounds for v_ε from Section 2.4 bound the derivatives of F through order four uniformly in smooth spherical charts, including charts at the equator. Each of the first two derivatives in t of λe^{-t} is bounded uniformly in $\lambda \in [0,1]$ for sufficiently large t. Differentiating under the integral and using the chain rule bounds R_a and its derivatives through order two by Ce^{-2t}. Taking the supremum on $[-J,J]$ after translating t gives the C^2 bound for $R_a(t+\cdot,\cdot)$.

Finally, using $\bar\theta = \sin\beta\,\vartheta$, we get

$$\begin{aligned}&\sin^{(n-2)/2}\beta\, e^{-t}(\overline{a}\cdot\bar\theta)h_-(t,\beta)\\ &\qquad = e^{-t}\sin^{n/2}\beta\, h_-(t,\beta)(\overline{a}\cdot\vartheta) = z_-(t,\beta)(\overline{a}\cdot\vartheta),\end{aligned}$$

where the last equality is the definition of z_- in (4.3). □

Marques [18, Theorem 2] obtains a relative error of order $O(|x|^\gamma)$ for some $\gamma > 1$ by using deformed Fowler solutions in the problem without boundary. The following proposition establishes a first-order obstruction to such an estimate in boundary Fermi coordinates. The proof uses the boundary trace to determine the possible approximating solution and then evaluates the error along the inward normal.

Proposition 5.3. *For every $n \ge 3$ and $\kappa > 0$, there exist $\rho > 0$, a smooth conformally flat metric g (Definition 2.1) written in boundary Fermi coordinates on B^+_ρ, and a positive solution U satisfying the hypotheses of Theorem D. Its Emden–Fowler transform w, defined by (2.20), satisfies, for every $J > 0$,*

$$\|w(t+\cdot,\cdot) - C_n\|_{C^2([-J,J]\times\mathbb{S}^{n-1}_+)} \longrightarrow 0 \qquad \text{as } t \to \infty.$$

There are no parameters $0 < \varepsilon \le 1$, $\tau \in \mathbb{R}$, $a = (\overline{a},0)$ with $\overline{a} \in \mathbb{R}^{n-1}$, and $\sigma > 1$ for which

$$U(y) = \Big(1 + O(|y|^\sigma)\Big)U_{*,a}(y)$$

as $y \to 0$ in $\mathbb{R}^n_+$. Here $U_{*,a}$ is obtained from $U_* = U_{\varepsilon,\tau}$ by Proposition 5.2.

Proof. Fix $\lambda \ne 0$ and choose $\rho > 0$ such that $1 + \lambda y_n > 0$ for $y \in \mathbb{R}^n_+$ with $|y| \le \rho$. Consider the metric

$$g = \mathrm{d}y_n^2 + (1+\lambda y_n)^2 \sum_{i=1}^{n-1} \mathrm{d}y_i^2.$$

The induced boundary metric is Euclidean, and the normal coordinate lines are unit-speed geodesics. Hence y is a boundary Fermi chart. The outward normal is $-\partial_{y_n}$ at $y_n = 0$, so $h_g(0) = -\lambda$. The change of variables

$$x_n = \lambda^{-1}\log(1+\lambda y_n), \qquad \overline{x} = \overline{y}$$

writes $g = e^{2\lambda x_n}\delta$. Define

$$U(y) = (1+\lambda y_n)^{-\frac{n-2}{2}} U_{1,0}\Big(\overline{y}, \lambda^{-1}\log(1+\lambda y_n)\Big),$$

where $U_{1,0}$ is the cylindrical solution in (5.1). Conformal covariance of L_g and B_g gives (2.4). The trace is $u(\overline{y}) = A_\kappa^{-1}|\overline{y}|^{-(n-2)/2}$, so (5.2) holds.

By homogeneity of $U_{1,0}$, for z in any fixed closed half-annulus and sufficiently small $r > 0$,

$$r^{\frac{n-2}{2}} U(rz) = (1+\lambda r z_n)^{-\frac{n-2}{2}} U_{1,0}\left(\overline{z}, \frac{\log(1+\lambda r z_n)}{\lambda r}\right).$$

The right-hand side tends to $U_{1,0}(z)$ in C^2 on that half-annulus. By (2.20), $w(t+\cdot,\cdot)$ therefore tends to C_n in $C^2([-J,J]\times\mathbb{S}^{n-1}_+)$ for every $J > 0$ as $t \to \infty$.

Suppose that the estimate in the statement holds for some ε, τ, a and $\sigma > 1$. The expansion in Proposition 5.2 gives, uniformly in $\vartheta \in \mathbb{S}^{n-2}$ as $r \to 0$,

$$A_\kappa r^{\frac{n-2}{2}} U_{*,a}(r\vartheta, 0) = \tilde{v}_\varepsilon(-\log r + \tau) + O(r).$$

The exact trace of U now implies that $\tilde{v}_\varepsilon(t+\tau) \to 1$ as $t \to \infty$. By periodicity, $\tilde{v}_\varepsilon \equiv 1$, which gives $\varepsilon = 1$. On the flat face,

$$\Big|\overline{y} - |\overline{y}|^2\overline{a}\Big|^2 = |\overline{y}|^2\Big(1 - 2\overline{a}\cdot\overline{y} + |\overline{a}|^2|\overline{y}|^2\Big).$$

Using the definition of $U_{*,a}$ and the trace of $U_{1,0}$, we get

$$\begin{aligned}\frac{U_{*,a}(\overline{y},0)}{U_{1,0}(\overline{y},0)} &= \Big(1 - 2\overline{a}\cdot\overline{y} + |\overline{a}|^2|\overline{y}|^2\Big)^{-\frac{n-2}{4}} \\ &= 1 + \frac{n-2}{2}\overline{a}\cdot\overline{y} + O(|\overline{y}|^2).\end{aligned}$$

If $\overline{a} \neq 0$, taking $\overline{y}$ in the direction of $\overline{a}$ gives a nonzero linear term, which contradicts $\sigma > 1$. Thus $a = 0$ and $U_{*,a} = U_{1,0}$. Along the inward normal, homogeneity of $U_{1,0}$ gives

$$\begin{aligned}\frac{U(0,s)}{U_{1,0}(0,s)} &= \left(\frac{s}{(1+\lambda s)\lambda^{-1}\log(1+\lambda s)}\right)^{\frac{n-2}{2}} \\ &= 1 - \frac{(n-2)\lambda}{4}s + O(s^2)\end{aligned}$$

as $s \to 0$ through positive values. The coefficient of s is nonzero, contradicting the assumed estimate. □

If $\mathrm{II}_g(p) = 0$, the expansion in (2.3) gives $g_{ij}(x) - \delta_{ij} = O(|x|^2)$ and $h_g(\overline{x}, 0) = O(|\overline{x}|)$. The rescaling argument in the proof of Proposition 2.4 then gives (2.28) with e^{-2t_0} in place of e^{-t_0}, as detailed in Section A.1. To obtain a higher-order expansion by this approach, the improved bound must be combined with estimates for the Jacobi equation in exponentially weighted spaces, separating the conformal fields of degree 1 from the other decaying fields.

Appendix A. Computations in cylindrical coordinates

Section A.1 expands the rescaled Laplacian in boundary Fermi coordinates. Section A.2 records the period shift of the auxiliary function used in the pairing argument. Section A.3 verifies the equations for the fields of degree 1 and estimates the endpoint term as $\beta \to 0$ in Picone's identity. Section A.4 computes the first and second derivatives of the Hamiltonian.

A.1. **Expansion of the rescaled metric Laplacian.** We compute the linear term of the rescaled Laplacian in the boundary Fermi chart and use it to improve the error estimates in (2.28) when $\mathrm{II}_g(p) = 0$.

In the boundary Fermi chart of (2.1), let $(g_{r_0})_{ij}(z) = g_{ij}(r_0 z)$. By (2.2), we have $(g_{r_0})_{nn} = 1$ and $(g_{r_0})_{an} = (g_{r_0})_{na} = 0$ for $1 \le a \le n-1$. Write $g_{r_0}^{ij}$ for the entries of the inverse matrix. The identity

$$\sum_{i=1}^{n} g_{r_0}^{ji}(g_{r_0})_{in} = g_{r_0}^{jn} = \delta_{jn}$$

gives $g_{r_0}^{nn} = 1$ and $g_{r_0}^{an} = g_{r_0}^{na} = 0$ for $1 \le a \le n-1$.

For $1 \le a, b \le n-1$, the expansion (2.3) gives

$$(g_{r_0})_{ab}(z) = \delta_{ab} - 2r_0 z_n \mathrm{II}_{ab}(p) + O(r_0^2).$$

The coefficient of r_0 is the derivative with respect to r_0 at $r_0 = 0$. To compute the corresponding coefficient for the inverse, differentiate $\sum_{d=1}^{n-1} g_{r_0}^{ad}(g_{r_0})_{db} = \delta_{ab}$ at $r_0 = 0$. Both matrices equal the identity there, so

$$0 = \partial_{r_0} g_{r_0}^{ab}(z)\Big|_{r_0=0} - 2z_n \mathrm{II}_{ab}(p).$$

Taylor's formula therefore gives

$$g_{r_0}^{ab}(z) = \delta_{ab} + 2r_0 z_n \mathrm{II}_{ab}(p) + O(r_0^2).$$

In the determinant expansion, each product other than the product of the diagonal entries contains at least two off-diagonal entries. Each such entry is $O(r_0)$. Thus

$$\begin{aligned}\det g_{r_0}(z) &= \prod_{a=1}^{n-1}(g_{r_0})_{aa}(z) + O(r_0^2)\\ &= 1 - 2r_0 z_n \sum_{a=1}^{n-1} \mathrm{II}_{aa}(p) + O(r_0^2)\\ &= 1 - 2r_0 z_n (n-1) h_g(p) + O(r_0^2).\end{aligned}$$

Since $\sqrt{\det g_{r_0}} + 1 = 2 + O(r_0)$, the identity

$$\sqrt{\det g_{r_0}} - 1 = \frac{\det g_{r_0} - 1}{\sqrt{\det g_{r_0}} + 1}$$

gives $\sqrt{\det g_{r_0}(z)} = 1 - r_0 z_n (n-1) h_g(p) + O(r_0^2)$. The remainders in these expansions satisfy the same $O(r_0^2)$ estimate after any fixed number of z-derivatives on a fixed half-annulus.

For a smooth function ψ in the rescaled Fermi coordinates, the product rule gives

$$\begin{aligned}\Delta_{g_{r_0}}\psi &= \frac{1}{\sqrt{\det g_{r_0}}} \sum_{i,j=1}^{n} \partial_{z_i}\Big(\sqrt{\det g_{r_0}}\, g_{r_0}^{ij} \partial_{z_j}\psi\Big)\\ &= \sum_{i,j=1}^{n} g_{r_0}^{ij} \partial_{z_i z_j}\psi\\ &\quad + \sum_{j=1}^{n} \left(\sum_{i=1}^{n} \partial_{z_i} g_{r_0}^{ij} + \frac{\sum_{i=1}^{n} g_{r_0}^{ij} \partial_{z_i}\sqrt{\det g_{r_0}}}{\sqrt{\det g_{r_0}}} \right) \partial_{z_j}\psi.\end{aligned}$$

For $i < n$, the coefficient of r_0 in $g_{r_0}^{ij}$ is independent of z_i. For $i = n$, we have $g_{r_0}^{nj} = \delta_{nj}$. Hence $\sum_{i=1}^{n} \partial_{z_i} g_{r_0}^{ij} = O(r_0^2)$. The expansion of $\sqrt{\det g_{r_0}}$ gives

$$\partial_{z_i}\sqrt{\det g_{r_0}} = -r_0 (n-1) h_g(p) \delta_{in} + O(r_0^2).$$

Multiplication by $g_{r_0}^{ij}$, summation over i, and division by $\sqrt{\det g_{r_0}} = 1 + O(r_0)$ give

$$\frac{\sum_{i=1}^{n} g_{r_0}^{ij} \partial_{z_i}\sqrt{\det g_{r_0}}}{\sqrt{\det g_{r_0}}} = -r_0 (n-1) h_g(p) \delta_{jn} + O(r_0^2).$$

It follows that

$$\Delta_{g_{r_0}} - \Delta_\delta = 2r_0 z_n \sum_{a,b=1}^{n-1} \mathrm{II}_{ab}(p)\partial_{z_a z_b} - r_0(n-1)h_g(p)\partial_{z_n} + O(r_0^2).$$

Here $O(r_0^2)$ is an operator of order at most two whose coefficients satisfy the preceding estimates on fixed half-annuli.

The change of variables $x = r_0 z$ gives

$$r_0^2(\Delta_g U)(r_0 z) = \big[\Delta_{g_{r_0}}(U(r_0\,\cdot))\big](z).$$

The same identity holds for Δ_δ. These are the rescaling identities used to estimate f_g in Proposition 2.4. If $\mathrm{II}_g(p) = 0$, the coefficient of r_0 in $\Delta_{g_{r_0}} - \Delta_\delta$ vanishes, and $h_g(p) = 0$. Fix $\ell > 0$ and write $t = t_0 + s$ and $r_0 = e^{-t_0}$. Smoothness gives $h_g(r_0 e^{-s}\vartheta, 0) = O(r_0)$ for $|s| \le \ell$, with the same bound for every fixed number of derivatives in (s, ϑ). Together with the uniform rescaled estimates in (2.25), these bounds improve both errors in (2.28) to $O(r_0^2)$.

A.2. **The shift of the auxiliary function.** Use d, E_ε, F_ε, and G_ε from the proof of Proposition 3.1. The T_ε-periodicity of E_ε and F_ε gives, for every $t \in \mathbb{R}$ and $\theta \in \mathbb{S}^{n-1}_+$,

$$\begin{aligned}
&(t+T_\varepsilon)E_\varepsilon(t+T_\varepsilon,\theta) + \frac{d}{2}(t+T_\varepsilon)^2 F_\varepsilon(t+T_\varepsilon,\theta) \\
&\qquad - tE_\varepsilon(t,\theta) - \frac{d}{2}t^2 F_\varepsilon(t,\theta) \\
&\quad = T_\varepsilon\Big(E_\varepsilon(t,\theta) + dtF_\varepsilon(t,\theta)\Big) + \frac{dT_\varepsilon^2}{2}F_\varepsilon(t,\theta) \\
&\quad = T_\varepsilon G_\varepsilon(t,\theta) + \frac{dT_\varepsilon^2}{2}F_\varepsilon(t,\theta).
\end{aligned}$$

At $t = 0$ and $\beta = L$, this difference equals $T_\varepsilon > 0$ by (3.4).

A.3. **The equations in degree 1 and Picone's identity.** We verify the interior equation and boundary condition for $z_\pm$ from (4.3) and compute the Taylor expansion of $h_\pm$ at $\beta = 0$. We then estimate the term at $\beta = \delta$ in the integration by parts used in Picone's identity.

Let v be the cylindrically symmetric solution considered in Proposition 4.1. In expressions involving $\Delta_{\mathbb{S}^{n+1}_+}$, a function of β is regarded as a cylindrically symmetric function on $\mathbb{S}^{n+1}_+$. For $0 < \beta < L$, its Laplacian on this hemisphere is $\partial_\beta^2 + n\cot\beta\,\partial_\beta$. The corresponding formula on

$\mathbb{S}^{n-1}_+$ and (2.10) give

$$\left(-\partial_t^2 - \Delta_{\mathbb{S}^{n+1}_+} + \frac{n^2}{4} - 1\right) v = (n-2)v - 2\cot\beta\,\partial_\beta v.$$

Differentiating the equation for v with respect to β gives

$$\begin{aligned}\left(-\partial_t^2 - \partial_\beta^2 - (n-2)\cot\beta\,\partial_\beta + \frac{(n-2)^2}{4}\right)\partial_\beta v \\ = -(n-2)\csc^2\beta\,\partial_\beta v.\end{aligned}$$

Using $(\cot\beta)' = -\csc^2\beta$ in the product rule, we obtain

$$\begin{aligned}\left(-\partial_t^2 - \Delta_{\mathbb{S}^{n+1}_+} + \frac{n^2}{4} - 1\right)\left(\cot\beta\,\partial_\beta v\right) \\ = 2\partial_\beta^2 v + (n-2)\cot\beta\,\partial_\beta v.\end{aligned}$$

Indeed, the coefficient of $\partial_\beta^2 v$ is $2(\csc^2\beta - \cot^2\beta) = 2$. The terms with $\cot\beta\,\csc^2\beta\,\partial_\beta v$ have coefficients $-(n-2)$, -2, and n, and their sum is zero. It follows from the equation for v that

$$\begin{aligned}\left(-\partial_t^2 - \Delta_{\mathbb{S}^{n+1}_+} + \frac{n^2}{4} - 1\right)\left(\frac{n-2}{2}v - \cot\beta\,\partial_\beta v\right) \\ = \frac{(n-2)^2}{2}v - 2(n-2)\cot\beta\,\partial_\beta v - 2\partial_\beta^2 v = 2\partial_t^2 v.\end{aligned}$$

Differentiating the identity for the operator applied to v with respect to t gives

$$\begin{aligned}\left(-\partial_t^2 - \Delta_{\mathbb{S}^{n+1}_+} + \frac{n^2}{4} - 1\right)\partial_t v \\ = 2\partial_t\left(\frac{n-2}{2}v - \cot\beta\,\partial_\beta v\right).\end{aligned}$$

Adding or subtracting this identity from the preceding formula and using the definition of $h_\pm$ in (4.3) gives (4.6) for $0 < \beta < L$.

The operator in (4.1) with $\ell = 1$ has potential $n(n-2)\csc^2\beta/4$. The identities

$$\begin{aligned}\frac{\partial_\beta(\sin^{n/2}\beta)}{\sin^{n/2}\beta} &= \frac{n}{2}\cot\beta, \\ \frac{\partial_\beta^2(\sin^{n/2}\beta)}{\sin^{n/2}\beta} &= \frac{n(n-2)}{4}\csc^2\beta - \frac{n^2}{4}\end{aligned}$$

show how this potential cancels when $\mathcal{L}_1$ acts on $\sin^{n/2}\beta\, h_\pm$. More precisely,

$$(\mathcal{L}_1 - 1)(\sin^{n/2}\beta\, h_\pm) = \sin^{n/2}\beta \left(-\partial_t^2 - \Delta_{\mathbb{S}^{n+1}_+} + \frac{n^2}{4} - 1\right) h_\pm = \pm 2 \sin^{n/2}\beta\, \partial_t h_\pm.$$

This is (4.5). By the product rule, applying $\mathcal{L}_1$ to $z_\pm$ gives

$$\mathcal{L}_1 z_\pm = e^{\pm t}(\mathcal{L}_1 - 1 \mp 2\partial_t)(\sin^{n/2}\beta\, h_\pm) = 0.$$

At $\beta = L$, the boundary condition in (2.10) gives $\partial_\beta v(t, L) = c\tilde{v}(t)^{n/(n-2)}$. Differentiating the formula for $h_\pm$ in (4.3) with respect to β and evaluating at L gives

$$\begin{aligned} h_\pm(t, L) &= \frac{n-2}{2}\tilde{v}(t) \pm \tilde{v}'(t), \\ \partial_\beta h_\pm(t, L) &= \frac{n}{2}\,\partial_\beta v(t, L) \pm \partial_t \partial_\beta v(t, L) \\ &= \frac{nc}{n-2}\tilde{v}(t)^{2/(n-2)} \left(\frac{n-2}{2}\tilde{v}(t) \pm \tilde{v}'(t)\right). \end{aligned}$$

The term $+\partial_\beta v(t, L)$ in the second line comes from differentiating $-\cot\beta\, \partial_\beta v$. Since $\sin^{n/2} L = 1$ and $\partial_\beta(\sin^{n/2}\beta)|_{\beta=L} = 0$, this gives the condition for $z_\pm$ in (4.1).

The even extensions of v and $h_\pm$ near $\beta = 0$ are described in Section 4.1. Taylor's formula for the even extension of v gives

$$\begin{aligned} v(t, \beta) &= v(t, 0) + \frac{\beta^2}{2}\partial_\beta^2 v(t, 0) + O(\beta^4), \\ \partial_t v(t, \beta) &= \partial_t v(t, 0) + \frac{\beta^2}{2}\partial_t \partial_\beta^2 v(t, 0) + O(\beta^4), \\ \partial_\beta v(t, \beta) &= \beta\, \partial_\beta^2 v(t, 0) + \frac{\beta^3}{6}\partial_\beta^4 v(t, 0) + O(\beta^5). \end{aligned}$$

Using $\cot\beta = \beta^{-1} - \beta/3 + O(\beta^3)$ in the last line gives

$$\cot\beta\, \partial_\beta v(t, \beta) = \partial_\beta^2 v(t, 0) + \beta^2 \left(\frac{1}{6}\partial_\beta^4 v(t, 0) - \frac{1}{3}\partial_\beta^2 v(t, 0)\right) + O(\beta^4).$$

Substitution in (4.3) yields

$$\begin{aligned} h_\pm(t, \beta) = &\left[\frac{n-2}{2} v - \partial_\beta^2 v \pm \partial_t v\right](t, 0) \\ &+ \beta^2 \left[\left(\frac{n-2}{4} + \frac{1}{3}\right)\partial_\beta^2 v - \frac{1}{6}\partial_\beta^4 v \pm \frac{1}{2}\partial_t \partial_\beta^2 v\right](t, 0) + O(\beta^4). \end{aligned}$$

More generally, for each integer $k \geq 0$, evenness gives

$$h_{\pm}(t,\beta) = \sum_{j=0}^{k} \frac{\partial_{\beta}^{2j} h_{\pm}(t,0)}{(2j)!}\beta^{2j} + O(\beta^{2k+2}).$$

In geodesic normal coordinates centered at $(0,\dots,0,1)$ on $\mathbb{S}^{n+1}_{+}$, β is the norm of the coordinate vector. Each power β^{2j} is therefore a polynomial in these coordinates. By Taylor's formula and the chain rule, the derivatives of the remainder of order $m \leq 2k+1$ in these coordinates are $O(\beta^{2k+2-m})$. Since k is arbitrary, the radial extension is smooth at the center of the chart, and (4.6) holds there by continuity.

In the integration by parts for (4.10) over $I \times [\delta, L]$, the lower endpoint contributes

$$-\int_{I} \frac{\partial_{\beta}\phi_0(t,\delta)}{\phi_0(t,\delta)} Z(t,\delta)^2 \, \mathrm{d}t, \qquad \phi_0 = \sqrt{z_+ z_-}.$$

For the smooth approximations in the proof of Proposition 4.2, this term vanishes when δ is sufficiently small. For the coefficient Z_1 of a Jacobi field, smoothness at $\beta = 0$ and the zero mean of a degree-1 spherical harmonic give $y_1(t,\beta) = O(\beta)$, as in Section 4.1. Multiplication by $\sin^{(n-2)/2}\beta$ gives $Z_1(t,\beta) = O(\beta^{n/2})$. The estimate is uniform for $t \in [0,T]$ in the periodic case. For $I = \mathbb{R}$, apply the same estimate to Z_1 after multiplication by a fixed smooth cutoff in t.

By (4.3), $\phi_0 = \sin^{n/2}\beta\sqrt{h_+ h_-}$. Differentiation gives

$$\frac{\partial_{\beta}\phi_0}{\phi_0} = \frac{n}{2}\cot\beta + \frac{\partial_{\beta}h_+}{2h_+} + \frac{\partial_{\beta}h_-}{2h_-}.$$

The Taylor expansions above give $\partial_{\beta}h_{\pm}(t,\beta) = O(\beta)$, and (4.4) bounds $h_{\pm}$ below by a positive constant. It follows that $\partial_{\beta}\phi_0/\phi_0 = O(\beta^{-1})$. Thus for $Z = Z_1$ in the periodic case, or for the cutoff of Z_1 when $I = \mathbb{R}$, the endpoint term satisfies

$$\left|\int_{I} \frac{\partial_{\beta}\phi_0(t,\delta)}{\phi_0(t,\delta)} Z(t,\delta)^2 \, \mathrm{d}t\right| \leq C\delta^{n-1} \longrightarrow 0.$$

A.4. **Differentiation of the Hamiltonian.** We record the first variation of the Hamiltonian (2.11) in an arbitrary direction and its second derivative at the cylindrical solution C_n.

Fix t and let v and y be smooth near $\{t\} \times \mathbb{S}^{n-1}_{+}$, with $\tilde{v}(t,\vartheta) > 0$ on $\partial\mathbb{S}^{n-1}_{+}$. By compactness of the boundary section, $\tilde{v} + s\tilde{y}$ remains positive there for sufficiently small $|s|$. Differentiating $s \mapsto H(v+sy)(t)$

at $s = 0$ gives

$$|\mathbb{S}^{n-2}|DH_v[y](t) = \int_{\mathbb{S}^{n-1}_+} \left(v_t y_t - \langle \nabla_{\mathbb{S}^{n-1}_+} v, \nabla_{\mathbb{S}^{n-1}_+} y \rangle - \frac{(n-2)^2}{4} vy \right) \mathrm{d}\theta + c \int_{\partial \mathbb{S}^{n-1}_+} \tilde{v}^{n/(n-2)} \tilde{y} \, \mathrm{d}\sigma.$$

The derivative of the boundary integral has coefficient c because the exponent $2(n-1)/(n-2)$ cancels the factor $(n-2)/(2(n-1))$ in (2.11).

Apply this formula to w from (2.20). Integrating by parts on $\mathbb{S}^{n-1}_+$ gives the boundary term $-\partial_{\nu_{\mathbb{S}^{n-1}_+}} w \, \tilde{y}$. Adding the derivative of the boundary integral and using (2.26) gives $-f_{\partial,g}\tilde{y}$. In the interior, the same equation gives $\Delta_{\mathbb{S}^{n-1}_+} w - (n-2)^2 w/4 = f_g - w_{tt}$. Thus

$$|\mathbb{S}^{n-2}|DH_w[y](t) = \int_{\mathbb{S}^{n-1}_+} (w_t y_t - w_{tt} y) \, \mathrm{d}\theta + \int_{\mathbb{S}^{n-1}_+} f_g y \, \mathrm{d}\theta - \int_{\partial \mathbb{S}^{n-1}_+} f_{\partial,g} \tilde{y} \, \mathrm{d}\sigma.$$

Taking $y = w_t$ recovers the derivative of $H(w)(t)$ used to prove (2.29). When the two errors vanish, the identity reduces to (4.13).

At C_n, the derivative vanishes by (4.13) and $\partial_t C_n = 0$. Since C_n has trace 1, differentiating twice in the directions y_1 and y_2 gives

$$|\mathbb{S}^{n-2}|D^2 H_{C_n}[y_1, y_2](t) = \int_{\mathbb{S}^{n-1}_+} \left(\partial_t y_1 \, \partial_t y_2 - \langle \nabla_{\mathbb{S}^{n-1}_+} y_1, \nabla_{\mathbb{S}^{n-1}_+} y_2 \rangle - \frac{(n-2)^2}{4} y_1 y_2 \right) \mathrm{d}\theta + \frac{nc}{n-2} \int_{\partial \mathbb{S}^{n-1}_+} \tilde{y}_1 \tilde{y}_2 \, \mathrm{d}\sigma.$$

For $y_1 = y_2 = y$, division by 2 gives the quadratic term $\mathcal{Q}_c[y](t)$ in (4.22).

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